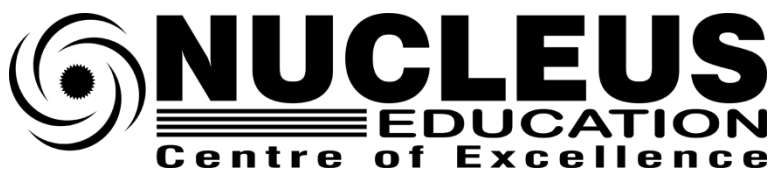


IIT MATHEMATICS

APPLICATION OF DERIVATIVE (SHEET & ADDITIONAL EXERCISE)



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RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
Gen. Category - 2014

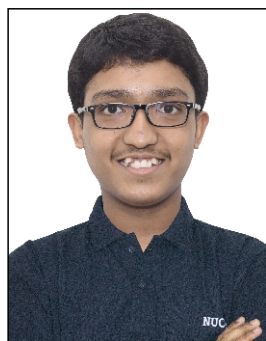


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NISHIT AGARWAL
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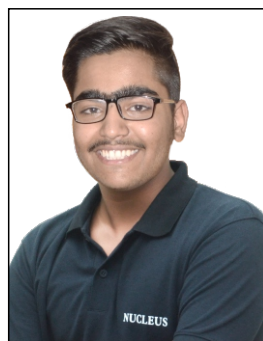


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AMEY GUPTA
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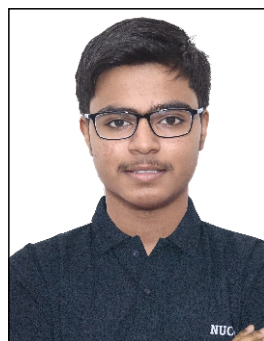
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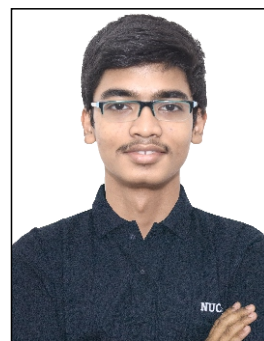
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(Physics)
DAKSH Khandelwal
2020



100 Percentile
(Maths & Physics)
VAIBHAV SAHA
2020



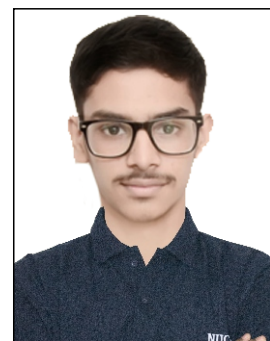
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ANISH MOHAN
2020



100 Percentile
(Maths)
ARCHIT PATNAIK
2020

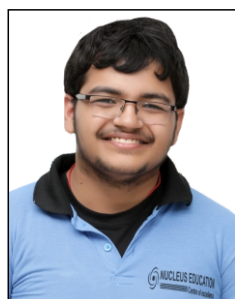


100 Percentile
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2020

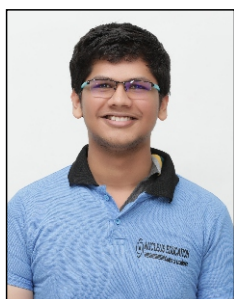


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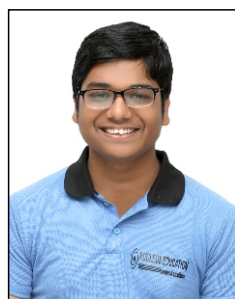
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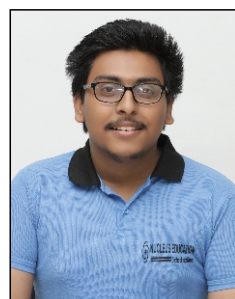
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LAKSHAY SHARMA
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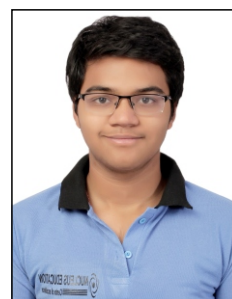
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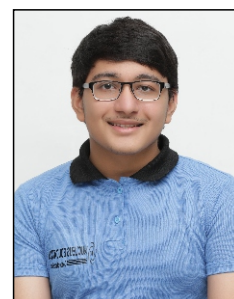
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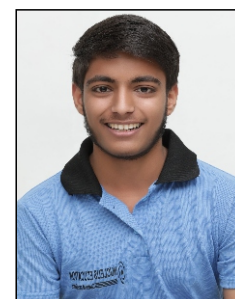
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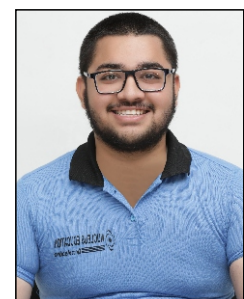
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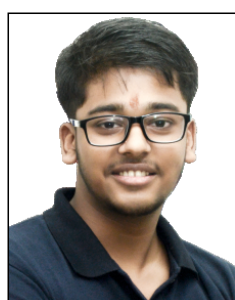
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SATVIK
Gen. Category - 2017



AIR-66
MAYANK DUBEY
Gen. Category - 2017



AIR-98
HRITHIK
Gen. Category - 2017



AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
Gen. Category - 2018



AIR-61
SIDDHANT CHOUDHARY
Gen. Category - 2018



AIR-67
ANISWAR S KRISHNAN
Gen. Category - 2018
DLP



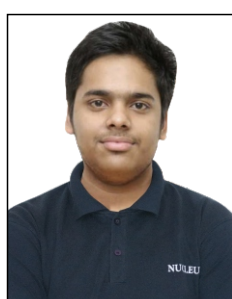
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AAYUSH KADAM
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AIR-61
SARTHAK BEHERA
Gen. Category - 2018
*SDCCP



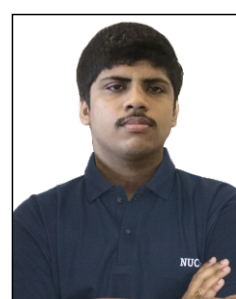
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ANDREWS G. VARGHESE
Gen. Category - 2018
*SDCCP



AIR-2
HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



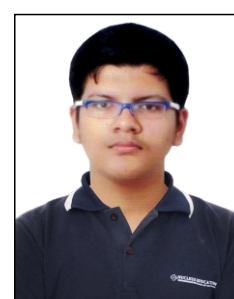
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AIR-33
S. PRAJEETH
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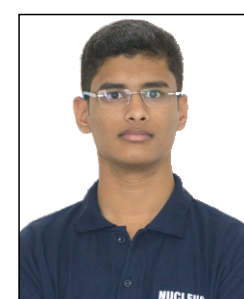
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AIR-51
SAYANTAN DHAR
Gen. Category - 2019
DLP



AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
DLP



AIR-86
SATVIK JAIN
Gen. Category - 2019

RATE MEASURE, TANGENT & NORMAL

1. RATE MEASUREMENT :

Whenever one quantity y varies with another quantity x , satisfying some rule $y = f(x)$, then $\frac{dy}{dx}$ (or $f'(x)$) represents the rate of change of y with respect to x and $\left. \frac{dy}{dx} \right|_{x=a}$ (or $f'(a)$) represents the rate of change of y with respect to x at $x = a$.

Illustration 1 : The volume of a cube is increasing at a rate of $9\text{cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 10cm ?

Solution : Let x be the length of side, V be the volume and S be the surface area of the cube. Then $V = x^3$ and $S = 6x^2$, where x is a function of time t .

$$\frac{dV}{dt} = 9\text{cm}^3/\text{s} = \frac{d}{dt}(x^3) = 3x^2 \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{3}{x^2}$$

$$\frac{dS}{dt} = \frac{d}{dt}(6x^2) = 12x \left(\frac{3}{x^2} \right) = \frac{36}{x}$$

$$\left. \frac{dS}{dt} \right|_{x=10\text{cm}} = 3.6 \text{ cm}^2/\text{s}.$$

Illustration 2 : x and y are the sides of two squares such that $y = x - x^2$. Find the rate of change of the area of the second square with respect to the first square.

Solution : Given x and y are sides of two squares. Thus the area of two squares are x^2 and y^2

$$\text{We have to obtain } \frac{d(y^2)}{d(x^2)} = \frac{2y \frac{dy}{dx}}{2x} = \frac{y}{x} \cdot \frac{dy}{dx} \quad \dots\dots\dots (i)$$

$$\text{where the given curve is, } y = x - x^2 \Rightarrow \frac{dy}{dx} = 1 - 2x \quad \dots\dots\dots (ii)$$

$$\text{Thus, } \frac{d(y^2)}{d(x^2)} = \frac{y}{x}(1 - 2x) \quad [\text{from (i) and (ii)}]$$

$$\text{or } \frac{d(y^2)}{d(x^2)} = \frac{(x - x^2)(1 - 2x)}{x} \Rightarrow \frac{d(y^2)}{d(x^2)} = (2x^2 - 3x + 1)$$

The rate of change of the area of second square with respect to first square is $(2x^2 - 3x + 1)$

Do yourself - 1 :

- (i) What is the rate of change of the area of a circle with respect to its radius r at $r = 6\text{cm}$.
- (ii) A stone is dropped into a quiet lake and waves move in circles at the speed of 5cm/s . At the instant when the radius of the circular wave is 8cm , how fast is the enclosed area increasing?

2. APPROXIMATION USING DIFFERENTIALS :

In order to calculate the approximate value of a function, differentials may be used where the differential of a function is equal to its derivative multiplied by the differential of the independent variable. In general $dy = f'(x)dx$ or $df(x) = f'(x)dx$

Note :

- (i) For the independent variable 'x', increment Δx and differential dx are equal but this is not the case with the dependent variable 'y' i.e. $\Delta y \neq dy$.

$\therefore$ Approximate value of y when increment Δx is given to independent variable x in $y = f(x)$ is

$$y + \Delta y = f(x + \Delta x) = f(x) + \frac{dy}{dx} \cdot \Delta x$$

- (ii) The relation $dy = f'(x) dx$ can be written as $\frac{dy}{dx} = f'(x)$; thus the quotient of the differentials of 'y' and 'x' is equal to the derivative of 'y' w.r.t. 'x'.

Illustration 3 : Find the approximate value of square root of 25.2.

Solution : Let $f(x) = \sqrt{x}$

$$\text{Now, } f(x + \Delta x) - f(x) = f'(x) \cdot \Delta x = \frac{\Delta x}{2\sqrt{x}}$$

we may write, $25.2 = 25 + 0.2$

Taking $x = 25$ and $\Delta x = 0.2$, we have

$$f(25.2) - f(25) = \frac{0.2}{2\sqrt{25}}$$

$$\text{or } f(25.2) - \sqrt{25} = \frac{0.2}{10} = 0.02 \Rightarrow f(25.2) = 5.02$$

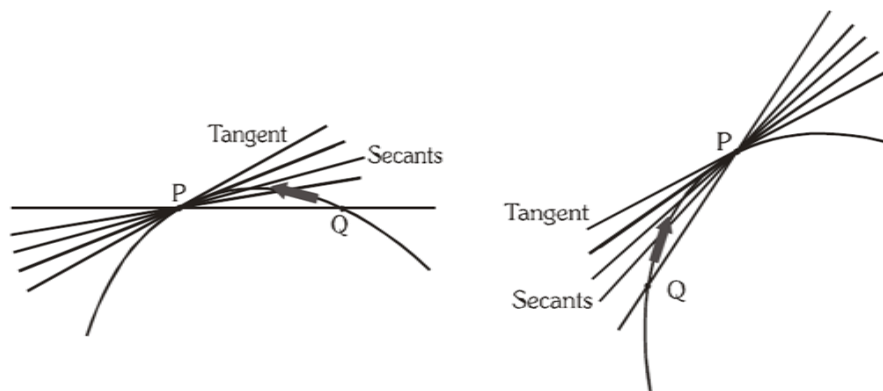
$$\text{or } \sqrt{(25.2)} = 5.02$$

Do yourself - 2 :

- (i) Find the approximate value of $(0.009)^{1/3}$.

3. TANGENT TO THE CURVE AT A POINT :

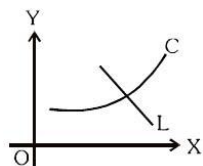
The tangent to the curve at 'P' is the line through P whose slope is limit of the secant slopes as $Q \rightarrow P$ from either side.



4. MYTHS ABOUT TANGENT :

- (a) **Myth :** A line meeting the curve only at one point is a tangent to the curve.

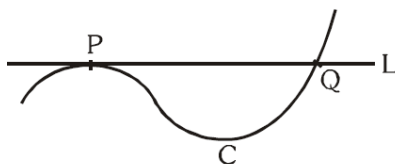
Explanation : A line meeting the curve in one point is not necessarily tangent to it.



Here L is not tangent to C

- (b) **Myth :** A line meeting the curve at more than one point is not a tangent to the curve.

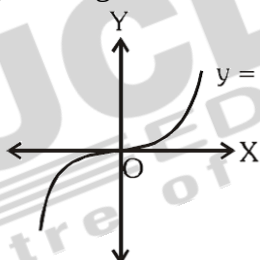
Explanation : A line may meet the curve at several points and may still be tangent to it at some point



Here L is tangent to C at P, and cutting it again at Q.

- (c) **Myth :** Tangent at a point to the curve can not cross it at the same point.

Explanation : A line may be tangent to the curve and also cross it.



Here X-axis is tangent to $y = x^3$ at origin.

5. NORMAL TO THE CURVE AT A POINT :

A line which is perpendicular to the tangent at the point of contact is called normal to the curve at that point.

6. POINTS TO REMEMBER :

- (a) The value of the derivative at $P(x_1, y_1)$ gives the slope of the tangent to the curve at P. Symbolically

$$f'(x_1) = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \text{Slope of tangent at } P(x_1, y_1) = m(\text{say}).$$

- (b) Equation of tangent at (x_1, y_1) is ; $y - y_1 = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} (x - x_1)$

- (c) Equation of normal at (x_1, y_1) is ; $y - y_1 = \left. \frac{-1}{\frac{dy}{dx}} \right|_{(x_1, y_1)} (x - x_1).$

Note :

- (i) The point P (x_1, y_1) will satisfy the equation of the curve & the equation of tangent & normal line.
- (ii) If the tangent at any point P on the curve is parallel to the axis of x then $dy/dx = 0$ at the point P.
- (iii) If the tangent at any point on the curve is parallel to the axis of y, then dy/dx not defined or $dx/dy = 0$.
- (iv) If the tangent at any point on the curve is equally inclined to both the axes then $dy/dx = \pm 1$.
- (v) If a curve passing through the origin be given by a rational integral algebraic equation, then the equation of the tangent (or tangents) at the origin is obtained by equating to zero the terms of the lowest degree in the equation. e.g. If the equation of a curve be $x^2 - y^2 + x^3 + 3x^2y - y^3 = 0$, the tangents at the origin are given by $x^2 - y^2 = 0$ i.e. $x + y = 0$ and $x - y = 0$

Illustration 4 : Find the equation of the tangent to the curve $y = (x^3 - 1)(x - 2)$ at the points where the curve cuts the x-axis.

Solution : The equation of the curve is $y = (x^3 - 1)(x - 2)$ (i)
It cuts x-axis at $y = 0$. So, putting $y = 0$ in (i), we get $(x^3 - 1)(x - 2) = 0$
 $\Rightarrow (x - 1)(x - 2)(x^2 + x + 1) = 0 \Rightarrow x - 1 = 0, x - 2 = 0 [\because x^2 + x + 1 \neq 0]$
 $\Rightarrow x = 1, 2$.
Thus, the points of intersection of curve (i) with x-axis are (1, 0) and (2, 0). Now,
 $y = (x^3 - 1)(x - 2) \Rightarrow \frac{dy}{dx} = 3x^2(x - 2) + (x^3 - 1) \Rightarrow \left(\frac{dy}{dx}\right)_{(1,0)} = -3$ and $\left(\frac{dy}{dx}\right)_{(2,0)} = 7$
The equations of the tangents at (1, 0) and (2, 0) are respectively
 $y - 0 = -3(x - 1)$ and $y - 0 = 7(x - 2) \Rightarrow y + 3x - 3 = 0$ and $7x - y - 14 = 0$ **Ans.**

Illustration 5 : The equation of the normal to the curve $y = x + \sin x \cos x$ at $x = \frac{\pi}{2}$ is -

- (A) $x = 2$ (B) $x = \pi$ (C) $x + \pi = 0$ (D) $2x = \pi$

Solution : $\because x = \frac{\pi}{2} \Rightarrow y = \frac{\pi}{2} + 0 = \frac{\pi}{2}$, so the given point = $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

Now from the given equation $\frac{dy}{dx} = 1 + \cos^2 x - \sin^2 x \Rightarrow \left(\frac{dy}{dx}\right)_{\left(\frac{\pi}{2}, \frac{\pi}{2}\right)} = 1 + 0 - 1 = 0$

$\Rightarrow$ The curve has vertical normal at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$.

the equation to this normal is $x = \frac{\pi}{2}$

$$x - \frac{\pi}{2} = 0 \Rightarrow 2x = \pi$$

Ans. (D)

Illustration 6 : The equation of normal to the curve $x + y = x^y$, where it cuts x - axis is –
(A) $y = x + 1$ (B) $y = -x + 1$ (C) $y = x - 1$ (D) $y = -x - 1$

Solution : Given curve is $x + y = x^y$ (i)

at x -axis $y = 0$,

$$\therefore x + 0 = x^0 \Rightarrow x = 1$$

$\therefore$ Point is A (1, 0)

Now to differentiate $x + y = x^y$ take log on both sides

$$\Rightarrow \log(x + y) = y \log x \quad \therefore \frac{1}{x + y} \left\{ 1 + \frac{dy}{dx} \right\} = y \cdot \frac{1}{x} + (\log x) \frac{dy}{dx}$$

$$\text{Putting } x = 1, y = 0 \left\{ 1 + \frac{dy}{dx} \right\} = 0 \Rightarrow \left(\frac{dy}{dx} \right)_{(1,0)} = -1$$

$\therefore$ slope of normal = 1

$$\text{Equation of normal is, } \frac{y - 0}{x - 1} = 1 \Rightarrow y = x - 1$$

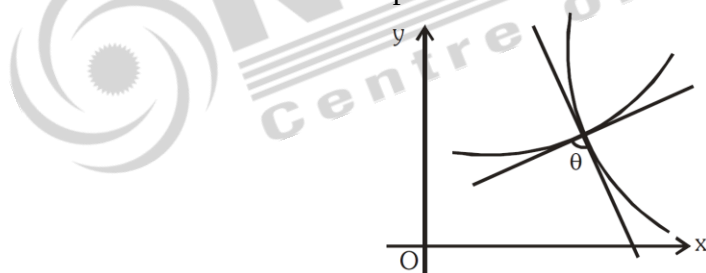
Ans.(C)

Do yourself - 3 :

- (i) Find the distance between the point (1, 1) and the tangent to the curve $y = e^{2x} + x^2$ drawn at the point where the curve cuts y -axis.
(ii) Find the equation of a line passing through (-2, 3) and parallel to tangent at origin for the circle $x^2 + y^2 + x - y = 0$.

7. ANGLE OF INTERSECTION BETWEEN TWO CURVES :

Angle of intersection between two curves is defined as the angle between the two tangents drawn to the two curves at their point of intersection.



Orthogonal curves :

If the angle between two curves at each point of intersection is 90° then they are called **orthogonal curves**.

For example, the curves $x^2 + y^2 = r^2$ & $y = mx$ are orthogonal curves.

Illustration 7 : The angle of intersection between the curve $x^2 = 32y$ and $y^2 = 4x$ at point (16, 8) is-

- (A) 60° (B) 90° (C) $\tan^{-1}\left(\frac{3}{5}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

Solution : $x^2 = 32y \Rightarrow \frac{dy}{dx} = \frac{x}{16} \Rightarrow y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y}$

$$\therefore \text{at } (16, 8), \left(\frac{dy}{dx} \right)_1 = 1, \left(\frac{dy}{dx} \right)_2 = \frac{1}{4}$$

$$\text{So required angle} = \tan^{-1} \left(\frac{1 - \frac{1}{4}}{1 + 1 \left(\frac{1}{4} \right)} \right) = \tan^{-1} \left(\frac{3}{5} \right)$$

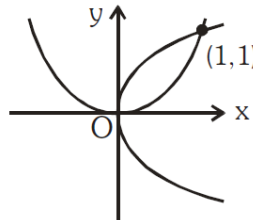
Ans. (C)

Illustration 8 : Check the orthogonality of the curves $y^2 = x$ & $x^2 = y$.

Solution : Solving the curves simultaneously we get points of intersection as (1, 1) and (0, 0).

$$\text{At (1,1) for first curve } 2y \left(\frac{dy}{dx} \right)_1 = 1 \Rightarrow m_1 = \frac{1}{2}$$

$$\text{and for second curve } 2x = \left(\frac{dy}{dx} \right)_2 \Rightarrow m_2 = 2$$



$m_1 m_2 \neq -1$ at (1, 1).

But at (0, 0) clearly x-axis and y-axis are their respective tangents hence they are orthogonal at (0,0) but not at (1, 1). Hence these curves are not said to be orthogonal.

Illustration 9 : If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is -

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 2 (D) 3

Solution :

$$\begin{aligned} y &= 1 - ax^2 \\ \Rightarrow \frac{dy}{dx} &= -2ax \\ y &= x^2 \\ \Rightarrow \frac{dy}{dx} &= 2x \end{aligned}$$

Two curves intersect orthogonally if $\left(\frac{dy}{dx} \right)_1 \left(\frac{dy}{dx} \right)_2 = -1$

$$\begin{aligned} \Rightarrow (-2ax)(2x) &= -1 \\ \Rightarrow 4ax^2 &= 1 \end{aligned} \quad \dots (i)$$

Now eliminating y from the given equations we have $1 - ax^2 = x^2$

$$\Rightarrow (1 + a)x^2 = 1 \quad \dots (ii)$$

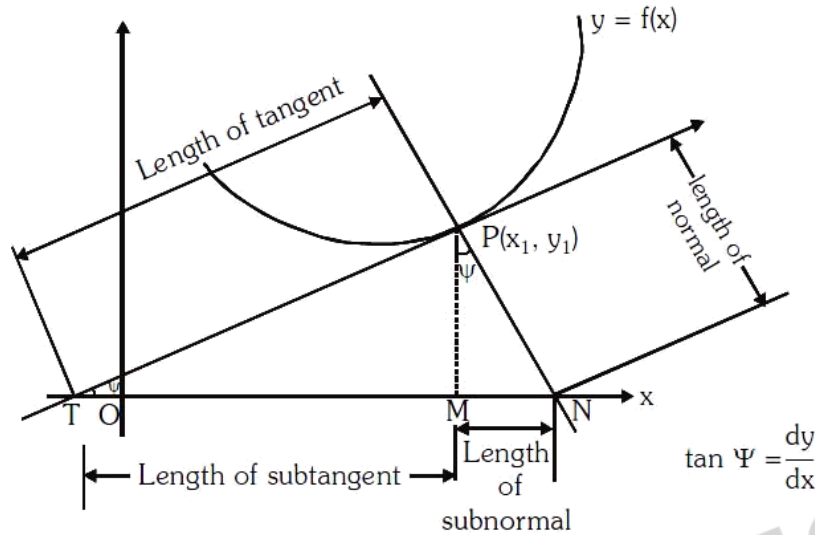
$$\text{Eliminating } x^2 \text{ from (i) and (ii) we get } \frac{4a}{1+a} = 1 \Rightarrow a = \frac{1}{3}$$

Ans. (B)

Do yourself -4 :

- (i) If two curves $y = a^x$ and $y = b^x$ intersect at an angle α , then find the value of $\tan \alpha$.
(ii) Find the angle of intersection of curves $y = 4 - x^2$ and $y = x^2$.

8. LENGTH OF TANGENT, SUBTANGENT, NORMAL AND SUBNORMAL :



(a) Length of the tangent (PT) = $\left| \frac{y_1 \sqrt{1 + [f'(x_1)]^2}}{f'(x_1)} \right|$

(b) Length of Sub tangent (MT) = $\left| \frac{y_1}{f'(x_1)} \right|$

(c) Length of Normal (PN) = $\left| y_1 \sqrt{1 + [f'(x_1)]^2} \right|$

(d) Length of Subnormal (MN) = $|y_1 f'(x_1)|$

Illustration 10 : The length of the normal to the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ at $\theta = \frac{\pi}{2}$ is –

- (A) $2a$ (B) $\frac{a}{2}$ (C) $\sqrt{2}a$ (D) $\frac{a}{\sqrt{2}}$

Solution : $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \tan \frac{\theta}{2} \Rightarrow \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}} = \tan\left(\frac{\pi}{4}\right) = 1$

Also at $\theta = \frac{\pi}{2}$, $y = a \left(1 - \cos \frac{\pi}{2}\right) = a$

$\therefore$ required length of normal = $y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = a \sqrt{1+1} = \sqrt{2}a$ **Ans.(C)**

Illustration 11: The length of the tangent to the curve $x = a \left(\cos t + \log \tan \frac{t}{2} \right)$, $y = a \sin t$ is

- (A) ax (B) ay (C) a (D) xy

Solution : $\frac{dy}{dx} = \left(\frac{dy}{dt} \right) / \left(\frac{dx}{dt} \right) = \frac{a \cos t}{a \left(-\sin t + \frac{1}{\sin t} \right)} = \tan t$

$$\therefore \text{length of the tangent} = y \frac{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}}{\left(\frac{dy}{dx} \right)} = a \sin t \frac{\sqrt{1 + \tan^2 t}}{\tan t} = a \sin t \left(\frac{\sec t}{\tan t} \right) = a$$

Ans.(C)

Do yourself -5 :

- (i) Prove that at any point of a curve, the product of the length of sub tangent and the length of subnormal is equal to square of the ordinates of point of contact.
- (ii) Find the length of subtangent to the curve $x^2 + y^2 + xy = 7$ at the point $(1, -3)$.

Miscellaneous Illustrations :

Illustration 12 : Find the slope of normal at the point with abscissa $x = -2$ of the graph of the function

$$f(x) = |x^2 - |x||$$

Solution : At $x = -2$, $f(x)$ becomes

$$f(x) = x^2 + x$$

$$\frac{dy}{dx} = 2x + 1 = -3$$

$$\text{Slope of normal} = \frac{1}{3}$$

Illustration 13 : If $y = 4x - 5$ is a tangent to the curve $y^2 = px^3 + q$ at $(2, 3)$ then

$$(A) p = 2, q = -7 \quad (B) p = -2, q = 7 \quad (C) p = -2, q = -7 \quad (D) p = 2, q = 7$$

Solution : $\frac{dy}{dx} = 4$ and $9 = 8p + q$

$$2y \frac{dy}{dx} = 3px^2$$

$$6 \frac{dy}{dx} = 3p(4) \Rightarrow \frac{dy}{dx} = 2p = 4 \Rightarrow p = 2 \text{ and } q = -7$$

ANSWERS FOR DO YOURSELF

- | | | | | |
|-----|------|--|------|--|
| 1 : | (i) | $12\pi \text{ cm}$ | (ii) | $80 \pi \text{ cm}^2/\text{s}$ |
| 2 : | (i) | 0.208 | | |
| 3 : | (i) | $\frac{2}{\sqrt{5}} \text{ units}$ | (ii) | $x - y + 5 = 0$ |
| 4 : | (i) | $\left \frac{\ell na - \ell nb}{1 + \ell na \ell nb} \right $ | (ii) | $\tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$ |
| 5 : | (ii) | 15 | | |

TANGENT & NORMAL

EXERCISE # 1

- If an equation of a tangent to the curve, $y = \cos(x + y)$, $-1 \leq x \leq 1 + \pi$, is $x + 2y = k$ then k is equal to: **[JEE-MAIN Online 2013]**
 (A) $\frac{\pi}{4}$ (B) 2 (C) $\frac{\pi}{2}$ (D) 1
- The slope of the curve $y = \sin x + \cos^2 x$ is zero at the point, where-
 (A) $x = \frac{\pi}{4}$ (B) $x = \frac{\pi}{2}$ (C) $x = \pi$ (D) No where
- The equation of tangent at the point (at^2, at^3) on the curve $ay^2 = x^3$ is-
 (A) $3tx - 2y = at^3$ (B) $tx - 3y = at^3$ (C) $3tx + 2y = at^3$ (D) None of these
- The equation of normal to the curve $y = x^3 - 2x^2 + 4$ at the point $x = 2$ is-
 (A) $x + 4y = 0$ (B) $4x - y = 0$ (C) $x + 4y = 18$ (D) $4x - y = 18$
- The equation of tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point (x_1, y_1) is-
 (A) $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \frac{1}{\sqrt{a}}$ (B) $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{a}$
 (C) $x\sqrt{x_1} + y\sqrt{y_1} = \sqrt{a}$ (D) None of these
- The slope of the normal to the curve $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ at point $\theta = \pi/2$ is-
 (A) 0 (B) 1 (C) -1 (D) $1/\sqrt{2}$
- The angle between the tangent lines to the graph of the function $f(x) = \int_2^x (2t - 5) dt$ at the points where the graph cuts the x-axis is
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
- The coordinates of the points on the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, where tangent is inclined at an angle $\pi/4$ to the x-axis are-
 (A) (a, a) (B) $\left(a\left(\frac{\pi}{2} - 1\right), a\right)$ (C) $\left(a\left(\frac{\pi}{2} + 1\right), a\right)$ (D) $\left(a, a\left(\frac{\pi}{2} + 1\right)\right)$

9. Consider the curve represented parametrically by the equation $x = t^3 - 4t^2 - 3t$ and $y = 2t^2 + 3t - 5$ where $t \in \mathbb{R}$.
If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical then
(A) $H = 2$ and $V = 1$ (B) $H = 1$ and $V = 2$
(C) $H = 2$ and $V = 2$ (D) $H = 1$ and $V = 1$
10. The line $x/a + y/b = 1$ touches the curve $y = be^{-x/a}$ at the point-
(A) $(0, a)$ (B) $(0, 0)$ (C) $(0, b)$ (D) $(b, 0)$
11. If the tangent to the curve $2y^3 = ax^2 + x^3$ at a point (a, a) cuts off intercepts p and q on the coordinates axes, where $p^2 + q^2 = 61$, then a equals-
(A) 30 (B) -30 (C) 0 (D) ± 30
12. The sum of the intercepts made by a tangent to the curve $\sqrt{x} + \sqrt{y} = 4$ at point $(4, 4)$ on coordinate axes is-
(A) $4\sqrt{2}$ (B) $6\sqrt{3}$ (C) $8\sqrt{2}$ (D) $\sqrt{256}$
13. The curve $x^2 - y^2 = 5$ and $\frac{x^2}{18} + \frac{y^2}{8} = 1$ cut each other at any common point at an angle-
(A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) None of these
14. The angle of intersection of curves $2y = x^3$ and $y^2 = 32x$ at the origin is-
(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/2$ (D) None of these
15. The lines tangent to the curve $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to-
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
16. The value of n for which the area of the triangle included between the co-ordinate axes and any tangent to the curve $xy^n = a^{n+1}$ is constant, is
(A) -1 (B) 0 (C) 1 (D) a
17. The angle of intersection between the curves $y = 4x^2$ and $y = x^2$ is-
(A) 0° (B) 30° (C) 45° (D) 90°

18. The angle of intersection between the curves $y^2 = 8x$ and $x^2 = 4y - 12$ at $(2, 4)$ is-
 (A) 90° (B) 60° (C) 45° (D) 0°
19. The length of subtangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, -3)$ is-
 (A) 3 (B) 5 (C) 15 (D) $3/5$
20. For a curve $\frac{(\text{length of normal})^2}{(\text{length of tangent})^2}$ is equal to -
 (A) (subnormal)/(subtangent) (B) (subtangent)/(subnormal)
 (C) (subtangent $\times$ subnormal) (D) constant
21. At any point of a curve (subtangent) $\times$ (subnormal) is equal to the square of the-
 (A) slope of the tangent at that point
 (B) slope of the normal at that point
 (C) abscissa of that point
 (D) ordinate of that point
22. If length of subnormal at any point of the curve $y^n = a^{n-1}x$ is constant, then n equals-
 (A) 1 (B) 3 (C) 2 (D) 0
23. Let S be a square with sides of length x . If we approximate the change in size of the area of S by $h \cdot \left. \frac{dA}{dx} \right|_{x=x_0}$, when the sides are changed from x_0 to $x_0 + h$, then the absolute value of the error in our approximation, is
 (A) h^2 (B) $2hx_0$ (C) x_0^2 (D) h
24. A Spherical balloon is being inflated at the rate of 35cc/min. The rate of increase in the surface area (in $\text{cm}^2/\text{min.}$) of the balloon when its diameter is 14 cm, is : **[JEE-MAIN Online 2013]**
 (A) $\sqrt{10}$ (B) $10\sqrt{10}$ (C) 100 (D) 10
25. If the surface area of a sphere of radius r is increasing uniformly at the rate $8\text{cm}^2/\text{s}$, then the rate of change of its volume is : **[JEE-MAIN Online 2013]**
 (A) proportional to r^2 (B) constant
 (C) proportional to r (D) proportional to $\sqrt{r}$

RATE MEASURE & TANGENT NORMAL

EXERCISE # 2

- Find the equation of the normal to the curve $y = (1 + x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$.
- Find all the lines that pass through the point $(1, 1)$ and are tangent to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$.
- The tangent to $y = ax^2 + bx + \frac{7}{2}$ at $(1, 2)$ is parallel to the normal at the point $(-2, 2)$ on the curve $y = x^2 + 6x + 10$. Find the value of a and b .
- A line is tangent to the curve $f(x) = \frac{41x^3}{3}$ at the point P in the first quadrant, and has a slope of 2009. This line intersects the y -axis at $(0, b)$. Find the value of ' b '.
- Find all the tangents to the curve $y = \cos(x + y)$, $-2\pi \leq x \leq 2\pi$, that are parallel to the line $x + 2y = 0$.
- The curve $y = ax^3 + bx^2 + cx + 5$, touches the x -axis at $P(-2, 0)$ & cuts the y -axis at a point Q where its gradient is 3. Find a, b, c .
- Find the gradient of the line passing through the point $(2, 8)$ and touching the curve $y = x^3$.
- The graph of a certain function f contains the point $(0, 2)$ and has the property that for each number ' p ' the line tangent to $y = f(x)$ at $(p, f(p))$ intersect the x -axis at $p + 2$. Find $f(x)$.
- Find the value of n so that the subnormal at any point on the curve $xy^n = a^{n+1}$ may be constant.
 - Show that in the curve $y = a \ln(x^2 - a^2)$, sum of the length of tangent & subtangent varies as the product of the coordinates of the point of contact.
- Show that the curves $\frac{x^2}{a^2 + K_1} + \frac{y^2}{b^2 + K_1} = 1$ and $\frac{x^2}{a^2 + K_2} + \frac{y^2}{b^2 + K_2} = 1$ intersect orthogonally.
 - If the two curves $C_1: x = y^2$ and $C_2: xy = k$ cut at right angles find the value of k .
- Show that the angle between the tangent at any point ' A ' of the curve $\ln(x^2 + y^2) = C \tan^{-1} \frac{y}{x}$ and the line joining A to the origin is independent of the position of A on the curve.
- Water is being poured on to a cylindrical vessel at the rate of $1 \text{ m}^3/\text{min}$. If the vessel has a circular base of radius 3 m, find the rate at which the level of water is rising in the vessel.
- A man 1.5 m tall walks away from a lamp post 4.5 m high at the rate of 4 km/hr.
 - how fast is the farther end of the shadow moving on the pavement ?
 - how fast is his shadow lengthening ?
- A particle moves along the curve $6y = x^3 + 2$. Find the points on the curve at which the y coordinate is changing 8 times as fast as the x coordinate.

15. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of $1.5 \text{ cm}^3/\text{min}$. Find the rate at which level of water in the cone is rising, when the depth of water is 4 cm.
16. Water is dripping out from a conical funnel of semi vertical angle $\pi/4$, at the uniform rate of $2 \text{ cm}^3/\text{sec}$ through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, find the rate of decrease of the slant height of the water.
17. An air force plane is ascending vertically at the rate of 100 km/h. If the radius of the earth is R Km, how fast the area of the earth, visible from the plane increasing at 3 min after it started ascending. Take visible area $A = \frac{2\pi R^2 h}{R + h}$ Where h is the height of the plane in kms above the earth.
18. If in a triangle ABC, the side 'c' and the angle 'C' remain constant, while the remaining elements are changed slightly, show that $\frac{da}{\cos A} + \frac{db}{\cos B} = 0$.
19. (i) Use differentials to approximate the values of ; (a) $\sqrt{36.6}$ and (b) $\sqrt[3]{26}$.
(ii) If the radius of a sphere is measured as 9 cm with an error of 0.03 cm, then find the approximate error in calculating its volume.
20. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of 1 cu. cm/sec. Water is poured into the tank at a constant rate of C cu. cm/sec. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep.
21. Sand is pouring from a pipe at the rate of 12 cc/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always 1/6th of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm.
22. A circular ink blot grows at the rate of 2 cm^2 per second. Find the rate at which the radius is increasing after $2\frac{6}{11}$ seconds. Use $\pi = \frac{22}{7}$.
23. Water is flowing out at the rate of $6 \text{ m}^3/\text{min}$ from a reservoir shaped like a hemispherical bowl of radius $R = 13 \text{ m}$. The volume of water in the hemispherical bowl is given by $V = \frac{\pi}{3} y^2 (3R - y)$ when the water is y meter deep. Find
(a) At what rate is the water level changing when the water is 8 m deep.
(b) At what rate is the radius of the water surface changing when the water is 8 m deep.
24. At time $t > 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 unit and at $t = 15$ the radius is 2 units.
(a) Find the radius of the sphere as a function of time t.
(b) At what time t will the volume of the sphere be 27 times its volume at $t = 0$.

EXERCISE # 3 (JM)

- The normal to the curve $x = a(1 + \cos\theta)$, $y = a\sin\theta$ at ' θ ' always passes through the fixed point-
(1) $(a, 0)$ (2) $(0, a)$ (3) $(0, 0)$ (4) (a, a) [AIEEE-2004]
- The normal to the curve $x = a(\cos\theta + \theta\sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ at any point θ is such that-
(1) it passes through the origin (2) it makes angle $\left(\frac{\pi}{2} + \theta\right)$ with the x-axis
(3) it passes through $\left(a\frac{\pi}{2}, -a\right)$ (4) it is a constant distance from the origin
- Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points $(2, 0)$ and $(3, 0)$ is-
(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{3}$ [AIEEE-2006]
- The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x-axis, is :-
(1) $y = 0$ (2) $y = 1$ (3) $y = 2$ (4) $y = 3$ [AIEEE-2010]
- The intercepts on x-axis made by tangents to the curve, $y = \int_0^x |t| dt$, $x \in \mathbb{R}$, which are parallel to the line $y = 2x$, are equal to [JEE-MAIN 2013]
(1) ± 1 (2) ± 2 (3) ± 3 (4) ± 4
- The normal to the curve, $x^2 + 2xy - 3y^2 = 0$, at $(1, 1)$: [JEE-MAIN 2015]
(1) meets the curve again in the third quadrant
(2) meets the curve again in the fourth quadrant
(3) does not meet the curve again
(4) meets the curve again in the second quadrant
- A wire of length 2 units is cut into two parts which are bent respectively to form a square of side = x units and a circle of radius = r units. If the sum of the areas of the square and the circle so formed is minimum, then : [JEE-MAIN 2016]
(1) $2x = r$ (2) $2x = (\pi + 4)r$ (3) $(4 - \pi)x = \pi r$ (4) $x = 2r$
- Consider $f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $x \in \left(0, \frac{\pi}{2}\right)$, A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point : [JEE-MAIN 2016]
(1) $\left(\frac{\pi}{4}, 0\right)$ (2) $(0, 0)$ (3) $\left(0, \frac{2\pi}{3}\right)$ (4) $\left(\frac{\pi}{6}, 0\right)$
- The normal to the curve $y(x - 2)(x - 3) = x + 6$ at the point where the curve intersects the y-axis passes through the point : [JEE-MAIN 2017]
(1) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (2) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ (3) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (4) $\left(\frac{1}{2}, -\frac{1}{3}\right)$

10. The tangent to the curve, $y = xe^{x^2}$ passing through the point (1, e) also passes through the point: [JEE-MAIN 2019]
 (1) (2, 3e) (2) (3, 6e) (3) $\left(\frac{4}{3}, 2e\right)$ (4) $\left(\frac{5}{3}, 2e\right)$
11. A helicopter is flying along the curve given by $y - x^{3/2} = 7$, ($x \geq 0$). A soldier positioned at the point $\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is: [JEE-MAIN 2019]
 (1) $\frac{\sqrt{5}}{6}$ (2) $\frac{1}{3}\sqrt{\frac{7}{3}}$ (3) $\frac{1}{2}$ (4) $\frac{1}{6}\sqrt{\frac{7}{3}}$
12. The tangent to the curve $y = x^2 - 5x + 5$, parallel to the line $2y = 4x + 1$, also passes through the point: [JEE-MAIN 2019]
 (1) $\left(\frac{1}{8}, -7\right)$ (2) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (3) $\left(-\frac{1}{8}, 7\right)$ (4) $\left(\frac{1}{4}, \frac{7}{2}\right)$
13. The shortest distance between the line $y = x$ and the curve $y^2 = x - 2$ is [JEE-MAIN 2019]
 (1) $\frac{11}{4\sqrt{2}}$ (2) $\frac{7}{4\sqrt{2}}$ (3) 2 (4) $\frac{7}{8}$
14. Let S be the set of all values of x for which the tangent to the curve $y = f(x) = x^3 - x^2 - 2x$ at (x, y) is parallel to the line segment joining the points (1, f(1)) and (-1, f(-1)), then S is equal to: [JEE-MAIN 2019]
 (1) $\left\{-\frac{1}{3}, -1\right\}$ (2) $\left\{-\frac{1}{3}, 1\right\}$ (3) $\left\{\frac{1}{3}, -1\right\}$ (4) $\left\{\frac{1}{3}, 1\right\}$
15. If the tangent to the curve, $y = x^3 + ax - b$ at the point (1, -5) is perpendicular to the line, $-x + y + 4 = 0$, then which one of the following points lies on the curve? [JEE-MAIN 2019]
 (1) (2, -1) (2) (2, -2) (3) (-2, 1) (4) (-2, 2)
16. If the tangent to the curve $y = \frac{x}{x^2 - 3}$, $x \in \mathbb{R}$, ($x \neq \pm\sqrt{3}$), at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line $2x + 6y - 11 = 0$, then : [JEE-MAIN 2019]
 (1) $|6\alpha + 2\beta| = 9$ (2) $|6\alpha + 2\beta| = 19$ (3) $|2\alpha + 6\beta| = 11$ (4) $|2\alpha + 6\beta| = 19$
17. A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is : [JEE-MAIN 2019]
 (1) $\frac{1}{18\pi}$ (2) $\frac{1}{36\pi}$ (3) $\frac{1}{9\pi}$ (4) $\frac{5}{6\pi}$
18. Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then |m| is equal to _____. [JEE-MAIN 2020]

EXERCISE # 4 (JA)

1. Find the equation of the straight line which is tangent at one point and normal at another point of the curve, $x = 3t^2$, $y = 2t^3$. [REE 2000 (Mains) 5 out of 100]
2. If the normal to the curve, $y = f(x)$ at the point (3, 4) makes an angle $\frac{3\pi}{4}$ with the positive x-axis. Then $f'(3) =$ [JEE 2000 (Scr.) 1 out of 35]
- (A) -1 (B) $-\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 1
3. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is(are)
- (A) $\left(\pm\frac{4}{\sqrt{3}}, -2\right)$ (B) $\left(\pm\sqrt{\frac{11}{3}}, 1\right)$ (C) $(0, 0)$ (D) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$
- [JEE 2002 (Scr.), 3]
4. Tangent to the curve $y = x^2 + 6$ at a point P (1, 7) touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q. Then the coordinates of Q are [JEE 2005 (Scr.), 3]
- (A) $(-6, -11)$ (B) $(-9, -13)$ (C) $(-10, -15)$ (D) $(-6, -7)$
5. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$ [JEE 2007, 3]
- (A) on the left of $x = c$ (B) on the right of $x = c$
- (C) at no point (D) at all points

EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- If tangent drawn to the curve $f(x) = x^3 - 9x - 1$ at $P(x_0, f(x_0))$ meets the curve again at Q, m_A denotes the slope of the tangent at A and m_{OB} denotes the slope of the line joining 'O' origin and a point B on the curve, then
 (A) $m_Q - 4m_P = 27$ (B) $m_Q - 4m_P = 9$
 (C) $\frac{m_{OP}}{m_{OQ}} = 2$, where $x_0 = 1$ (D) $\frac{m_{OP}}{m_{OQ}} = \frac{1}{2}$, where $x_0 = 1$
- The abscissa of a point on the curve $xy = (a + x)^2$, the normal at which cuts off numerically equal intercepts from the coordinate axes, is
 (A) $-\frac{a}{\sqrt{2}}$ (B) $\sqrt{2}a$ (C) $\frac{\sqrt{2}a}{2}$ (D) $-\sqrt{2}a$
- For function $f(x) = \frac{\ln x}{x}$, which of the following statements are true.
 (A) $f(x)$ has horizontally tangent at $x = e$
 (B) $f(x)$ cuts the x-axis only at one point
 (C) $f(x)$ is many-one function
 (D) $f(x)$ has one vertical tangent
- The equation of tangent drawn to the curve $y = (x + 1)^3$, from origin is
 (A) $y = 3x$ (B) $y = -3x$ (C) $4y = 27x$ (D) $y = 0$
- For the curve represented parametrically by the equations, $x = 2\ln \cot t + 1$ and $y = \tan t + \cot t$ where t is parameter
 (A) tangent at $t = \frac{\pi}{4}$ is parallel to x – axis
 (B) normal at $t = \frac{\pi}{4}$ is parallel to y – axis
 (C) tangent at $t = \frac{\pi}{4}$ is parallel to the line $y = x$
 (D) tangent and normal intersect at the point (2,1)

6. The equation of the tangents to the curve $y = x^4$ from the point $(2, 0)$ not on the curve, are given by

(A) $y = 0$

(B) $y - 1 = 5(x - 1)$

(C) $y - \frac{4096}{81} = \frac{2048}{27} \left(x - \frac{8}{3} \right)$

(D) $y - \frac{32}{243} = \frac{64}{81} \left(x - \frac{2}{3} \right)$

7. A curve $f(x)$ is given by.

$$f(x) = \begin{cases} x^2 - 2x + 3, & x \geq 0 \\ -x^2 - 4x - \frac{11}{2}, & x < 0 \end{cases}$$

If the line $y = mx + c$ touches the $f(x)$ at two points then.

(A) $m = 1$

(B) $m = -7$

(C) $c = \frac{3}{4}$

(D) $c = -\frac{13}{4}$

8. If $y = mx + c$ is tangent at one point (x_1, y_1) and normal at other point (x_2, y_2) to the curve $y = 8x^3 - 2x$, then which of the following is true

(A) $x_2 = -2x_1$

(B) $x_1^2 = \frac{5 \pm \sqrt{5}}{96}$

(C) sum of slopes of all such lines is -3

(D) sum of slopes of all such lines is $-\frac{3}{2}$

9. The Angles of intersection between curves $y = |x - 1|$ and $y = |x^2 - 3|$ are

(A) $\tan^{-1} \frac{3}{5}$

(B) $\tan^{-1} 3$

(C) $\tan^{-1} \left(\frac{\sqrt{17}}{\sqrt{17} - 2} \right)$

(D) $\tan^{-1} \left(\frac{\sqrt{17}}{\sqrt{17} + 2} \right)$

10. If $\frac{x}{a} + \frac{y}{b} = 1$ is a tangent to the curve $x = Kt, y = \frac{K}{t}, K > 0$ then :

(A) $a > 0, b > 0$

(B) $a > 0, b < 0$

(C) $a < 0, b > 0$

(D) $a < 0, b < 0$

11. The co-ordinates of the point(s) on the graph of the function, $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 7x - 4$

where the tangent drawn cut off intercepts from the co-ordinate axes which are equal in magnitude but opposite in sign, is

(A) $(2, 8/3)$

(B) $(3, 7/2)$

(C) $(1, 5/6)$

(D) none

EXERCISE # 6

- Find the equations of the tangents drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from the point (1, 2).
- Find the point of intersection of the tangents drawn to the curve $x^2y = 1 - y$ at the points where it is intersected by the curve $xy = 1 - y$.
- A function is defined parametrically by the equations

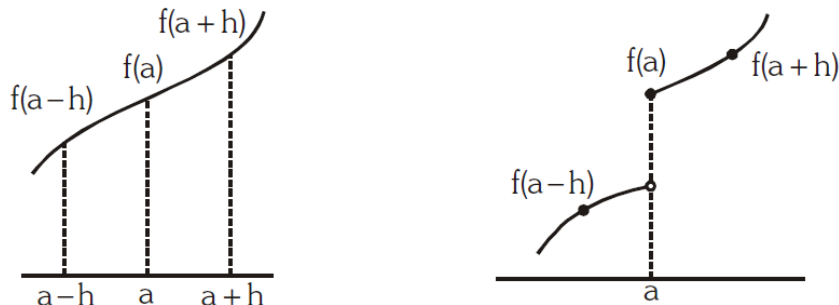
$$f(t) = x = \begin{cases} 2t + t^2 \sin \frac{1}{t} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases} \text{ and } g(t) = y = \begin{cases} \frac{1}{t} \sin t^2 & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases}$$

Find the equation of the tangent and normal at the point for $t = 0$ if exist.
- There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -8/x$ where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points respectively then find the value of (p + q).
- Tangent at a point P_1 [other than (0,0)] on the curve $y = x^3$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 & so on. Show that the abscissae of $P_1, P_2, P_3, \dots, P_n$, form a GP. Also find the ratio $\frac{\text{area of } \Delta(P_1 P_2 P_3)}{\text{area of } \Delta(P_2 P_3 P_4)}$.
- The chord of the parabola $y = -a^2x^2 + 5ax - 4$ touches the curve $y = \frac{1}{1-x}$ at the point $x = 2$ and is bisected by that point. Find 'a'.
- A variable ΔABC in the xy plane has its orthocentre at vertex 'B', a fixed vertex 'A' at the origin and the third vertex 'C' restricted to lie on the parabola $y = 1 + \frac{7x^2}{36}$. The point B starts at the point (0, 1) at time $t = 0$ and moves upward along the y axis at a constant velocity of 2 cm/sec. How fast is the area of the triangle increasing when $t = \frac{7}{2}$ sec.
- Show that the condition that the curves $x^{2/3} + y^{2/3} = c^{2/3}$ & $(x^2/a^2) + (y^2/b^2) = 1$ may touch if $c = a + b$.
- Prove that the segment of the normal to the curve $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ contained between the co-ordinate axes is equal to $2a$.

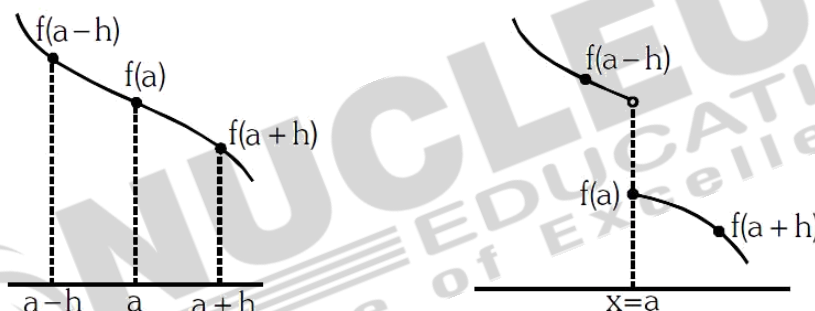
MONOTONICITY

1. MONOTONICITY AT A POINT :

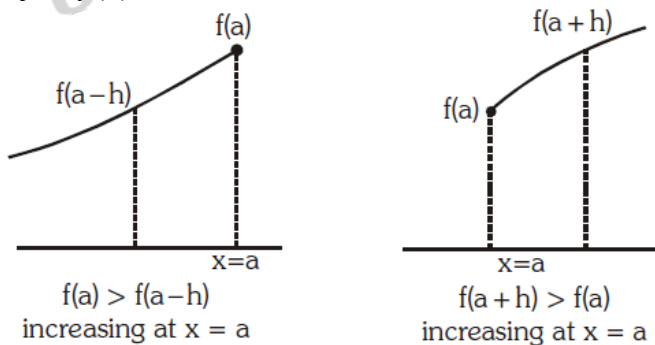
- (a) A function $f(x)$ is called an increasing function at point $x = a$, if in a sufficiently small neighbourhood of $x = a$; $f(a - h) < f(a) < f(a + h)$



- (b) A function $f(x)$ is called a decreasing function at point $x = a$, if in a sufficiently small neighbourhood of $x = a$; $f(a - h) > f(a) > f(a + h)$



Note : If $x = a$ is a boundary point, then use the appropriate one sides inequality to test monotonicity of $f(x)$.



(c) Testing of monotonicity of differentiable function at a point.

- (i) If $f'(a) > 0$, then $f(x)$ is increasing at $x = a$.
- (ii) If $f'(a) < 0$, then $f(x)$ is decreasing at $x = a$.
- (iii) If $f'(a) = 0$, then examine the sign of $f'(a^+)$ and $f'(a^-)$.
 - (1) If $f'(a^+) > 0$ and $f'(a^-) > 0$, then increasing
 - (2) If $f'(a^+) < 0$ and $f'(a^-) < 0$, then decreasing
 - (3) otherwise neither increasing nor decreasing.

Illustration 1 : Let $f(x) = x^3 - 3x + 2$. Examine the nature of function at points $x = 0, 1$ & 2 .

Solution :

$$f(x) = x^3 - 3x + 2$$

$$f'(x) = 3(x^2 - 1) = 0$$

$$\Rightarrow x = \pm 1$$

$$(i) f'(0) = -3 \Rightarrow \text{decreasing at } x = 0$$

$$(ii) f'(1) = 0$$

$$\text{also, } f'(1^+) = \text{positive and } f'(1^-) = \text{negative}$$

$$\Rightarrow \text{neither increasing nor decreasing at } x = 0.$$

$$(iii) f'(2) = 9 \Rightarrow \text{increasing at } x = 2$$



Do yourself - 1 :

- (i) If function $f(x) = x^3 + \lambda x^2 - \lambda x + 1$ is increasing at $x = 0$ & decreasing at $x = 1$, then find the greatest integral value of λ .

2. MONOTONICITY OVER AN INTERVAL :

- (a) A function $f(x)$ is said to be monotonically increasing (MI) in (a, b) if $f'(x) \geq 0$ where equality holds only for discrete values of x i.e. $f'(x)$ does not identically become zero for $x \in (a, b)$ or any sub interval.
- (b) $f(x)$ is said to be monotonically decreasing (MD) in (a, b) if $f'(x) \leq 0$ where equality holds only for discrete values of x i.e. $f'(x)$ does not identically become zero for $x \in (a, b)$ or any sub interval.

Note :

- (i) A function is said to be monotonic if it's either increasing or decreasing.
- (ii) If a function is invertible it has to be either increasing or decreasing.

Illustration 2 : Prove that the function $f(x) = \log(x^3 + \sqrt{x^6 + 1})$ is entirely increasing.

Solution : Now, $f(x) = \log(x^3 + \sqrt{x^6 + 1})$

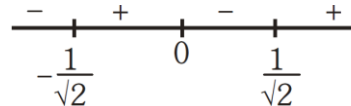
$$f'(x) = \frac{1}{x^3 + \sqrt{x^6 + 1}} \left(3x^2 + \frac{6x^5}{2\sqrt{x^6 + 1}} \right) = \frac{3x^2}{\sqrt{x^6 + 1}} > 0$$

$$\Rightarrow f(x) \text{ is increasing.}$$

Illustration 3 : Find the intervals of monotonicity of the function $y = x^2 - \log_e|x|$, ($x \neq 0$).

Solution :

$$\text{Let } y = f(x) = x^2 - \log_e|x|$$



$$f'(x) = 2x - \frac{1}{x}; \text{ for all } x (x \neq 0)$$

$$f'(x) = \frac{2x^2 - 1}{x} \Rightarrow f'(x) = \frac{(\sqrt{2}x - 1)(\sqrt{2}x + 1)}{x}$$

So $f'(x) > 0$ when $x \in \left(-\frac{1}{\sqrt{2}}, 0\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$ and $f'(x) < 0$

when $x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(0, \frac{1}{\sqrt{2}}\right)$

$\therefore f(x)$ is increasing when $x \in \left(-\frac{1}{\sqrt{2}}, 0\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$

and decreasing when $x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(0, \frac{1}{\sqrt{2}}\right)$

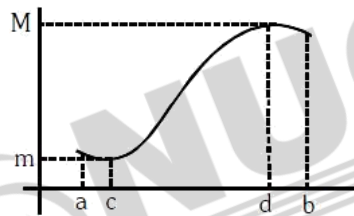
Ans.

Do yourself - 2 :

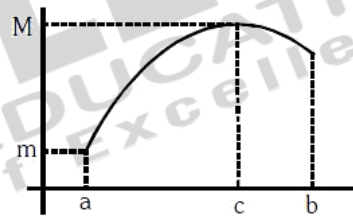
- (i) If $f(x) = \sin x + \ln |\sec x + \tan x| - 2x$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then check the monotonicity of $f(x)$
- (ii) Prove that $y = e^x + \sin x$ is increasing in $x \in \mathbb{R}^+$

3. GREATEST AND LEAST VALUE OF A FUNCTION :

- (a) **Extreme value Theorem :** If f is continuous on $[a, b]$ then f takes on, a least value m and a greatest value M on this interval.



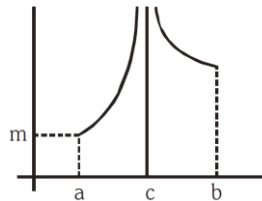
Minimum value 'm' occurs at $x = c$ and maximum value M occurs at $x = d$, $c, d \in (a, b)$



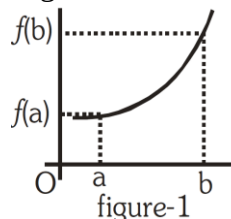
Minimum value 'm' occurs at the end point $x = a$ and maximum value M occurs at $x = c$, $c \in (a, b)$

NOTE :

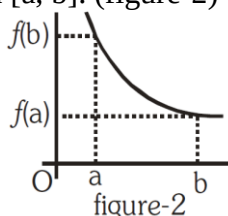
- (a) Continuity through the interval $[a, b]$ is essential for the validity of this theorem. There is a discontinuity at $x = c$, $c \in [a, b]$. The function has a minimum value at the $x = a$ and has no maximum value.



- (b) If a continuous function $y = f(x)$ is increasing in the closed interval $[a, b]$, then $f(a)$ is the least value and $f(b)$ is the greatest value of $f(x)$ in $[a, b]$ (figure-1)



- (c) If a continuous function $y = f(x)$ is decreasing in $[a, b]$, then $f(b)$ is the least and $f(a)$ is the greatest value of $f(x)$ in $[a, b]$. (figure-2)



- (d) If a continuous function $y = f(x)$ is increasing/decreasing in the (a, b) , then no greatest and least value exist.

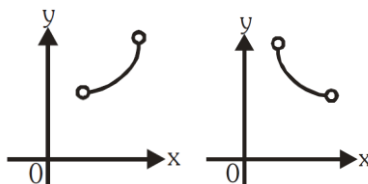


Illustration 4 : Show that $f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x$ is decreasing in $x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$. Also find its range.

Solution : $f(x) = \sin^{-1} \frac{x}{\sqrt{1+x^2}} - \ln x = \tan^{-1} x - \ln x \Rightarrow f'(x) = \frac{1}{1+x^2} - \frac{1}{x} = \frac{-(1+x^2-x)}{x(1+x^2)}$

$$\therefore f'(x) < 0 \quad \forall x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$$

$\Rightarrow f(x)$ is decreasing.

$$f(x)|_{\max} = f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} + \frac{1}{2} \ln 3 \quad \text{and} \quad f(x)|_{\min} = f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln 3$$

$$\therefore \text{Range of } f(x) = \left[\frac{\pi}{3} - \frac{1}{2} \ln 3, \frac{\pi}{6} + \frac{1}{2} \ln 3 \right]$$

Ans.

Illustration 5 : Find the greatest and least value of $f(x) = x^3 + 5x + e^x$ in $[1, 3]$

Solution : $f'(x) = 3x^2 + 5 + e^x \Rightarrow f(x)$ is always increasing.

$$\text{Least value} = f(1) = 6 + e$$

$$\text{greatest value} = f(3) = (42 + e^3)$$

Do yourself - 3 :

- (i) Let $f(x) = \frac{x^3}{3} - \frac{x^2}{2} + 2$ in $[-2, 2]$. Find the greatest and least value of $f(x)$ in $[-2, 2]$

4. PROVING INEQUALITIES USING MONOTONICITY :

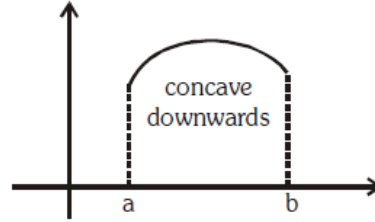
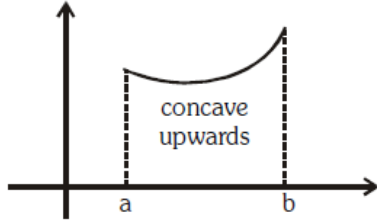
Comparison of two functions $f(x)$ and $g(x)$ can be done by analysing their monotonic behaviour.

5. SIGNIFICANCE OF THE SIGN OF IInd ORDER DERIVATIVE :

The sign of the 2nd order derivative determines the concavity of the curve.

If $f''(x) > 0 \quad \forall x \in (a, b)$ then graph of $f(x)$ is concave upward in (a, b) .

Similarly if $f''(x) < 0 \quad \forall x \in (a, b)$ then graph of $f(x)$ is concave downward in (a, b) .

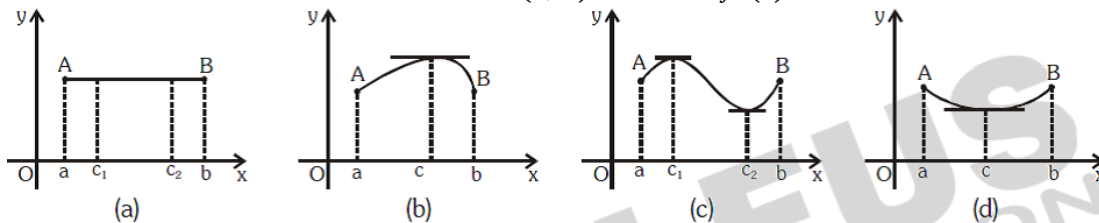


6. ROLLE'S THEOREM :

Let f be a function that satisfies the following three conditions:

- (a) f is continuous on the closed interval $[a, b]$.
- (b) f is differentiable on the open interval (a, b)
- (c) $f(a) = f(b)$

Then there exist at least one number c in (a, b) such that $f'(c) = 0$.



Note : If f is differentiable function then between any two consecutive roots of $f(x) = 0$, there is at least one root of the equation $f'(x) = 0$.

(d) Geometrical Interpretation :

Geometrically, the Rolle's theorem says that somewhere between A and B the curve has at least one tangent parallel to x-axis.

Illustration 6 : Verify Rolle's theorem for the function $f(x) = x^3 - 3x^2 + 2x$ in the interval $[0, 2]$.

Solution : Here we observe that

(a) $f(x)$ is polynomial and since polynomial are always continuous, as well as differentiable. Hence $f(x)$ is continuous in the $[0, 2]$ and differentiable in the $(0, 2)$.

&

(b) $f(0) = 0, f(2) = 2^3 - 3 \cdot (2)^2 + 2(2) = 0$

$\therefore f(0) = f(2)$

Thus, all the condition of Rolle's theorem are satisfied.

So, there must exists some $c \in (0, 2)$ such that $f'(c) = 0$

$$\Rightarrow f'(c) = 3c^2 - 6c + 2 = 0 \Rightarrow c = 1 \pm \frac{1}{\sqrt{3}}$$

where both $c = 1 \pm \frac{1}{\sqrt{3}} \in (0, 2)$ thus Rolle's theorem is verified.

Illustration 7 : Let Rolle's theorem holds for $f(x) = x^3 + bx^2 + ax$, when $1 \leq x \leq 2$ at the point $c = \frac{4}{3}$, then find $a + b$.

Solution :

$$f(1) = f(2) \Rightarrow 1 + b + a = 8 + 4b + 2a$$

$$a + 3b + 7 = 0$$

.....(1)

$$f'(c) = 3x^2 + 2bx + a = 0$$

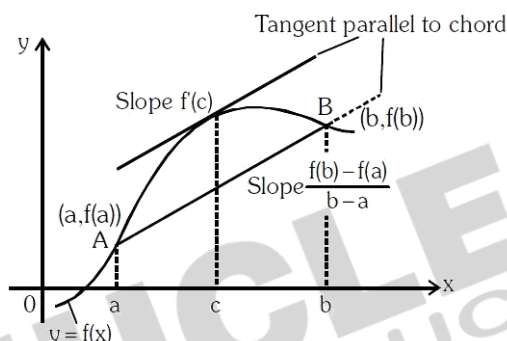
$$\frac{16}{3} + \frac{8b}{3} + a = 0 \Rightarrow 3a + 8b + 16 = 0 \quad \dots\dots\dots(2)$$

By solving $a = 8, b = -5$

Do yourself - 4 :

- (i) Verify Rolle's theorem for $y = 1 - x^{4/3}$ on the interval $[-1, 1]$
- (ii) (a) Let $f(x) = 1 - x^{2/3}$. Show that $f(-1) = f(1)$ but there is no number c in $(-1, 1)$ such that $f'(c) = 0$.
Why does this not contradict Rolle's Theorem ?
- (b) Let $f(x) = (x - 1)^{-2}$. Show that $f(0) = f(2)$ but there is no number c in $(0, 2)$ such that $f'(c) = 0$.
Why does this not contradict Rolle's Theorem ?

7. LAGRANGE'S MEAN VALUE THEOREM (LMVT) :



Let f be a function that satisfies the following conditions:

- (i) f is continuous in $[a, b]$
- (ii) f is differentiable in (a, b) .

Then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

(a) Geometrical Interpretation :

Geometrically, the Mean Value Theorem says that somewhere between A and B the curve has at least one tangent parallel to chord AB.

(b) Physical Interpretations :

If we think of the number $(f(b) - f(a))/(b - a)$ as the average change in f over $[a, b]$ and $f'(c)$ as an instantaneous change, then the Mean Value Theorem says that at some interior point the instantaneous change must equal the average change over the entire interval.

Illustration 8 : Find c of the Lagrange's mean value theorem for the function $f(x) = 3x^2 + 5x + 7$ in the interval $[1, 3]$.

Solution : Given $f(x) = 3x^2 + 5x + 7$ (i)
 $\Rightarrow f(1) = 3 + 5 + 7 = 15$ and $f(3) = 27 + 15 + 7 = 49$
 Again $f'(x) = 6x + 5$
 Here $a = 1, b = 3$

Now from Lagrange's mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow 6c + 5 = \frac{f(3) - f(1)}{3 - 1} = \frac{49 - 15}{2} = 17 \text{ or } c = 2.$$

Illustration 9 : If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is -
(A) 7 (B) 9 (C) 15 (D) 21

Solution :

Apply LMVT

$$f'(x) = \frac{f(5) - f(-2)}{5 - (-2)} \text{ for some } x \text{ in } (-2, 5)$$

$$\text{Now, } -4 \leq \frac{f(5) - f(-2)}{7} \leq 3$$

$$-28 \leq f(5) - f(-2) \leq 21$$

$\therefore$ Greatest possible value of $f(5) - f(-2)$ is 21.

Do yourself - 5 :

- (i) If $f(x) = x^2$ in $[a, b]$, then show that there exist atleast one c in (a, b) such that a, c, b are in A.P.
(ii) Find C of LMVT for $f(x) = |x|^3$ in $[2, 5]$.

8. SPECIAL NOTE :

Use of Monotonicity in identifying the number of roots of the equation in a given interval.
Suppose a and b are two real numbers such that,

(a) Let $f(x)$ is differentiable & either MI or MD for $0 \leq x \leq b$.
&

(b) $f(a)$ and $f(b)$ have opposite signs.

Then there is one & only one root of the equation $f(x) = 0$ in (a, b) .

Miscellaneous Illustrations :

Illustration 10 : If $g(x) = f(x) + f(1 - x)$ and $f''(x) < 0$; $0 \leq x \leq 1$, show that $g(x)$ increasing in $x \in (0, 1/2)$ and decreasing in $x \in (1/2, 1)$

Solution :

$$\therefore f''(x) < 0 \Rightarrow f'(x) \text{ is decreasing function.}$$

$$\text{Now, } g(x) = f(x) + f(1 - x)$$

$$\therefore g'(x) = f'(x) - f'(1 - x) \quad \dots\dots\dots (i)$$

Case I : If $x > (1 - x)$

$$\Rightarrow x > 1/2$$

$$\therefore f'(x) < f'(1 - x)$$

$$\Rightarrow f'(x) - f'(1 - x) < 0$$

$$\Rightarrow g'(x) < 0$$

$$\therefore g(x) \text{ decreases in } x \in \left(\frac{1}{2}, 1\right)$$

Case II : If $x < (1 - x) \Rightarrow x < 1/2$

$$\therefore f'(x) > f'(1 - x)$$

$$\Rightarrow f'(x) - f'(1 - x) > 0$$

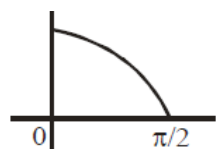
$$\Rightarrow g'(x) > 0$$

$$\therefore g(x) \text{ increases in } x \in (0, 1/2)$$

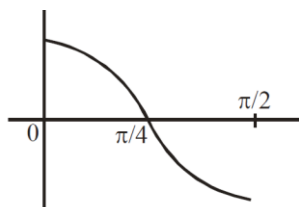
Illustration 11 : Which of the following functions are decreasing on $\left(0, \frac{\pi}{2}\right)$

- (A) $\cos x$ (B) $\cos 2x$ (C) $\cos 3x$ (D) $\tan x$

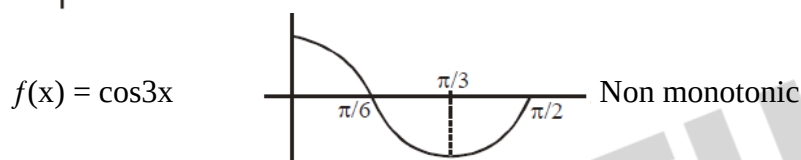
Solution :



$$f(x) = \cos x$$



$$f(x) = \cos 2x$$



$$f(x) = \cos 3x$$

$f(x) = \tan x$ is increasing in $\left(0, \frac{\pi}{2}\right)$ Option A and B are correct.

Illustration 12 : Prove that the equation $e^{(x-1)} + x = 2$ has one solution

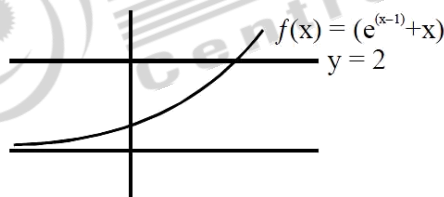
Solution :

$$\text{Let } f(x) = e^{(x-1)} + x$$

$$f'(x) = e^{(x-1)} + 1$$

$f(x)$ is always an increasing function

$$\lim_{x \rightarrow \infty} f(x) = \infty \text{ and } \lim_{x \rightarrow -\infty} f(x) = 0$$



$f(x) = 2$ has exactly one solution.

ANSWERS FOR DO YOURSELF

- 1 : (i) -4
 2 : (i) Increasing
 3 : (i) Greatest is $\frac{8}{3}$ and least value is $-\frac{8}{3}$.
 4 : (i) Rolles theorem is valid
 (ii) (a) $f(x)$ is non-differentiable at $x = 0$ in $(-1, 1)$
 (b) $f(x)$ is discontinuous at $x = 1$ in $(0, 2)$
 5 : (ii) $C = \sqrt{\frac{117}{9}}$

MONOTONOCITY

EXERCISE # 1

- If the function $f(x) = 2x^2 - kx + 5$ is increasing in $[1, 2]$, then 'k' lies in the interval
(A) $(-\infty, 4)$ (B) $(4, \infty)$ (C) $(-\infty, 8]$ (D) $(8, \infty)$
- $f(x) = x + 1/x$, $x \neq 0$ is monotonic increasing when-
(A) $|x| < 1$ (B) $|x| > 1$ (C) $|x| < 2$ (D) $|x| > 2$
- The function x^x decreases on the interval-
(A) $(0, e)$ (B) $(0, 1)$ (C) $\left(0, \frac{1}{e}\right)$ (D) None of these
- Function $f(x) = x^2(x-2)^2$ is-
(A) increasing in $(0, 1) \cup (2, \infty)$ (B) decreasing in $(0, 1) \cup (2, \infty)$
(C) decreasing function (D) increasing function
- If f and g are two decreasing function such that $f \circ g$ is defined, then $f \circ g$ will be-
(A) increasing function (B) decreasing function
(C) neither increasing nor decreasing (D) None of these
- If function $f(x) = 2x^2 + 3x - m \log x$ is monotonic decreasing in the interval $(0, 1)$, then the least value of the parameter m is-
(A) 7 (B) $\frac{15}{2}$ (C) $\frac{31}{4}$ (D) 8
- If $f(x) = \frac{x}{2} + \frac{2}{x}$ for $-7 \leq x \leq 7$, then $f(x)$ is monotonic increasing function of x in the interval-
(A) $[7, 0]$ (B) $[2, 7]$ (C) $[-2, 2]$ (D) $[0, 7]$
- If $f(x) = x^3 - 10x^2 + 200x - 10$, then $f(x)$ is-
(A) decreasing in $(-\infty, 10]$ and increasing in $(10, \infty)$
(B) increasing in $(-\infty, 10]$ and decreasing in $(10, \infty)$
(C) increasing for every value of x
(D) decreasing for every value of x
- Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?
(A) f is not differentiable at $x = 0$ (B) f is monotonic
(C) f is even (D) f has an extremum
- Number of critical points of the function, $f(x) = \frac{2}{3} \sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2} \cos 2t - \sqrt{t} \right) dt$ which lie in the interval $[-2\pi, 2\pi]$ is :
(A) 2 (B) 4 (C) 6 (D) 8

11. The value of K in order that $f(x) = \sin x - \cos x - Kx + b$ decreases for all real values is given by-
 (A) $K < 1$ (B) $K \geq 1$ (C) $K \geq \sqrt{2}$ (D) $K < \sqrt{2}$
12. When $0 \leq x \leq 1$, $f(x) = |x| + |x - 1|$ is-
 (A) increasing (B) decreasing (C) constant (D) None of these
13. Let $f(x)$ and $g(x)$ be two continuous functions defined from $\mathbb{R} \rightarrow \mathbb{R}$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2)$, $\forall x_1 > x_2$, then solution set of $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ is
 (A) $\mathbb{R}$ (B) $\emptyset$ (C) $(1, 4)$ (D) $\mathbb{R} - [1, 4]$
14. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval-
 (A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) none
15. If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$ has a positive root $x = \alpha$, then the equation $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root, which is-
 (A) Smaller than α (B) Greater than α
 (C) Equal to α (D) Greater than or equal to α
16. A value of C for which the conclusion of Mean values theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is-
 (A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$
17. The value of c in Lagrange's theorem for the function $f(x) = \begin{cases} x \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ in the interval $[-1, 1]$ is-
 (A) 0 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) Non existent in the interval
18. If the function $f(x) = x^3 - 6x^2 + ax + b$ defined on $[1, 3]$, satisfies the Rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$ then-
 (A) $a = 11, b = 6$ (B) $a = -11, b = 6$ (C) $a = 11, b \in \mathbb{R}$ (D) None of these
19. Given: $f(x) = 4 - \left(\frac{1}{2} - x\right)^{2/3}$ $g(x) = \begin{cases} \frac{\tan[x]}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$
 $h(x) = \{x\}$ $k(x) = 5^{\log_2(x+3)}$
 then in $[0, 1]$, Lagrange's Mean Value Theorem is NOT applicable to
 (A) f, g, h (B) h, k (C) f, g (D) g, h, k
 where $[x]$ and $\{x\}$ denotes the greatest integer and fractional part function.

20. The function $f : [a, \infty) \rightarrow \mathbb{R}$ where $\mathbb{R}$ denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided
(A) $a \geq 1$ (B) $a \geq 0$ (C) $a \leq 0$ (D) $a \leq 1$
21. The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is -
(A) increasing in its domain
(B) decreasing in its domain
(C) decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$
(D) increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$
22. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x) \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals
(A) $-1/4$ (B) $-1/8$ (C) $1/4$ (D) $1/8$
23. **Statement-1 :** The function $x^2(e^x + e^{-x})$ is increasing for all $x > 0$. [JEE-MAIN Online 2013]
Statement-2 : The functions x^2e^x and x^2e^{-x} are increasing for all $x > 0$ and the sum of two increasing functions in any interval (a, b) is an increasing function in (a, b) .
(A) Statement-1 is false; Statement-2 is true.
(B) Statement-1 is true; Statement-2 is false.
(C) Statement-1 is true; Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
(D) Statement-1 is true ; Statement-2 is true; Statement 2 is not a correct explanation for Statement-1.
24. Let $f(x)$ and $g(x)$ are two function which are defined and differentiable for all $x \geq x_0$. If $f(x_0) = g(x_0)$ and $f'(x) > g'(x)$ for all $x > x_0$ then
(A) $f(x) < g(x)$ for some $x > x_0$
(B) $f(x) = g(x)$ for some $x > x_0$
(C) $f(x) > g(x)$ only for some $x > x_0$
(D) $f(x) > g(x)$ for all $x > x_0$
25. The set of value(s) of 'a' for which the function $f(x) = \frac{ax^3}{3} + (a+2)x^2 + (a-1)x + 2$ posses a negative point of inflection.
(A) $(-\infty, -2) \cup (0, \infty)$ (B) $\{-4/5\}$ (C) $(-2, 0)$ (D) empty set
26. If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$, then the value of 'a' is equal to
(A) 3 (B) 4 (C) 6 (D) 1

EXERCISE # 2

1. Find the intervals of monotonicity for the following functions & represent your solution set on the number line.

(a) $f(x) = 2 \cdot e^{x^2-4x}$ (b) $f(x) = e^x/x$ (c) $f(x) = x^2 e^{-x}$ (d) $f(x) = 2x^2 - \ln |x|$

Also plot the graphs in each case & state their range.

2. Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality,
 $1 - f(x) - f^3(x) > f(1 - 5x)$

3. Find the intervals of monotonicity of the functions in $[0, 2\pi]$

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$

(b) $g(x) = 2 \sin x + \cos 2x$ in $(0 \leq x \leq 2\pi)$.

(c) $f(x) = \frac{4 \sin x - 2x - x \cos x}{2 + \cos x}$

4. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max\{f(t) : 0 \leq t \leq x\}, & 0 \leq x \leq 1 \\ 3 - x, & 1 < x \leq 2 \end{cases}$

Discuss the continuity & differentiability of $g(x)$ in the interval $(0, 2)$.

5. Find the greatest & the least values of the following functions in the given interval if they exist.

(a) $f(x) = \sin^{-1} \frac{x}{\sqrt{x^2+1}} - \ln x$ in $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$

(b) $f(x) = 12x^{4/3} - 6x^{1/3}$, $x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1$ in $[-1, 2]$

6. If $b > a$, find the minimum value of $|(x-a)^3| + |(x-b)^3|$, $x \in \mathbb{R}$.

7. If $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ monotonically increases for every $x \in \mathbb{R}$ then find the range of values of 'a'.

8. Find the set of values of x for which the inequality $\ln(1+x) > x/(1+x)$ is valid.
9. (a) Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.
 (b) Show that exactly two real values of x satisfy the equation $x^2 = x \sin x + \cos x$.
 (c) Prove that inequality $e^x > (1+x)$ for all $x \in \mathbb{R}_0$ and use it to determine which of the two numbers e^π and π^e is greater.
10. Verify Rolles theorem for $f(x) = (x-a)^m(x-b)^n$ on $[a, b]$; m, n being positive integer.
11. Let $f(x) = 4x^3 - 3x^2 - 2x + 1$, use Rolle's theorem to prove that there exist $c, 0 < c < 1$ such that $f(c) = 0$.
12. Assume that f is continuous on $[a, b]$, $a > 0$ and differentiable on an open interval (a, b) . Show that if $\frac{f(a)}{a} = \frac{f(b)}{b}$, then there exist $x_0 \in (a, b)$ such that $x_0 f'(x_0) = f(x_0)$.
13. $f(x)$ and $g(x)$ are differentiable functions for $0 \leq x \leq 2$ such that $f(0) = 5, g(0) = 0, f(2) = 8, g(2) = 1$. Show that there exists a number c satisfying $0 < c < 2$ and $f'(c) = 3g'(c)$.
14. For what value of a, m and b does the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval $[0, 2]$.
15. Let f be continuous on $[a, b]$ and assume the second derivative f'' exists on (a, b) . Suppose that the graph of f and the line segment joining the point $(a, f(a))$ and $(b, f(b))$ intersect at a point $(x_0, f(x_0))$ where $a < x_0 < b$. Show that there exists a point $c \in (a, b)$ such that $f''(c) = 0$.

EXERCISE # 3 (JM)

- $f(x)$ and $g(x)$ are two differentiable function in $[0, 2]$ such that $f''(x) - g''(x) = 0$, $f'(1) = 2$, $g'(1) = 4$, $f(2) = 3$, $g(2) = 9$, then $f(x) - g(x)$ at $x = 3/2$ is- **[AIEEE-2002]**

(1) 0 (2) 2 (3) 10 (4) -5
- A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function, is-

(1) $(x - 1)^2$ (2) $(x - 1)^3$ (3) $(x + 1)^3$ (4) $(x + 1)^2$ **[AIEEE-2004]**
- A function is matched below against an interval where it is supposed to be increasing. which of the following pairs is incorrectly matched ? **[AIEEE-2005]**

	interval	function
(1)	$(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(2)	$[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(3)	$\left[-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(4)	$(-\infty, -4)$	$x^3 + 6x^2 + 6$
- The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in- **[AIEEE-2007]**

(1) $(\pi/4, \pi/2)$ (2) $(-\pi/2, \pi/4)$ (3) $(0, \pi/2)$ (4) $(-\pi/2, \pi/2)$
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function defined by $f(x) = \frac{1}{e^x + 2e^{-x}}$

Statement-1 : $f(c) = \frac{1}{3}$, for some $c \in \mathbb{R}$.

Statement-2 : $0 < f(x) \leq \frac{1}{2\sqrt{2}}$, for all $x \in \mathbb{R}$. **[AIEEE-2010]**

(1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.

(3) Statement-1 is true, Statement-2 is false.

(4) Statement-1 is false, Statement-2 is true.
- If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in]0, 1[$: **[JEE-MAIN 2014]**

(1) $2f'(c) = g'(c)$ (2) $2f'(c) = 3g'(c)$ (3) $f'(c) = g'(c)$ (4) $f'(c) = 2g'(c)$

7. Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$, $x \in \mathbb{R}$, where a , b and d are non-zero real constants.

Then:

- (1) f is an increasing function of x
- (2) f is a decreasing function of x
- (3) f is neither increasing nor decreasing function of x
- (4) f' is not a continuous function of x

[JEE-MAIN 2019]

8. If the function f given by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in \mathbb{R}$ is increasing in $(0, 1]$ and decreasing in $[1, 5)$, then a root of the equation, $\frac{f(x)-14}{(x-1)^2} = 0$ ($x \neq 1$) is:

[JEE-MAIN 2019]

- (1) 6
- (2) 7
- (3) 5
- (4) -7

9. Let $f: [0, 2] \rightarrow \mathbb{R}$ be a twice differentiable function such that $f''(x) > 0$, for all $x \in (0, 2)$.

If $\phi(x) = f(x) + f(2-x)$, then ϕ is

[JEE-MAIN 2019]

- (1) decreasing on $(0, 2)$
- (2) increasing on $(0, 2)$
- (3) increasing on $(0, 1)$ and decreasing on $(1, 2)$
- (4) decreasing on $(0, 1)$ and increasing on $(1, 2)$

10. If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$; then the set

$$S = \{x \in \mathbb{R} : f(x) = f(0)\}$$

contains exactly :

[JEE-MAIN 2019]

- (1) four irrational numbers
- (2) two irrational and one rational number
- (3) two irrational and two rational numbers
- (4) four rational numbers

11. Let $f(x) = e^x - x$ and $g(x) = x^2 - x$, $\forall x \in \mathbb{R}$. Then the set of all $x \in \mathbb{R}$, where the function $h(x) = (f \circ g)(x)$ is increasing is :

[JEE-MAIN 2019]

- (1) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
- (2) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$
- (3) $[0, \infty)$
- (4) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$

12. Let the function, $f: [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f , $f(-1) + f(0)$ lies in the interval :

[JEE-MAIN 2020]

- (1) $[-3, 11]$
- (2) $(-\infty, 20]$
- (3) $(-\infty, 11]$
- (4) $[-6, 20]$

13. The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is :

[JEE-MAIN 2020]

- (1) $\frac{2}{3}$
- (2) $\frac{\sqrt{7}-2}{3}$
- (3) $\frac{4-\sqrt{5}}{3}$
- (4) $\frac{4-\sqrt{7}}{3}$

14. Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. [JEE-MAIN 2020]

If $\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3} \right) = 4$, then which one of the following is **NOT** true ?

- (1) $x = 1$ is a point of maxima and $x = -1$ is a point of minimum of f .
- (2) $x = 1$ is a point of minima and $x = -1$ is a point of maxima of f .
- (3) f is an odd function
- (4) $f(1) - 4f(-1) = 4$

15. Let $f(x) = x \cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ then which of the following is true?

- (1) $f'(0) = -\frac{\pi}{2}$
- (2) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$
- (3) f is not differentiable at $x = 0$
- (4) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$ [JEE-MAIN 2020]

16. If c is a point at which Rolle's theorem holds for the function, $f(x) = \log_e \left(\frac{x^2 + \alpha}{7x} \right)$ in the interval $[3, 4]$, where $\alpha \in \mathbb{R}$, then $f''(c)$ is equal to : [JEE-MAIN 2020]

- (1) $\frac{1}{12}$
- (2) $-\frac{1}{12}$
- (3) $\frac{\sqrt{3}}{7}$
- (4) $-\frac{1}{24}$

17. Let S be the set of all functions $f: [0, 1] \rightarrow \mathbb{R}$, which are continuous on $[0, 1]$ and differentiable on $(0, 1)$ then for every f in S there exists a $c \in (0, 1)$ depending of f , such that.

- (1) $\frac{f(1) - f(c)}{1 - c} = f'(c)$
- (2) $|f(c) + f(1)| < (1 + c) |f'(c)|$
- (3) $|f(c) - f(1)| < (1 - c) |f'(c)|$
- (4) $|f(c) - f(1)| < |f'(c)|$ [JEE-MAIN 2020]

18. Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b) . If for all $x \in (a, b)$, $f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c) - f(a)}{f(b) - f(c)}$ is greater than [JEE-MAIN 2020]

- (1) $\frac{b - c}{c - a}$
- (2) 1
- (3) $\frac{a + b}{b - a}$
- (4) $\frac{c - a}{b - c}$

19. Let a function $f: [0, 5] \rightarrow \mathbb{R}$ continuous, $f(1) = 3$ and F be defined as : [JEE-MAIN 2020]

$F(x) = \int_1^x t^2 g(t) dt$, where $g(t) = \int_1^t f(u) du$. Then for the function F , the point $x = 1$ is :

- (1) a point of local minima.
- (2) not a critical point.
- (3) a point of inflection.
- (4) a point of local maxima.

EXERCISE # 4 (JA)

SECTION-1

1. (a) For all $x \in (0, 1)$:
 (A) $e^x < 1 + x$ (B) $\log_e (1 + x) < x$ (C) $\sin x > x$ (D) $\log_e x > x$
 (b) Consider the following statements S and R :
 S : Both $\sin x$ & $\cos x$ are decreasing functions in the interval $(\pi/2, \pi)$.
 R : If a differentiable function decreases in an interval (a, b) , then its derivative also decreases in (a, b) .
 Which of the following is true ?
 (A) both S and R are wrong
 (B) both S and R are correct, but R is not the correct explanation for S
 (C) S is correct and R is the correct explanation for S
 (D) S is correct and R is wrong.
 (c) Let $f(x) = \int e^x (x-1)(x-2) dx$ then f decreases in the interval :
 (A) $(-\infty, 2)$ (B) $(-2, -1)$ (C) $(1, 2)$ (D) $(2, +\infty)$
[JEE 2000 (Scr.) 1+1+1 out of 35]
2. (a) If $f(x) = xe^{x(1-x)}$, then $f(x)$ is
 (A) increasing on $\left(-\frac{1}{2}, 1\right)$ (B) decreasing on $\left[-\frac{1}{2}, 1\right]$
 (C) increasing on R (D) decreasing on R
 (b) Let $-1 \leq p \leq 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval $\left[\frac{1}{2}, 1\right]$ and identify it.
[JEE 2001, 1 + 5]
3. The length of a longest interval in which the function $f(x) = 3 \sin x - 4 \sin^3 x$ is increasing, is
 (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) π
[JEE 2002 (Screening), 3]
4. (a) Let $f(x) = \begin{cases} x^\alpha \ln x, & x > 0 \\ 0, & x = 0 \end{cases}$. Rolle's theorem is applicable to f for $x \in [0, 1]$, if $\alpha =$
 (A) -2 (B) -1 (C) 0 (D) 1/2
 (b) If f is a strictly increasing function, then $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is equal to
 (A) 0 (B) 1 (C) -1 (D) 2 **[JEE 2004 (Scr)]**
6. If $f(x)$ is a twice differentiable function and given that $f(1) = 1, f(2) = 4, f(3) = 9$, then
 (A) $f''(x) = 2$, for $\forall x \in (1, 3)$ (B) $f''(x) = f'(x) = 2$, for some $x \in (2, 3)$
 (C) $f''(x) = 3$, for $\forall x \in (2, 3)$ (D) $f''(x) = 2$, for some $x \in (1, 3)$
[JEE 2005 (Scr), 3]

7. Let $f(x) = 2 + \cos x$ for all real x .

Statement-1: For each real t , there exists a point ' c ' in $[t, t + \pi]$ such that $f'(c) = 0$.

because

Statement-2: $f(t) = f(t + 2\pi)$ for each real t .

(A) Statement-1 is true, statement-2 is true; statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true; statement-2 is NOT a correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

[JEE 2007, 3]

8. Let $f : (0,1) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{b-x}{1-bx}$,

where b is a constant such that $0 < b < 1$. Then

(A) f is not invertible on $(0,1)$

(B) $f \neq f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$

(C) $f = f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$

(D) f^{-1} is differentiable on $(0,1)$ [JEE 2011, 4M]

9. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is [JEE 2013, 2M]

(A) 6

(B) 4

(C) 2

(D) 0

10. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and

$$f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1, \text{ then}$$

[JEE-Advanced 2017]

(A) $f'(1) > 1$

(B) $f'(1) \leq 0$

(C) $\frac{1}{2} < f'(1) \leq 1$

(D) $0 < f'(1) \leq \frac{1}{2}$

SECTION-2

11. (a) Using the relation $2(1 - \cos x) < x^2$, $x \neq 0$ or otherwise, prove that

$$\sin(\tan x) \geq x, \forall x \in \left[0, \frac{\pi}{4}\right].$$

(b) Let $f : [0, 4] \rightarrow \mathbb{R}$ be a differentiable function.

(i) Show that there exist $a, b \in [0, 4]$, $(f(4))^2 - (f(0))^2 = 8 f'(a) f(b)$

(ii) Show that there exist α, β with $\alpha, \beta \in (0, 2)$ such that

$$\int_0^4 f(t) dt = 2(\alpha f(\alpha^2) + \beta f(\beta^2)) \quad [\text{JEE 2003 (Mains), 4 + 4 out of 60}]$$

12. If $p(x) = 51x^{101} - 2323x^{100} - 45x + 1035$, using Rolle's theorem, prove that at least one root of $p(x)$ lies between $(45^{(1/100)}, 46)$. [JEE 2004, 2 out of 60]

13. **Paragraph**

[JEE 2007, 4+4+4]

If a continuous function f defined on the real line $\mathbf{R}$, assumes positive and negative values in $\mathbf{R}$ then the equation $f(x) = 0$ has a root in $\mathbf{R}$. For example, if it is known that a continuous function f on $\mathbf{R}$ is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in $\mathbf{R}$.

Consider $f(x) = ke^x - x$ for all real x where k is a real constant.

- (i) The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at
 (A) no point (B) one point
 (C) two points (D) more than two points
- (ii) The positive value of k for which $ke^x - x = 0$ has only one root is
 (A) $1/e$ (B) 1 (C) e (D) $\log_e 2$
- (iii) For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is
 (A) $(0, 1/e)$ (B) $(1/e, 1)$ (C) $(1/e, \infty)$ (D) $(0, 1)$

14. **Match the column.**

[JEE 2007, 6]

In the following $[x]$ denotes the greatest integer less than or equal to x .

Match the functions in **Column I** with the properties in **Column II**.

- | Column I | Column II |
|-------------------------|---|
| (A) $x x $ | (P) continuous in $(-1, 1)$ |
| (B) $\sqrt{ x }$ | (Q) differentiable in $(-1, 1)$ |
| (C) $x + [x]$ | (R) strictly increasing in $(-1, 1)$ |
| (D) $ x - 1 + x + 1 $ | (S) non differentiable at least at one point in $(-1, 1)$ |

15. (a) Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is
 (A) even and is strictly increasing in $(0, \infty)$
 (B) odd and is strictly decreasing in $(-\infty, \infty)$
 (C) odd and is strictly increasing in $(-\infty, \infty)$
 (D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

- (b) Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'(1/4) = 0$. Then

- (A) $f''(x)$ vanishes at least twice on $[0, 1]$
 (B) $f'(1/2) = 0$

(C) $\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x \, dx = 0$

(D) $\int_0^{1/2} f(t) e^{\sin \pi t} \, dt = \int_{1/2}^1 f(1-t) e^{\sin \pi t} \, dt$

[JEE 2008, 3 + 4]

16. For the function $f(x) = x \cos \frac{1}{x}, x \geq 1$,
- (A) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$
- (B) $\lim_{x \rightarrow \infty} f'(x) = 1$
- (C) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$
- (D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$ [JEE 2009, 4]
17. Let f be a real-valued function defined on the interval $(0, \infty)$ by $f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} dt$.
- Then which of the following statement(s) is/(are) true? [JEE 10, 3M]
- (A) $f''(x)$ exists for all $x \in (0, \infty)$
- (B) $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$
- (C) there exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
- (D) there exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (0, \infty)$
18. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is [JEE 2011, 4M]
- Paragraph for questions 19 and 20**
- Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$, and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t) dt$ for all $x \in (1, \infty)$
19. Consider the statements :
- P** : There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1 + x^2)$
- Q** : There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1 + x)$ Then [JEE 2012, 3M, -1M]
- (A) both **P** and **Q** are true (B) **P** is true and **Q** is false
- (C) **P** is false and **Q** is true (D) both **P** and **Q** are false
20. Which of the following is true ? [JEE 2012, 3M, -1M]
- (A) g is increasing on $(1, \infty)$
- (B) g is decreasing on $(1, \infty)$
- (C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
- (D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

21. If $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$ for all $x \in (0, \infty)$ then -

[JEE 2012, 4M]

- (A) f has a local maximum at $x = 2$
- (B) f is decreasing on $(2, 3)$
- (C) there exists some $c \in (0, \infty)$ such that $f''(c) = 0$
- (D) f has a local minimum at $x = 3$

22. Let $f(x) = x \sin \pi x$, $x > 0$. Then for all natural numbers n , $f'(x)$ vanishes at -

[JEE 2013, 4M, -1M]

- (A) a unique point in the interval $\left(n, n + \frac{1}{2}\right)$
- (B) a unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
- (C) a unique point in the interval $(n, n + 1)$
- (D) two points in the interval $(n, n + 1)$

23. Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous function which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table :

	$X = -1$	$X = 0$	$X = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is(are)

[JEE 2015, 4M, -2M]

- (A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
- (B) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$
- (C) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(0, 2)$
- (D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$

Answer Q. 24, Q.25 and Q.26 by appropriately matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$

- Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.
- Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.
- Column 3 contains information about increasing/decreasing nature of $f(x)$, and $f'(x)$.

Column 1	Column 2	Column 3
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

24. Which of the following options is the only CORRECT combination ? **[JEE-Advanced 2017]**
 (A) (IV) (i) (S) (B) (I) (ii) (R) (C) (III) (iv) (P) (D) (II) (iii) (S)
25. Which of the following options is the only CORRECT combination ? **[JEE-Advanced 2017]**
 (A) (III) (iii) (R) (B) (I) (i) (P) (C) (IV) (iv) (S) (D) (II) (ii) (Q)
26. Which of the following options is the only INCORRECT combination? **[JEE-Advanced 2017]**
 (A) (II) (iii) (P) (B) (II) (iv) (Q) (C) (I) (iii) (P) (D) (III) (i) (R)
27. Let $f : \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$? **[JEE-Advanced 2017]**
 (A) $e^x - \int_0^x f(t) \sin t \, dt$ (B) $x^9 - f(x)$
 (C) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t \, dt$ (D) $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t \, dt$
28. For every twice differentiable function $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE? **[JEE-Advanced 2018]**
 (A) There exist $r, s \in \mathbb{R}$ where $r < s$, such that f is one-one on the open interval (r, s)
 (B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$
 (C) $\lim_{x \rightarrow \infty} f(x) = 1$
 (D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(x)$ is a differentiable function and $f'(x) < 0$ and $h(x) = f(x) f^{-1}(x) - x^2 + 16$, ($f^{-1}(x)$ denotes the inverse of $f(x)$), then
 (A) equation $h(x) = 0$ has at least one real root
 (B) equation $h(x) = 0$ has no real root
 (C) equation $h(x) = 0$ has at least two real roots
 (D) $f(x) f^{-1}(x)$ is a decreasing function
- Given a function 'g' which has a derivative $g'(x)$ for every real x and satisfies $g'(0) = 2$ and $g(x+y) = e^y g(x) + e^x g(y)$ for all x and y then
 (A) $\lim_{x \rightarrow \infty} \frac{g(x)}{x} = 2$
 (B) $g(x)$ is increasing function $\forall x \in \mathbb{R}$
 (C) $g(x)$ is one- one function $\forall x \in [-1, \infty)$
 (D) Range of $g(x)$ is $\left[-\frac{2}{e}, \infty\right)$
- Let $y = f(x)$ be a bijective function and differentiable $\forall x \in \mathbb{R}$ then which of the following is/are correct?
 (A) $y = f^{-1}(x)$ is differentiable if $y = f(x)$ is always concave up
 (B) $y = f^{-1}(x)$ is differentiable if $y = f(x)$ is always concave down
 (C) $y = f^{-1}(x)$ is differentiable if $f'(x) = 0$ has no real roots
 (D) None of these
- Let $f(x)$ is a derivable function, which is increasing for all $x \in \mathbb{R}$ (having no critical point), then:
 (A) $f(3-4x)$ is an increasing function for all x .
 (B) $f(3-4x)$ is a decreasing function for all x .
 (C) $f(x^2-x)$ increasing for $x > \frac{1}{2}$
 (D) $(f(x))^3$ is an increasing function for all x .
- Let $g'(x) > 0$ and $f'(x) < 0$, $\forall x \in \mathbb{R}$, then
 (A) $g(f(x+1)) > g(f(x-1))$
 (B) $f(g(x-1)) > f(g(x+1))$
 (C) $g(f(x+1)) < g(f(x-1))$
 (D) $g(g(x+1)) < g(g(x-1))$
- If $f(x) = x^3 - x^2 + 100x + 1001$, then
 (A) $f(2000) > f(2001)$
 (B) $f\left(\frac{1}{1999}\right) > f\left(\frac{1}{2000}\right)$
 (C) $f(x+1) > f(x-1)$
 (D) $f(3x-5) > f(3x)$

7. If the derivative of an odd cubic polynomial vanishes at two different value of 'x' then
 (A) coefficient of x^3 and x in the polynomial must be same in sign
 (B) coefficient of x^3 and x in the polynomial must be different in sign
 (C) the values of 'x' where derivative vanishes are closer to origin as compared to the respective roots on either side of origin.
 (D) the values of 'x' where derivative vanishes are far from origin as compared to the respective roots on either side of origin.
8. Let $h(x) = f(x) - (f(x))^2 + (f(x))^3$ for every real number 'x' and $f(x)$ is a differentiable function, then
 (A) 'h' is increasing whenever 'f' is increasing
 (B) 'h' is increasing whenever 'f' is decreasing
 (C) 'h' is decreasing whenever 'f' is decreasing
 (D) Nothing can be said in general
9. The set of all x for which the function $h(x) = -\log_2(-2x - 3 + x^2)$ is defined and monotonic, is
 (A) (1, 3) (B) $(-\infty, -1)$ (C) $(-1, 1)$ (D) $(3, \infty)$
10. If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3 \end{cases}$
 Then -
 (A) $f(x)$ is increasing on $[-1, 2]$ (B) $f(x)$ is continuous on $[-1, 3]$
 (C) $f'(2)$ doesn't exist (D) $f(x)$ has the maximum value at $x = 2$
11. Which of the following is/are correct ?
 (A) Between any two roots of $e^x \cos x = 1$, there exists at least one root of $\tan x = 1$.
 (B) Between any two roots of $e^x \sin x = 1$, there exists at least one root of $\tan x = -1$.
 (C) Between any two roots of $e^x \cos x = 1$, there exists at least one root of $e^x \sin x = 1$.
 (D) Between any two roots of $e^x \sin x = 1$, there exists at least one root of $e^x \cos x = -1$.
12. The function $f(x) = x^{1/3}(x - 1)$
 (A) has 2 inflection points.
 (B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$.
 (C) is concave down in $(-1/2, 0)$.
 (D) Area enclosed by the curve lying in the fourth quadrant is $9/28$.
13. The function $f(x) = \int_0^x \sqrt{1-t^4} dt$ is such that :
 (A) it is defined on the interval $[-1, 1]$ (B) it is an increasing function
 (C) it is an odd function (D) the point (0, 0) is the point of inflection

EXERCISE # 6

- Let $f(x)$ be an increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. Find the range of a .
- Find all the values of the parameter ' a ' for which the function ;
 $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$ increasing & has no critical points for all $x \in \mathbb{R}$.
- Find the values of ' a ' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ increases throughout the number line.
- Find the minimum value of the function $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x .
- Find the set of values of ' a ' for which the function,
 $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is increasing at every point of its domain.
- If f, ϕ, ψ are continuous in $[a, b]$ and derivable in $]a, b[$ then show that there is a value of c lying between a & b such that,

$$\begin{vmatrix} f(a) & f(b) & f'(c) \\ \phi(a) & \phi(b) & \phi'(c) \\ \psi(a) & \psi(b) & \psi'(c) \end{vmatrix} = 0$$
- Let $a + b = 4$, where $a < 2$ and let $g(x)$ be a differentiable function. If $\frac{dg}{dx} > 0$ for all x , prove that $\int_0^a g(x) dx + \int_0^b g(x) dx$ increases as $(b - a)$ increases.
- Find the value of $x > 1$ for which the function $F(x) = \int_x^{x^2} \frac{1}{t} \ln\left(\frac{t-1}{32}\right) dt$ is increasing and decreasing.
- Prove that, $x^2 - 1 > 2x \ln x > 4(x - 1) - 2 \ln x$ for $x > 1$.
- Prove that if f is differentiable on $[a, b]$ and if $f(a) = f(b) = 0$ then for any real α there is an $x \in (a, b)$ such that $\alpha f(x) + f'(x) = 0$.
- Let $a > 0$ and f be continuous in $[-a, a]$. Suppose that $f'(x)$ exists and $f'(x) \leq 1$ for all $x \in (-a, a)$. If $f(a) = a$ and $f(-a) = -a$, show that $f(0) = 0$.
- Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = a$ and $f(b) = b$, show that there exist distinct c_1, c_2 in (a, b) such that $f'(c_1) + f'(c_2) = 2$.

MAXIMA-MINIMA

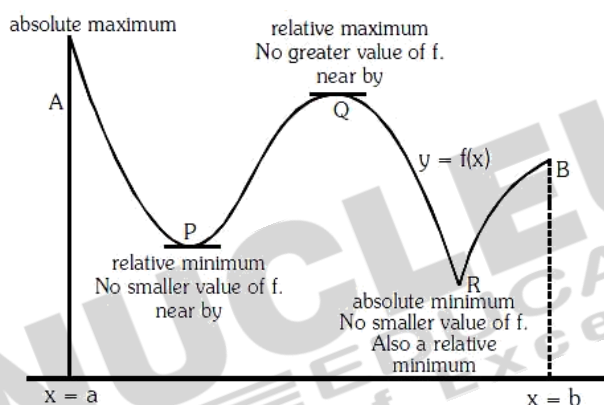
1. INTRODUCTION :

Some of the most important applications of differential calculus are optimization problems, in which we are required to find the optimal (best) way of doing something. Here are examples of such problems that we will solve in this chapter

- What is the shape of a vessel that can with-stand maximum pressure ?
- What is the maximum acceleration of a space shuttle ? (This is an important question to the astronauts who have to withstand the effects of acceleration)
- What is the radius of a contracted windpipe that expels air most rapidly during a cough?

These problems can be reduced to finding the maximum or minimum values of a function. Let's first explain exactly what we mean by maxima and minima.

(a) Maxima (Local/Relative maxima) :



A function $f(x)$ is said to have a maxima at $x = a$ if there exist a neighbourhood $(a - h, a + h) - \{a\}$ such that $f(a) > f(x) \forall x \in (a - h, a + h) - \{a\}$

(b) Minima (Local/Relative minima) :

A function $f(x)$ is said to have a minima at $x = a$ if there exist a neighbourhood $(a - h, a + h) - \{a\}$ such that $f(a) < f(x) \forall x \in (a - h, a + h) - \{a\}$

(c) Absolute maxima (Global maxima) :

A function f has an absolute maxima (or global maxima) at c if $f(c) \geq f(x)$ for all x in D , where D is the domain of f . The number $f(c)$ is called the maximum value of f on D .

(d) Absolute minima (Global minima) :

A function f has an absolute minima at c if $f(c) \leq f(x)$ for all x in D and the number $f(c)$ is called the minimum value of f on D . The maximum and minimum values of f are called the extreme values of f .

Note :

- the maximum & minima values of a function are also known as **local/relative maxima or local/relative minima** as these are the greatest & least values of the function relative to some neighbourhood of the point in question.
- the term 'extrema' is used both for maxima or minima.
- a maximum (minimum) value of a function may not be the greatest (least) value in a finite interval.

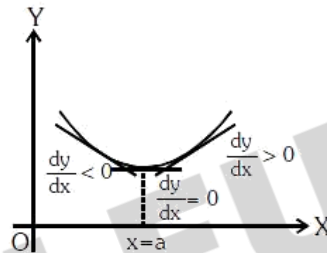
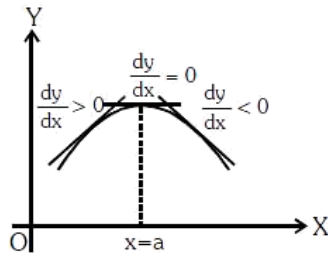
- (iv) a function can have several extreme values & a local minimum value may even be greater than a local maximum value.
- (v) local maximum & local minimum values of a continuous function occur alternately & between two consecutive local maximum values there is a local minimum value & vice versa.

2. DERIVATIVE TEST FOR ASCERTAINING MAXIMA/MINIMA :

(a) First derivative test :

If $f'(x) = 0$ at a point (say $x = a$) and

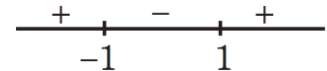
- (i) If $f'(x)$ changes sign from positive to negative in the neighbourhood of $x = a$ then $x = a$ is said to be a point **local maxima**.
- (ii) If $f'(x)$ changes sign from negative to positive in the neighbourhood of $x = a$ then $x = a$ is said to be a point **local minima**.



Note : If $f'(x)$ does not change sign i.e. has the same sign in a certain complete neighbourhood of a , then $f(x)$ is either increasing or decreasing throughout this neighbourhood implying that $x=a$ is not a point of extremum of f .

Illustration1 : Let $f(x) = x + \frac{1}{x}$; $x \neq 0$. Discuss the local maximum and local minimum values of $f(x)$.

Solution : Here, $f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2} = \frac{(x-1)(x+1)}{x^2}$



Using number line rule, $f(x)$ will have local maximum at $x = -1$ and local minimum at $x = 1$

$\therefore$ local maximum value of $f(x) = -2$ at $x = -1$
and local minimum value of $f(x) = 2$ at $x = 1$

Ans.

Illustration2 : If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3, \end{cases}$ then

(A) $f(x)$ is increasing on $[-1, 2)$

(B) $f(x)$ is continuous on $[-1, 3]$

(C) $f'(x)$ does not exist at $x = 2$

(D) $f(x)$ has the maximum value at $x = 2$

Solution : Given, $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3, \end{cases}$

$$\Rightarrow f'(x) = \begin{cases} 6x + 12, & -1 \leq x \leq 2 \\ -1, & 2 < x \leq 3, \end{cases}$$

(A) which shows $f'(x) > 0$ for $x \in [-1, 2)$

So, $f(x)$ is increasing on $[-1, 2)$

Hence, (A) is correct.

- (B) for continuity of $f(x)$. (check at $x = 2$)
 $RHL = 35$, $LHL = 35$ and $f(2) = 35$
 So, (B) is correct
- (C) $Rf'(2) = -1$ and $Lf'(2) = 24$
 so, not differentiable at $x = 2$.
 Hence, (C) is correct.
- (D) we know $f(x)$ is increasing on $[-1, 2)$ and decreasing on $(2, 3]$,
 Thus maximum at $x = 2$,
 Hence, (D) is correct.
 $\therefore$ (A), (B), (C), (D) all are correct.

Ans.

Do yourself - 1 :

- (i) Find local maxima and local minima for the function $f(x) = x^3 - 3x$.
 (ii) If function $f(x) = x^3 - 62x^2 + ax + 9$ has local maxima at $x = 1$, then find the value of a .

(b) Second derivative test :

If $f(x)$ is continuous and differentiable at $x = a$ where $f'(a) = 0$ and $f''(a)$ also exists then for ascertaining maxima/minima at $x = a$, 2nd derivative test can be used -

- (i) If $f''(a) > 0 \Rightarrow x = a$ is a point of local minima
 (ii) If $f''(a) < 0 \Rightarrow x = a$ is a point of local maxima
 (iii) If $f''(a) = 0 \Rightarrow$ second derivative test fails. To identify maxima/minima at this point either first derivative test or higher derivative test can be used.

Illustration 3 : If $f(x) = 2x^3 - 3x^2 - 36x + 6$ has local maximum and minimum at $x = a$ and $x = b$ respectively, then ordered pair (a, b) is -

- (A) $(3, -2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(-3, 2)$

Solution :

$$f(x) = 2x^3 - 3x^2 - 36x + 6$$

$$f'(x) = 6x^2 - 6x - 36 \quad \& \quad f''(x) = 12x - 6$$

$$\text{Now } f'(x) = 0 \Rightarrow 6(x^2 - x - 6) = 0 \Rightarrow (x - 3)(x + 2) = 0 \Rightarrow x = -2, 3$$

$$f''(-2) = -30$$

$\therefore x = -2$ is a point of local maximum

$$f''(3) = 30 \quad \therefore x = 3 \text{ is a point of local minimum}$$

Hence, $(-2, 3)$ is the required ordered pair.

Ans. (C)

Illustration 4 : Find the point of local maxima of $f(x) = \sin x (1 + \cos x)$ in $x \in (0, \pi/2)$.

Solution : Let $f(x) = \sin x (1 + \cos x) = \sin x + \frac{1}{2} \sin 2x$

$$\Rightarrow f'(x) = \cos x + \cos 2x$$

$$f''(x) = -\sin x - 2\sin 2x$$

$$\text{Now } f'(x) = 0 \Rightarrow \cos x + \cos 2x = 0 \Rightarrow \cos 2x = -\cos x \Rightarrow \cos 2x = \cos(\pi - x) \Rightarrow x = \pi/3$$

$$\text{Also } f''(\pi/3) = -\sqrt{3}/2 - \sqrt{3} < 0 \therefore f(x) \text{ has a maxima at } x = \pi/3$$

Ans.

Illustration 5 : Find the global maximum and global minimum of $f(x) = \frac{e^x + e^{-x}}{2}$ in $[-\log_e 2, \log_e 7]$.

Solution : $f(x) = \frac{e^x + e^{-x}}{2}$ is differentiable at all x in its domain.

$$\text{Then } f'(x) = \frac{e^x - e^{-x}}{2}, \quad f''(x) = \frac{e^x + e^{-x}}{2}$$

$$f'(x) = 0 \Rightarrow \frac{e^x - e^{-x}}{2} = 0 \Rightarrow e^{2x} = 1 \Rightarrow x = 0$$

$f''(0) = 1 \therefore x = 0$ is a point of local minimum

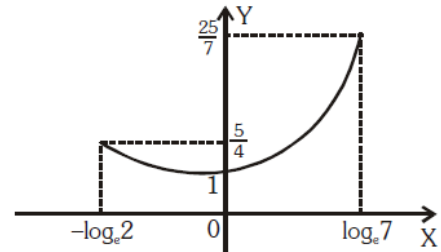
Points $x = -\log_e 2$ and $x = \log_e 7$ are extreme points.

Now, check the value of $f(x)$ at all these three points $x = -\log_e 2, 0, \log_e 7$

$$\Rightarrow f(-\log_e 2) = \frac{e^{-\log_e 2} + e^{+\log_e 2}}{2} = \frac{5}{4}$$

$$f(0) = \frac{e^0 + e^{-0}}{2} = 1$$

$$f(\log_e 7) = \frac{e^{\log_e 7} + e^{-\log_e 7}}{2} = \frac{25}{7}$$



$\therefore x = 0$ is absolute minima & $x = \log_e 7$ is absolute maxima

Hence, absolute/global minimum value of $f(x)$ is 1 at $x = 0$

and absolute/global maximum value of $f(x)$ is $\frac{25}{7}$ at $x = \log_e 7$

Ans.

Do yourself - 2 :

(i) Find local maximum value of function $f(x) = \frac{\ln x}{x}$

(ii) If $f(x) = x^2 e^{-2x}$ ($x > 0$), then find the local maximum value of $f(x)$.

Illustration 6 : Identify a point of maxima/minima in $f(x) = (x + 1)^4$.

Solution :

$$f(x) = (x + 1)^4$$

$$f'(x) = 4(x + 1)^3$$

$$\begin{array}{c} \text{---ve} \quad \quad \quad \text{+ve} \\ \text{---} \quad \quad \quad \text{---} \\ \quad \quad \quad -1 \quad \quad \quad \frac{dy}{dx} \end{array}$$

at $x = -1$ $f(x)$ is having local minima

$\therefore$ at $x = -1$ $f(x)$ has point of minima.

Illustration 7 : Find point of local maxima and minima of $f(x) = x^5 - 5x^4 + 5x^3 - 1$

Solution :

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

$$f'(x) = 5x^4 - 20x^3 + 15x^2 = 5x^2(x^2 - 4x + 3)$$

$$= 5x^2(x - 1)(x - 3)$$

$$f'(x) = 0 \Rightarrow x = 0, 1, 3$$

$$f''(x) = 10x(2x^2 - 6x + 3)$$

But at $x = 0$, derivative sign is positive in its neighbourhood.

Now $f''(1) < 0 \Rightarrow$ Maxima at $x = 1$

$f''(3) > 0 \Rightarrow$ Minima at $x = 3$

$\Rightarrow$ Neither maxima nor minima at $x = 0$.

Do yourself - 3 :

(i) Identify the point of local maxima/minima in $f(x) = (x - 3)^{10}$.

3. **USEFUL FORMULAE OF MENSURATION TO REMEMBER :**

- (a) Volume of a cuboid = ℓbh .
- (b) Surface area of a cuboid = $2 (\ell b + bh + h\ell)$.
- (c) Volume of a prism = area of the base $\times$ height.
- (d) Lateral surface area of prism = perimeter of the base $\times$ height.
- (e) Total surface area of a prism = lateral surface area + 2 area of the base
(Note that lateral surfaces of a prism are all rectangles).
- (f) Volume of a pyramid = $\frac{1}{3}$ area of the base $\times$ height.
- (g) Curved surface area of a pyramid = $\frac{1}{2}$ (perimeter of the base) $\times$ slant height.
(Note that slant surfaces of a pyramid are triangles).
- (h) Volume of a cone = $\frac{1}{3} \pi r^2 h$.
- (i) Curved surface area of a cylinder = $2 \pi rh$.
- (j) Total surface area of a cylinder = $2 \pi rh + 2 \pi r^2$.
- (k) Volume of a sphere = $\frac{4}{3} \pi r^3$.
- (l) Surface area of a sphere = $4 \pi r^2$.
- (m) Area of a circular sector = $\frac{1}{2} r^2 \theta$, when θ is in radians.

4. **SUMMARY OF WORKING RULES FOR SOLVING REAL LIFE OPTIMIZATION**

PROBLEM :

First : When possible, draw a figure to illustrate the problem & label those parts that are important in the problem. Constants & variables should be clearly distinguished.

Second : Write an equation for the quantity that is to be maximized or minimized. If this quantity is denoted by 'y', it must be expressed in terms of a single independent variable x. This may require some algebraic manipulations.

Third : If $y = f(x)$ is a quantity to be maximum or minimum, find those values of x for which $dy/dx = f'(x) = 0$.

Fourth : Using derivative test, test each value of x for which $f'(x) = 0$ to determine whether it provides a maximum or minimum or neither.

Fifth : If the derivative fails to exist at some point, examine this point as possible maximum or minimum.

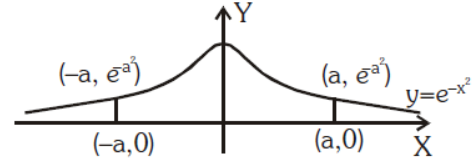
Sixth : If the function $y = f(x)$ is defined only for $x \in [a, b]$ then examine $x = a$ & $x = b$ for possible extreme values.

Illustration 8 : Determine the largest area of the rectangle whose base is on the x-axis and two of its vertices lie on the curve $y = e^{-x^2}$.

Solution : Area of the rectangle will be $A = 2a \cdot e^{-a^2}$
For max. area, $\frac{dA}{da} = \frac{d}{da} (2a e^{-a^2}) = e^{-a^2} [2 - 4a^2]$

$$\frac{dA}{da} = 0 \Rightarrow a = \pm \frac{1}{\sqrt{2}}$$

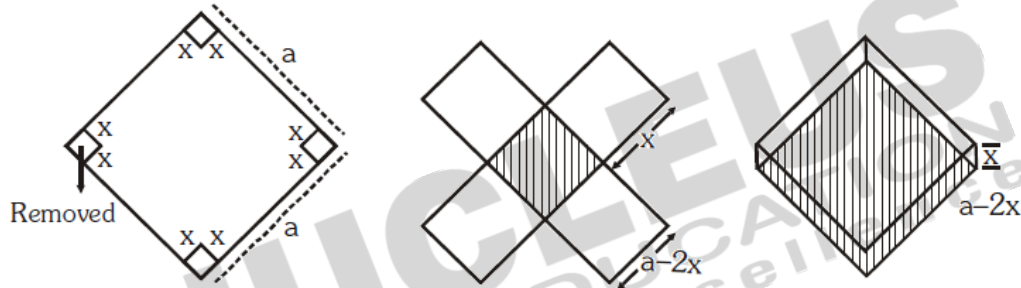
& sign of $\frac{dA}{da}$ changes from positive to negative at $a = +\frac{1}{\sqrt{2}}$



$$\Rightarrow x = \frac{1}{\sqrt{2}} \text{ are points of maxima} \Rightarrow A_{\max} = \frac{2}{\sqrt{2}} \cdot e^{-\left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\sqrt{2}}{e^{1/2}} \text{ sq units. Ans.}$$

Illustration 9 : A box of maximum volume with top open is to be made by cutting out four equal squares from four corners of a square tin sheet of side length a ft, and then folding up the flaps. Find the side of the square base cut off.

Solution : Volume of the box is, $V = x(a - 2x)^2$ i.e., squares of side x are cut out then we will get a box with a square base of side $(a - 2x)$ and height x .



$$\therefore \frac{dV}{dx} = (a - 2x)^2 + x \cdot 2(a - 2x)(-2)$$

$$\frac{dV}{dx} = (a - 2x)(a - 6x)$$

$$\text{For } V \text{ to be extremum } \frac{dV}{dx} = 0 \Rightarrow x = a/2, a/6$$

But when $x = a/2$; $V = 0$ (minimum) and we know minimum and maximum occurs alternately in a continuous function.

Hence, V is maximum when $x = a/6$.

Ans.

Illustration 10 : If a right circular cylinder is inscribed in a given cone. Find the dimension of the cylinder such that its volume is maximum.

Solution : Let x be the radius of cylinder and y be its height

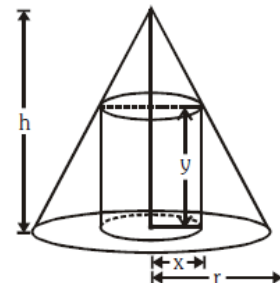
$$V = \pi x^2 y$$

x, y can be related by using similar triangles

$$\frac{y}{r - x} = \frac{h}{r} \Rightarrow y = \frac{h}{r}(r - x)$$

$$\Rightarrow V(x) = \pi r^2 \frac{h}{r}(r - x) \Rightarrow V(x) = \frac{\pi h}{r}(rx^2 - x^3)$$

$$V'(x) = \frac{\pi h}{r}(2rx - 3x^2)$$



$$V'(x) = 0 \Rightarrow x = 0, \frac{2r}{3}$$

$$V''(x) = 2\pi h \Rightarrow x = 0 \text{ is point of minima}$$

$$V''(x) = \frac{\pi h}{r} (2r - 6x)$$

$$V''\left(\frac{2r}{3}\right) = -2\pi h \Rightarrow x = \frac{2r}{3} \text{ is point of maxima}$$

$$\text{Thus volume is maximum at } x = \left(\frac{2r}{3}\right) \text{ and } y = \frac{h}{3}.$$

Do yourself - 4 :

- (i) Find the two positive numbers x & y such that their sum is 60 and xy^3 is maximum.
- (ii) If from a wire of length 36 metre, a rectangle of greatest area is made, then find its two adjacent sides in metre.
- (iii) If $ab = 2a + 3b$ where $a > 0$, $b > 0$, then find the minimum value of ab .
- (iv) Of all closed right circular cylinders of a given volume of 100 cubic centimetres, find the dimensions of cylinder which has minimum surface area.

Important note :

- (i) If the sum of two real numbers x and y is constant then their product is maximum if they are equal.

$$\text{i.e. } xy = \frac{1}{4} [(x+y)^2 - (x-y)^2]$$

- (ii) If the product of two positive numbers is constant then their sum is least if they are equal.

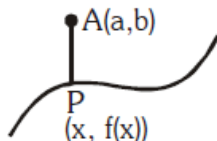
$$\text{i.e. } (x+y)^2 = (x-y)^2 + 4xy$$

5. LEAST/GREATEST DISTANCE BETWEEN TWO CURVES :

Least/Greatest distance between two non-intersecting curves always lies along the common normal. (Wherever defined)



Note : Given a fixed point $A(a, b)$ and a moving point $P(x, f(x))$ on the curve $y = f(x)$. Then AP will be maximum or minimum if it is normal to the curve at P .



$$\text{Proof : } F(x) = (x-a)^2 + (f(x)-b)^2 \Rightarrow F'(x) = 2(x-a) + 2(f(x)-b) \cdot f'(x)$$

$$\therefore f'(x) = -\left(\frac{x-a}{f(x)-b}\right). \text{ Also } m_{AP} = \frac{f(x)-b}{x-a}. \text{ Hence } f'(x) \cdot m_{AP} = 1.$$

Illustration 11 : Find the co-ordinates of the point on the curve $x^2 = 4y$, which is at least distance from the line $y = x - 4$.

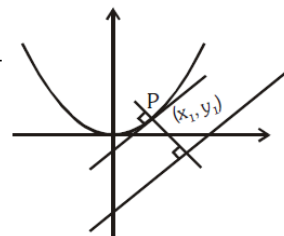
Solution : Let $P(x_1, y_1)$ be a point on the curve $x^2 = 4y$ at which normal is also a perpendicular to the line $y = x - 4$.

$$\text{Slope of the tangent at } (x_1, y_1) \text{ is } 2x = 4 \frac{dy}{dx} \Rightarrow \frac{dy}{dx}_{(x_1, y_1)} = \frac{x_1}{2}$$

$$\therefore \frac{x_1}{2} = 1 \Rightarrow x_1 = 2$$

$$\therefore x_1^2 = 4y_1 \Rightarrow y_1 = 1$$

Hence required point is (2, 1)



Do yourself - 5 :

- (i) Find the coordinates of point on the curve $y^2 = 8x$, which is at minimum distance from the line $x + y = -2$.

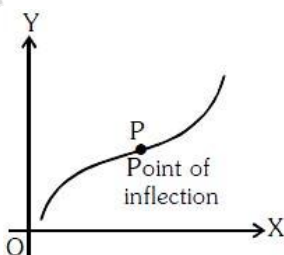
6. SOME SPECIAL POINTS ON A CURVE :

(a) **Critical points :** The points of domain for which $f'(x)$ is equal to zero or doesn't exist are called critical points.

(b) **Stationary points:** The stationary points are the points of domain where $f'(x) = 0$.

Note : Every stationary point is a critical point but vice-versa is not true.

(c) **Point of inflection :** A point where the graph of a function has a tangent line and where the concavity changes is called a point of inflection. If function $y = f(x)$ is double differentiable then the point at which $\frac{d^2y}{dx^2} = 0$ & changes its sign is the point of inflection.



Note : If at any point $\frac{d^2y}{dx^2}$ does not exist but sign of $\frac{d^2y}{dx^2}$ changes about this point then it is also called point of inflection.

Illustration 12 : Find the critical point(s) & stationary point(s) of the function $f(x) = (x-2)^{2/3}(2x+1)$

Solution : $f(x) = (x-2)^{2/3}(2x+1)$

$$f'(x) = (x-2)^{2/3} \cdot 2 + (2x+1) \frac{2}{3} (x-2)^{-1/3} = 2(x-2)^{2/3} + \frac{2}{3} (2x+1) \frac{1}{(x-2)^{1/3}}$$

$$= \left[2(x-2) + \frac{2}{3}(2x+1) \right] \frac{1}{(x-2)^{1/3}} = \frac{2(5x-5)}{3(x-2)^{1/3}}$$

$f'(x)$ does not exist at $x = 2$ and $f'(x) = 0$ at $x = 1$
 $\therefore x = 1, 2$ are critical points and $x = 1$ is stationary point.

Illustration 13 : The point of inflection for the curve $y = x^{\frac{5}{3}}$ is -

- (A) (1, 1) (B) (0, 0) (C) (1, 0) (D) (0, 1)

Solution : Here $\frac{d^2y}{dx^2} = \frac{10}{9x^{1/3}}$

From the given points we find that (0, 0) is the point of the curve where $\frac{d^2y}{dx^2}$ does

not exist but sign of $\frac{d^2y}{dx^2}$ changes about this point.

$\therefore$ (0, 0) is the required point

Ans. (B)

Illustration 14 : Find the inflection point of $f(x) = 3x^4 - 4x^3$. Also draw the graph of $f(x)$ giving due importance to maxima, minima and concavity.

Solution :

$$f(x) = 3x^4 - 4x^3$$

$$f'(x) = 12x^3 - 12x^2$$

$$f'(x) = 12x^2(x - 1)$$

$$f'(x) = 0 \Rightarrow x = 0, 1$$

examining sign change of $f'(x)$

thus $x = 1$ is a point of local minima

$$f''(x) = 12(3x^2 - 2x)$$

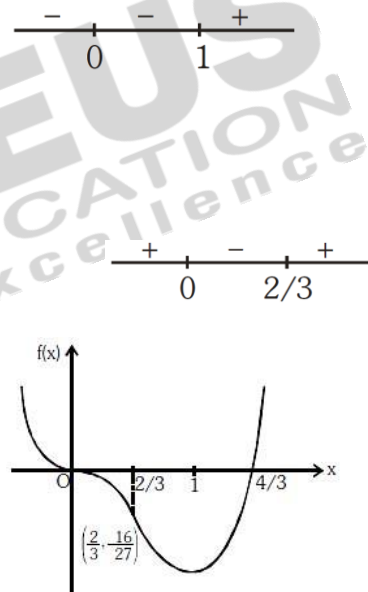
$$f''(x) = 12x(3x - 2)$$

$$f''(x) = 0$$

$$\Rightarrow x = 0, \frac{2}{3}$$

Again examining sign of $f''(x)$ thus $x = 0, \frac{2}{3}$

are the inflection points Hence the graph of $f(x)$ is



Do yourself - 6 :

- (i) Find the critical points and stationary point of the function $f(x) = \frac{e^x}{x}$
- (ii) Find the point of inflection for the curve $y = x^3 - 6x^2 + 12x + 5$
- (iii) Find the intervals for $f(x) = \frac{x^4}{12} - \frac{5x^3}{6} + 3x^2 + 7$ in which it is
 - (a) concave upward.
 - (b) concave downward.
 - (c) Hence find the points of inflection of $f(x)$.

Miscellaneous Illustrations :

Illustration 15 : Let a cuboid having square base has area 6. Then find its maximum volume.

Solution :

$$\text{Total area} = 2a^2 + 4ah = 6$$

$$h = \frac{(6 - 2a^2)}{4a} \Rightarrow V = a^2h = \frac{a^2(6 - 2a^2)}{4a}$$

$$\frac{dv}{da} = 6 - 6a^2 = 0$$

$$a = 1 \text{ and } \frac{d^2v}{da^2} = -ve$$

$$\text{Minimum } V = 1$$



ANSWERS FOR DO YOURSELF

- 1 : (i) local max. at $x = -1$, local min. at $x = 1$ (ii) 121
- 2 : (i) $1/e$ (ii) $1/e^2$
- 3 : (i) local minima at $x = 3$
- 4 : (i) $x = 15$ & $y = 45$ (ii) 9 & 9 (iii) 24 (iv) $r = \left(\frac{50}{\pi}\right)^{1/3}$ cm. and $h = 2\left(\frac{50}{\pi}\right)^{1/3}$ cm.
- 5 : (i) (2, -4)
- 6 : (i) $x = 1$ is a critical point as well as stationary point (Note $x = 0$ is not in the domain of $f(x)$)
(ii) $x = 2$ (iii) (a) $(-\infty, 2) \cup (3, \infty)$ (b) (2, 3) (c) $x = 2$ & $x = 3$

MAXIMA - MINIMA

EXERCISE # 1

- If $f(x) = |x| + |x - 1| + |x - 2|$, then-
 (A) $f(x)$ has minima at $x = 1$
 (B) $f(x)$ has maxima at $x = 0$
 (C) $f(x)$ has neither maxima nor minima at $x = 3$
 (D) None of these
- The maximum area of a right angled triangle with hypotenuse h is :-
[JEE-MAIN Online 2013]
 (A) $\frac{h^2}{\sqrt{2}}$ (B) $\frac{h^2}{2}$ (C) $\frac{h^2}{4}$ (D) $\frac{h^2}{2\sqrt{2}}$
- The cost of running a bus from A to B, is Rs. $\left(av + \frac{b}{v}\right)$, where v km/h is the average speed of the bus. When the bus travels at 30 km/h, the cost comes out to be Rs. 75 while at 40 km/h, it is Rs. 65. Then the most economical speed (in km/h) of the bus is: **[JEE-MAIN Online 2013]**
 (A) 40 (B) 60 (C) 45 (D) 50
- Point 'A' lies on the curve $y = e^{-x^2}$ and has the coordinate (x, e^{-x^2}) where $x > 0$. Point B has the coordinates $(x, 0)$. If 'O' is the origin then the maximum area of the triangle AOB is
 (A) $\frac{1}{\sqrt{2e}}$ (B) $\frac{1}{\sqrt{4e}}$ (C) $\frac{1}{\sqrt{e}}$ (D) $\frac{1}{\sqrt{8e}}$
- The greatest value of $x^3 - 18x^2 + 96x$ in the interval $(0, 9)$ is-
 (A) 128 (B) 60 (C) 160 (D) 120
- Difference between the greatest and the least values of the function $f(x) = x(\ln x - 2)$ on $[1, e^2]$ is
 (A) 2 (B) e (C) e^2 (D) 1
- A maximum point of $3x^4 - 2x^3 - 6x^2 + 6x + 1$ in $[0, 2]$ is-
 (A) $x = 0$ (B) $x = 1$ (C) $x = 1/2$ (D) does not exist
- The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :
 (A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $5\pi/12$
- If $xy = c^2$ then the minimum value of $ax + by$ ($a > 0, b > 0, c > 0$) is-
 (A) $c\sqrt{ab}$ (B) $-c\sqrt{ab}$ (C) $2c\sqrt{ab}$ (D) $-2c\sqrt{ab}$

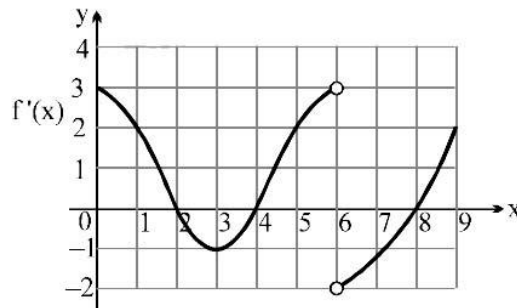
10. Consider the function $f(x) = x \cos x - \sin x$, then identify the statement which is correct .
 (A) f is neither odd nor even (B) f is monotonic decreasing at $x = 0$
 (C) f has a maxima at $x = \pi$ (D) f has a minima at $x = -\pi$
11. If $f(x) = x^3 + ax^2 + bx + c$ is minimum at $x = 3$ and maximum at $x = -1$, then-
 (A) $a = -3, b = -9, c = 0$ (B) $a = 3, b = 9, c = 0$
 (C) $a = -3, b = -9, c \in \mathbb{R}$ (D) none of these
12. If $f(x) = \int_x^{x^2} (t-1) dt, 1 \leq x \leq 2$, then global maximum value of $f(x)$ is
 (A) 1 (B) 2 (C) 4 (D) 5
13. Minimum value of the function $f(x) = \sum_{k=1}^5 (x-k)^2$ is at-
 (A) $x = 2$ (B) $x = 5/2$ (C) $x = 3$ (D) $x = 5$
14. Range of the function $f(x) = \frac{\ln x}{\sqrt{x}}$ is
 (A) $(-\infty, e)$ (B) $(-\infty, e^2)$ (C) $\left(-\infty, \frac{2}{e}\right]$ (D) $\left(-\infty, \frac{1}{e}\right]$
15. If $ax + \frac{b}{x} \geq c$ for all positive x , where $a, b > 0$, then-
 (A) $ab < \frac{c^2}{4}$ (B) $ab \geq \frac{c^2}{4}$ (C) $ab \geq \frac{c}{4}$ (D) None of these
16. Two sides of a triangle are to have lengths 'a' cm & 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is
 (A) $\frac{1}{2}\sqrt{a^2 + b^2}$ (B) $\frac{2a+b}{3}$ (C) $\sqrt{\frac{a^2 + b^2}{2}}$ (D) $\frac{a+2b}{3}$
17. $f(x) = 1 + [\cos x]x$, in $0 \leq x \leq \frac{\pi}{2}$
 (where $[.]$ denotes greatest integer function)
 (A) has a minimum value 0
 (B) has a maximum value 2
 (C) is continuous in $\left[0, \frac{\pi}{2}\right]$
 (D) is not differentiable at $x = \frac{\pi}{2}$

18. A rectangle has one side on the positive y-axis and one side on the positive x - axis. The upper right hand vertex of the rectangle lies on the curve $y = \frac{\ln x}{x^2}$. The maximum area of the rectangle is
(A) e^{-1} (B) $e^{-\frac{1}{2}}$ (C) 1 (D) $e^{\frac{1}{2}}$
19. If $(x - a)^{2m}(x - b)^{2n+1}$, where m and n are positive integers and $a > b$, is the derivative of a function f, then-
(A) $x = a$ gives neither a maximum, nor a minimum
(B) $x = a$ gives a maximum
(C) $x = b$ gives neither a maximum nor a minimum
(D) None of these
20. Consider the following behaviours of function in $(-1, 1)$
I. Increasing **II.** Decreasing **III.** Continuity **IV.** Derivability
Which one of the following four functions exhibits atleast three of the four mentioned above ?
(A) $\max\{x, x^3\}$ (B) $\max\{x, x^2\}$ (C) $\max\{x, |x|\}$ (D) $\max\{x, [x]\}$
where [] denotes greatest integer function.
21. The minimum value of $a \sec x + b \operatorname{cosec} x$, $0 < a < b$, $0 < x < \pi/2$ is-
(A) $a + b$ (B) $a^{2/3} + b^{2/3}$ (C) $(a^{2/3} + b^{2/3})^{3/2}$ (D) None of these
22. P is a point on positive x-axis, Q is a point on the positive y-axis and 'O' is the origin. If the line passing through P and Q is tangent to the curve $y = 3 - x^2$ then the minimum area of the triangle OPQ, is
(A) 2 (B) 4 (C) 8 (D) 9
23. A minimum value of $\sin x \cos 2x$ is-
(A) 1 (B) -1 (C) $-2/3 \sqrt{6}$ (D) None of these
24. The rate of change of the function $f(x) = 3x^5 - 5x^3 + 5x - 7$ is minimum when
(A) $x = 0$ (B) $x = 1/\sqrt{2}$ (C) $x = -1/\sqrt{2}$ (D) $x = \pm 1/\sqrt{2}$
25. For the function $f(x) = \int_0^x \frac{\sin t}{t} dt$, where $x > 0$,
(A) maximum occurs at $x = n\pi$, n is even (B) minimum occurs at $x = n\pi$, n is odd
(C) maximum occurs at $x = n\pi$, n is odd (D) None of these
26. The least area of a circle circumscribing any right triangle of area S is :
(A) πS (B) $2\pi S$ (C) $\sqrt{2} \pi S$ (D) $4 \pi S$

EXERCISE # 2

1. A cubic $f(x)$ vanishes at $x = -2$ & has relative minimum/maximum at $x = -1$ and $x = 1/3$.
If $\int_{-1}^1 f(x) dx = \frac{14}{3}$, find the cubic $f(x)$.
2. Investigate for maxima & minima for the function, $f(x) = \int_1^x [2(t-1)(t-2)^3 + 3(t-1)^2(t-2)^2] dt$
3. Find the greatest & least value for the function ;
(a) $y = x + \sin 2x$, $0 \leq x \leq 2\pi$ (b) $y = 2 \cos 2x - \cos 4x$, $0 \leq x \leq \pi$
4. Suppose $f(x)$ is a function satisfying the following conditions :
(i) $f(0) = 2$, $f(1) = 1$ (ii) f has a minimum value at $x = \frac{5}{2}$ and
(iii) for all x , $f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$
Where a, b are some constants. Determine the constants a, b & the function $f(x)$.
5. Let $P(x)$ be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4$.
If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set
 $A = \{x | x^2 + 6 \leq 5x\}$ then find $\frac{m}{M}$.
6. The length of three sides of a trapezium are equal, each being 10 cms. Find the maximum area of such a trapezium.
7. The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of length ' $2r$ ' and width ' s '. If the surface area A of the pool is fixed, for what value of ' r ' and ' s ' the perimeter ' P ' of the pool is minimum.
8. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2+3}$, find the equations of the tangent lines of minimum and maximum slope.
9. By the post office regulations, the combined length & girth of a parcel must not exceed 3 metre. Find the volume of the biggest cylindrical (right circular) packet that can be sent by the parcel post.

10. A running track of 440 ft. is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, find the length of its sides.
Use : $\pi \approx 22/7$.
11. A closed rectangular box with a square base is to be made to contain 1000 cubic feet. The cost of the material per square foot for the bottom is 15 paise, for the top 25 paise and for the sides 20 paise. The labour charges for making the box are Rs. 3/-. Find the dimensions of the box when the cost is minimum.
12. Find the area of the largest rectangle with lower base on the x-axis & upper vertices on the curve $y = 12 - x^2$.
13. If $y = \frac{ax+b}{(x-1)(x-4)}$ has a turning value at (2, -1) find a & b and show that the turning value is a maximum.
14. What are the dimensions of the rectangular plot of the greatest area which can be laid out within a triangle of base 36 ft. & altitude 12 ft ? Assume that one side of the rectangle lies on the base of the triangle.
15. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has a local minimum at $x = 1$. If $f(-1) = 10$ and $f(3) = -22$, then find the distance between its two horizontal tangents.
16. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$. If
- f is monotonic increasing in the interval $[a,b) \cup (c,d) \cup (e,f]$ and decreasing in $(p,q) \cup (r,s)$.
 - f has a local minima at $x = x_1$ and $x = x_2$.
 - f is concave up in $(l,m) \cup (n,t]$
 - f has inflection point at $x = k$
 - number of critical points of $y = f(x)$ is 'w'.



Find the value of $(a + b + c + d + e) + (p + q + r + s) + (\ell + m + n) + (x_1 + x_2) + (k + w)$.

EXERCISE # 3 (JM)

- If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals-
[AIEEE-2003]
(1) $1/2$ (2) 3 (3) 1 (4) 2
- The real number x when added to its inverse gives the minimum value of the sum at x equal to-
[AIEEE-2003]
(1) -2 (2) 2 (3) 1 (4) -1
- If $u = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$ then the difference between the maximum and minimum values of u^2 is given by-
[AIEEE-2004]
(1) $2(a^2 + b^2)$ (2) $2\sqrt{a^2 + b^2}$ (3) $(a + b)^2$ (4) $(a - b)^2$
- The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at-
[AIEEE-2006]
(1) $x = -2$ (2) $x = 0$ (3) $x = 1$ (4) $x = 2$
- A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x . The maximum area enclosed by the park is-
[AIEEE-2006]
(1) $\sqrt{\frac{x^3}{8}}$ (2) $\frac{1}{2}x^2$ (3) πx^2 (4) $\frac{3}{2}x^2$
- If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is-
[AIEEE-2007]
(1) 2 (2) $\frac{1}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$
- Suppose the cubic $x^3 - px + q$ has three real roots where $p > 0$ and $q > 0$. Then which of the following holds ?
[AIEEE-2008]
(1) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$
(2) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$
(3) The cubic has minima at both $-\sqrt{\frac{p}{3}}$ and $\sqrt{\frac{p}{3}}$
(4) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

8. Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x=0$ is the only real root of $P'(x) = 0$.
If $P(-1) < P(1)$, then in the interval $[-1, 1]$:-
(1) $P(-1)$ is the minimum but $P(1)$ is not the maximum of P . [AIEEE-2009]
(2) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P
(3) $P(-1)$ is the minimum and $P(1)$ is the maximum of P
(4) $P(-1)$ is not minimum but $P(1)$ is the maximum of P
9. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is :- [AIEEE-2009]
(1) $\frac{3\sqrt{2}}{5}$ (2) $\frac{\sqrt{3}}{4}$ (3) $\frac{3\sqrt{2}}{8}$ (4) $\frac{2\sqrt{3}}{8}$
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} k - 2x, & \text{if } x \leq -1 \\ 2x + 3, & \text{if } x > -1 \end{cases}$ [AIEEE-2010]
If f has a local minimum at $x = -1$, then a possible value of k is :
(1) 1 (2) 0 (3) $-\frac{1}{2}$ (4) -1
11. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t \, dt$, then f has :- [AIEEE-2011]
(1) local minimum at π and local maximum at 2π
(2) local maximum at π and local minimum at 2π
(3) local maximum at π and 2π
(4) local minimum at π and 2π
12. Let f be a function defined by $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$
Statement – 1 : $x = 0$ is point of minima of f .
Statement – 2 : $f'(0) = 0$. [AIEEE-2011]
(1) Statement-1 is false, statement-2 is true.
(2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.
(3) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.
(4) Statement-1 is true, statement-2 is false.
13. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :
(1) 9/2 (2) 9/7 (3) 7/9 (4) 2/9 [AIEEE-2012]

14. Let $a, b \in \mathbb{R}$ be such that the function f given by $f(x) = \ln |x| + bx^2 + ax$, $x \neq 0$ has extreme values at $x = -1$ and $x = 2$.

Statement-1 : f has local maximum at $x = -1$ and at $x = 2$.

Statement-2 : $a = \frac{1}{2}$ and $b = \frac{-1}{4}$.

[AIEEE-2012]

- (1) Statement-1 is true, Statement-2 is false.
(2) Statement-1 is false, Statement-2 is true.
(3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
(4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.

15. The real number k for which the equation $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$
(1) lies between 1 and 2. (2) lies between 2 and 3. [JEE-MAIN 2013]
(3) lies between -1 and 0 (4) does not exist

16. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$ then : [JEE-MAIN 2014]
(1) $\alpha = -6, \beta = \frac{1}{2}$ (2) $\alpha = -6, \beta = -\frac{1}{2}$ (3) $\alpha = 2, \beta = -\frac{1}{2}$ (4) $\alpha = 2, \beta = \frac{1}{2}$

17. Let $f(x)$ be a polynomial of degree four having extreme values at $x = 1$ and $x = 2$.
If $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$ is equal to : [JEE-MAIN 2015]
(1) 0 (2) 4 (3) -8 (4) -4

18. Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower-bed is : [JEE-MAIN 2017]
(1) 30 (2) 12.5 (3) 10 (4) 25

19. Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$, $x \in \mathbb{R} - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of $h(x)$ is : [JEE-MAIN 2018]
(1) $2\sqrt{2}$ (2) 3 (3) -3 (4) $-2\sqrt{2}$

20. The maximum volume (in cu. m) of the right circular cone having slant height 3m is :
(1) $3\sqrt{3}\pi$ (2) 6π (3) $2\sqrt{3}\pi$ (4) $\frac{4}{3}\pi$

[JEE-MAIN 2019]

21. The shortest distance between the point $\left(\frac{3}{2}, 0\right)$ and the curve $y = \sqrt{x}$, ($x > 0$) is:
(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{5}{4}$ (3) $\frac{3}{2}$ (4) $\frac{\sqrt{5}}{2}$ [JEE-MAIN 2019]

22. If S_1 and S_2 are respectively the sets of local minimum and local maximum points of the function $f(x) = 9x^4 + 12x^3 - 36x^2 + 25$, $x \in \mathbb{R}$, then: [JEE-MAIN 2019]
 (1) $S_1 = \{-1\}$; $S_2 = \{0, 2\}$ (2) $S_1 = \{-2\}$; $S_2 = \{0, 1\}$
 (3) $S_1 = \{-2, 1\}$; $S_2 = \{0\}$ (4) $S_1 = \{-2, 0\}$; $S_2 = \{1\}$
23. The height of a right circular cylinder of maximum volume inscribed in a sphere of radius 3 is :
 (1) $\frac{2}{3}\sqrt{3}$ (2) $\sqrt{6}$ (3) $\sqrt{3}$ (4) $2\sqrt{3}$ [JEE-MAIN 2019]
24. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10 m; is [JEE-MAIN 2019]
 (1) $1/10\pi$ (2) $1/15\pi$ (3) $1/5\pi$ (4) $2/\pi$
25. The maximum value of the function $f(x) = 3x^3 - 18x^2 + 27x - 40$ on the set $S = \{x \in \mathbb{R} : x^2 + 30 \leq 11x\}$ is : [JEE-MAIN 2019]
 (1) 222 (2) 122 (3) -122 (4) -222
26. The maximum area (in sq. units) of a rectangle having its base on the x-axis and its other two vertices on the parabola, $y = 12 - x^2$ such that the rectangle lies inside the parabola is : [JEE-MAIN 2019]
 (1) 32 (2) $20\sqrt{2}$ (3) 36 (4) $18\sqrt{3}$
27. Let $f: (1, 3) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. then the range of f is. [JEE-MAIN 2020]
 (1) $\left(\frac{3}{5}, \frac{4}{5}\right)$ (2) $\left(\frac{2}{5}, \frac{3}{5}\right) \cup \left[\frac{3}{4}, \frac{4}{5}\right)$ (3) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$ (4) $\left(\frac{2}{5}, \frac{4}{5}\right]$
28. Let $f(x)$ be a polynomial of degree 3 such that $f(-1) = 10$, $f(1) = -6$, $f(x)$ has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then $f(x)$ has local minima at $x =$ _____. [JEE-MAIN 2020]
29. A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness that melts at a rate of $50\text{cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which of the thickness of ice decreases, is : [JEE-MAIN 2020]
 (1) $\frac{1}{36\pi}$ (2) $\frac{5}{6\pi}$ (3) $\frac{1}{54\pi}$ (4) $\frac{1}{18\pi}$

EXERCISE # 4 (JA)

SECTION-1

- Let $f(x) = \begin{cases} |x| & \text{for } 0 < |x| \leq 2 \\ 1 & \text{for } x = 0 \end{cases}$. Then at $x = 0$, 'f' has : **[JEE 2000 Screening, 1 out of 35]**
 (A) a local maximum (B) no local maximum
 (C) a local minimum (D) no extremum.
- (a) Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is
 (A) $[0, 1]$ (B) $\left(0, \frac{1}{2}\right]$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $(0, 1]$
 (b) The maximum value of $(\cos \alpha_1) \cdot (\cos \alpha_2) \dots (\cos \alpha_n)$, under the restrictions $0 \leq \alpha_1, \alpha_2, \dots, \alpha_n \leq \frac{\pi}{2}$ and $\cot \alpha_1 \cdot \cot \alpha_2 \dots \cot \alpha_n = 1$ is
 (A) $\frac{1}{2^{n/2}}$ (B) $\frac{1}{2^n}$ (C) $\frac{1}{2n}$ (D) 1
[JEE 2001 Screening, 1 + 1 out of 35]
- If $a_1, a_2, \dots, a_n$ are positive real numbers whose product is a fixed number e , the minimum value of $a_1 + a_2 + a_3 + \dots + a_{n-1} + 2a_n$ is **[JEE 2002 Screening]**
 (A) $n(2e)^{1/n}$ (B) $(n+1)e^{1/n}$ (C) $2ne^{1/n}$ (D) $(n+1)(2e)^{1/n}$
- Let $f(x) = x^3 + bx^2 + cx + d$, $0 < b^2 < c$. Then f
 (A) is bounded (B) has a local maxima
 (C) has a local minima (D) is strictly increasing **[JEE 2004 (Scr.)]**
- The total number of local maxima and local minima of the function $f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$ is **[JEE 2008, 3]**
 (A) 0 (B) 1 (C) 2 (D) 3
- Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0, 1]$, then
 (A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$
 (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$ **[JEE 2010, 3]**

SECTION-2

7. Find the area of the right angled triangle of least area that can be drawn so as to circumscribe a rectangle of sides 'a' and 'b', the right angle of the triangle coinciding with one of the angles of the rectangle.
[REE 2001 Mains, 5 out of 100]
8. (a) Find a point on the curve $x^2 + 2y^2 = 6$ whose distance from the line $x + y = 7$, is minimum.
(b) For a circle $x^2 + y^2 = r^2$, find the value of 'r' for which the area enclosed by the tangents drawn from the point P(6, 8) to the circle and the chord of contact is maximum.
[JEE 2003, Mains, 2 + 2 out of 60]
9. Prove that $\sin x + 2x \geq \frac{3x(x+1)}{\pi}, \forall x \in \left[0, \frac{\pi}{2}\right]$. (Justify the inequality, if any used).
[JEE 2004, 4 out of 60]
10. If P(x) be a polynomial of degree 3 satisfying $P(-1) = 10$, $P(1) = -6$ and P(x) has maximum at $x = -1$ and P'(x) has minima at $x = 1$. Find the distance between the local maximum and local minimum of the curve.
[JEE 2005 (Mains), 4 out of 60]
11. (a) If $f(x)$ is cubic polynomial which has local maximum at $x = -1$. If $f(2) = 18$, $f(1) = -1$ and $f'(x)$ has local minima at $x = 0$, then
(A) the distance between $(-1, 2)$ and $(a, f(a))$, where $x = a$ is the point of local minima is $2\sqrt{5}$.
(B) $f(x)$ is increasing for $x \in (1, 2\sqrt{5}]$
(C) $f(x)$ has local minima at $x = 1$
(D) the value of $f(0) = 5$
- (b) $f(x) = \begin{cases} e^x & 0 \leq x \leq 1 \\ 2 - e^{x-1} & 1 < x \leq 2 \\ x - e & 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t) dt, x \in [1, 3]$ then $g(x)$ has
(A) local maxima at $x = 1 + \ln 2$ and local minima at $x = e$
(B) local maxima at $x = 1$ and local minima at $x = 2$
(C) no local maxima
(D) no local minima
[JEE 2006, 5 marks each]
- (c) If $f(x)$ is twice differentiable function such that $f(a) = 0$, $f(b) = 2$, $f(c) = -1$, $f(d) = 2$, $f(e) = 0$, where $a < b < c < d < e$, then find the minimum number of zeros of $g(x) = (f'(x))^2 + f(x) \cdot f''(x)$ in the interval $[a, e]$.
[JEE 2006, 6]

12. **Comprehension:**

Consider the function $f : (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

(i) Which of the following is true?

- (A) $(2+a)^2 f''(1) + (2-a)^2 f''(-1) = 0$ (B) $(2-a)^2 f''(1) - (2+a)^2 f''(-1) = 0$
(C) $f'(1) f'(-1) = (2-a)^2$ (D) $f'(1) f'(-1) = -(2+a)^2$

(ii) Which of the following is true?

- (A) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$
(B) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$
(C) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$.
(D) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$.

(iii) Let $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$

Which of the following is true?

- (A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$
(B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
(C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
(D) $g'(x)$ does not change sign on $(-\infty, \infty)$

[JEE 2008, 4 + 4 + 4]

13. (a) Let $p(x)$ be a polynomial of degree 4 having extremum at $x = 1, 2$ and $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2} \right) = 2$.

Then the value of $p(2)$ is

(b) The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set

$A = \{x \mid x^2 + 20 \leq 9x\}$ is

[JEE 2009, 4 + 4]

14. Let f be a function defined on $\mathbf{R}$ (the set of all real numbers) such that

$$f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4, \text{ for all } x \in \mathbf{R}.$$

If g is a function defined on $\mathbf{R}$ with values in the interval $(0, \infty)$ such that

$$f(x) = \ln(g(x)), \text{ for all } x \in \mathbf{R},$$

then the number of points in $\mathbf{R}$ at which g has a local maximum is

[JEE 2010, 3]

15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is **[JEE 2012, 4M]**
16. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is **[JEE 2012, 4M]**
17. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio of 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are **[JEE 2013, 4M, -1M]**
(A) 24 (B) 32 (C) 45 (D) 60
18. The function $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ has a local minimum or a local maximum at $x =$ **[JEE 2013, 3M, -1M]**
(A) -2 (B) $-\frac{2}{3}$ (C) 2 (D) $\frac{2}{3}$
19. A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container. If the volume of the material used to make the container is minimum when the inner radius of the container is 10mm, then the value of $\frac{V}{250\pi}$ is **[JEE 2015, 4M, 0M]**
20. If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then **[JEE- Advanced 2017]**
(A) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$
(B) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$
(C) $f(x)$ attains its maximum at $x = 0$
(D) $f(x)$ attains its minimum at $x = 0$

EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- Let $f(x) = \frac{e^x}{1+x^2}$ and $g(x) = f'(x)$ then
 (A) $g(x)$ has two local maxima and two local minima points
 (B) $g(x)$ has exactly one local maxima and one local minima point
 (C) $x = 1$ is a point of local maxima for $g(x)$
 (D) There is a point of local maxima for $g(x)$ in the interval $(-1, 0)$
- Let $F(x) = (f(x))^2 + (f'(x))^2$, $F(0) = 6$ where $f(x)$ is a thrice differentiable function such that $|f(x)| \leq 1 \forall x \in [-1, 1]$, then choose the correct statement(s)
 (A) there is at least one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $|f'(x)| \leq 2$
 (B) there is at least one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $F(x) \leq 5$
 (C) there is no point of local maxima of $F(x)$ in $(-1, 1)$
 (D) for some $c \in (-1, 1)$, $F(c) \geq 6$, $F'(c) = 0$ and $F''(c) \leq 0$
- Suppose that $f(x)$ is a differentiable invertible function $f'(x) \neq 0$ and $h(x) = \int_1^x f(t) dt$.
 Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$. Let $G(x) = x^2 g(x) - xh(g(x)) \forall x \in \mathbb{R}$.
 Which of the following are correct
 (A) $G'(1) = 2$ (B) $G'(1) = 3$ (C) $G''(1) = 2$ (D) $G''(1) = 3$
- Let $y = f(x)$ be a twice differentiable function such that $f''(x) > 0 \forall x \in \mathbb{R}$.
 Let $g(x) = f^{-1}(x)$ ($f(x)$ is invertible).
 $f(x) = g(x)$ has two distinct real roots $x = 1$ and 2 only. If $\int_1^2 f(x) dx = 1$ then which of the following must be correct?
 (A) $g''(x) < 0 \forall x \in \mathbb{R}$ (B) $\int_1^2 g(x) dx = 2$
 (C) $g'(x) < 0 \forall x \in \mathbb{R}$ (D) $f'(x) > 0 \forall x \in \mathbb{R}$
- Let $f(x) = (x-1)^m (x-2)^n$, $x \in \mathbb{R}$, then each critical point of $f(x)$ is either local maximum or local minimum where
 (A) $m=2, n=3$ (B) $m=2, n=4$
 (C) $m=3, n=4$ (D) $m=4, n=2$

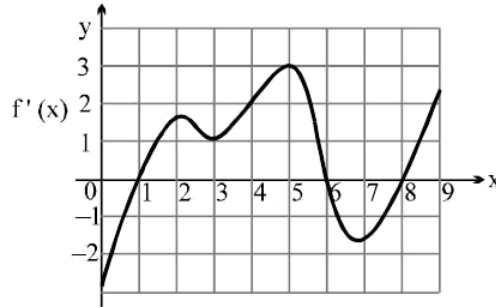
6. Given $f(x) = x^4 \cdot e^{-x^2}$; $x \in \mathbb{R}$; then
 (A) Minimum value of $f(x)$ is 0
 (B) Maximum value of $f(x)$ is $4e^{-2}$
 (C) Line $y = 4$ cuts $f(x)$ at 2 points
 (D) $f(x)$ is symmetric about y-axis
7. If $f(x) = (\log x)^n - x$, then -
 (A) $f(x) = 0$ has exactly two solutions for $n = 5$
 (B) $f(x) = 0$ has exactly one solution for $n = 5$
 (C) $f(x) = 0$ has no solutions if $n \in (0, 1)$
 (D) $f(x) = 0$ has exactly two solutions if $n \in (0, 1)$
8. Let $f(x) = \int_0^x \frac{\sin t}{t} dt$, $x > 0$. Then
 (A) $f(x)$ has local maxima at $x = n\pi$, where $n = 2k$, $k \in \mathbb{I}^+$.
 (B) $f(x)$ has local minima at $x = n\pi$, where $n = 2k$, $k \in \mathbb{I}^+$.
 (C) $f(x)$ has neither maxima nor minima at $x = n\pi$, where $n = k$, $k \in \mathbb{I}^+$.
 (D) $f(x)$ has local maxima at $x = n$, where $n = 2k-1$, $k \in \mathbb{I}^+$.
9. Let $f(x) = \sin^{-1}(\cos x) \cos^{-1}(\sin x)$, $\forall x \in [0, 2\pi]$ then which of the following statement(s) is/are correct?
 (A) $f(x)$ is continuous and differentiable in $[0, 2\pi]$.
 (B) Range of $f(x)$ is $\left[-\frac{\pi^2}{4}, \frac{\pi^2}{4}\right]$.
 (C) $x = \pi$ is a point of global minima as well as local minima.
 (D) $\int_0^\pi f(x) dx = 0$
10. Let $f'(x) = (x^2 - x + 2)(x^2 - x - 2)(x^2 - x - 6)(x^2 - x - 12)$, $x \in \mathbb{R}$, then
 (A) the equation of normal to $f(x)$ at $x = -2$ is $x = -2$.
 (B) the sum of values of x at which $f(x)$ has local maximum is -1 .
 (C) the minimum number of stationary points of $f'(x)$ is 5.
 (D) the value of $\lim_{x \rightarrow \infty} \left(\frac{f(x)}{2x^9} \right)$ equals $\frac{\pi}{2}$.
11. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ ('a' is a finite quantity), where $[\cdot]$ denotes greatest integer function and $f(x)$ is a non constant continuous function, then
 (A) $\lim_{x \rightarrow a} f(x)$ is an integer.
 (B) $\lim_{x \rightarrow a} f(x)$ need not be an integer.
 (C) $f(x)$ has a local minimum at $x = a$
 (D) $f(x)$ has a local maximum at $x = a$.

12. Let $f(x)$ be a cubic polynomial such that it has point of inflection at $x = 2$ and local minima at $x = 4$, then-
- (A) $f(x)$ has local minima at $x = 0$ (B) $f(x)$ has local maxima at $x = 0$
 (C) $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$ (D) $\lim_{x \rightarrow \infty} f(x) \rightarrow -\infty$
13. If $f(x) = x^3 + bx^2 + cx + d$ and $I = \int_{x_1}^{x_2} f(x) dx$, then which of the following option(s) is/are correct: (where $x_1 < x_2$, and $b, c, d \in \mathbb{R}$)
- (A) If $f(x_1) > 0$, $f(x_2) < 0$ and $I = 0$, then $f(x) = 0$ has exactly one real root in (x_1, x_2)
 (B) If $f(x_1) < 0$, $f(x_2) < 0$ and $I = 0$, then $f(x) = 0$ has exactly two real roots in (x_1, x_2)
 (C) If $f(x_1) < 0$, $f(x_2) > 0$ and $I = 0$, then $f(x) = 0$ has atleast one real root in (x_1, x_2)
 (D) If $f(x_1) < 0$, $f(x_2) > 0$ and $I = 0$, then $f(x) = 0$ has exactly one real root in (x_1, x_2)
14. Let $f(x) = 2x^3 - 15x^2 + 36x - 48$, then-
- (A) $f(x)$ is increasing function in $(-\infty, 2) \cup (3, \infty)$
 (B) graph of $y = f(x)$ is concave upwards in $\left(\frac{5}{2}, \infty\right)$
 (C) Point of inflection of $f(x)$ at $x = \frac{5}{2}$
 (D) Range of $f(x)$ for $x \in [2, 4]$ is $[-21, -16]$
15. If $g(x) = 7x^2 \cdot e^{-x^2} \forall x \in \mathbb{R}$, then $g(x)$ has
- (A) local maxima at $x = 0$ (B) local minima at $x = 0$
 (C) local maxima at $x = -1$ (D) two local maxima and one local minima
16. Which of the following statements is/are correct?
- (A) $\lim_{x \rightarrow 0} \frac{\int_0^x x e^{t^2} dt}{1 + x - e^x} = -2$
- (B) The points L and M are on the curve $14x^2 - 7xy + y^2 - 2 = 0$, each having x-coordinate 1. Then the y-coordinate of point of intersection of tangents at L and M is 4
- (C) Let $f(x) = |x - a_1| + |x - a_2| + \dots + |x - a_n|$, $a_i \in \mathbb{R}$ and $a_i < a_{i+1} \forall i \in \mathbb{N}$, $i \leq n-1$ if n is odd integer then $f(x)$ has minimum value exactly at one point
- (D) If LMVT is known to be applicable for a quadratic function $y = px^2 + qx + r$, $x \in [x_1, x_2]$ then 'c' of LMVT occurs at $c = \left(\frac{x_1 + x_2}{3}\right)$

EXERCISE # 6

1. Find the maximum perimeter of a triangle on a given base 'a' and having the given vertical angle α .
2. A statue 4 metres high sits on a column 5.6 metres high. How far from the column must a man, whose eye level is 1.6 metres from the ground, stand in order to have the most favourable view of statue.
3. A perpendicular is drawn from the centre to a tangent to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the greatest value of the intercept between the point of contact and the foot of the perpendicular.
4. A beam of rectangular cross section must be sawn from a round log of diameter d. What should the width x and height y of the cross section be for the beam to offer the greatest resistance (a) to compression; (b) to bending. Assume that the compressive strength of a beam is proportional to the area of the cross section and the bending strength is proportional to the product of the width of section by the square of its height.
5. Given two points A (-2, 0) & B (0, 4) and a line $y = x$. Find the co-ordinates of a point M on this line so that the perimeter of the ΔAMB is least.
6. A given quantity of metal is to be casted into a half cylinder i.e. with a rectangular base and semicircular ends. Show that in order that total surface area may be minimum, the ratio of the height of the cylinder to the diameter of the semi circular ends is $\pi/(\pi + 2)$.
7. Consider the function $f(x) = \begin{cases} \sqrt{x} \ln x & \text{when } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$
 - (a) Find whether f is continuous at $x = 0$ or not.
 - (b) Find the minima and maxima if they exist.
 - (c) Does $f'(0)$ exist? Find $\lim_{x \rightarrow 0} f'(x)$.
 - (d) Find the inflection points of the graph of $y = f(x)$.
8. Consider the function $y = f(x) = \ln(1 + \sin x)$ with $-2\pi \leq x \leq 2\pi$. Find
 - (a) the zeroes of $f(x)$
 - (b) inflection points if any on the graph
 - (c) local maxima and minima of $f(x)$
 - (d) asymptotes of the graph
 - (e) sketch the graph of $f(x)$ and compute the value of the definite integral $\int_{-\pi/2}^{\pi/2} f(x) dx$.

9. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$
- On what intervals is f increasing or decreasing?
 - At what values of x does f have a local maximum or minimum?
 - On what intervals is f concave upward or downward?
 - State the x -coordinate(s) of the point(s) of inflection.
 - Assuming that $f(0) = 0$, sketch a graph of f .



10. Find the set of value of m for the cubic $x^3 - \frac{3}{2}x^2 + \frac{5}{2} = \log_{1/4}(m)$ has 3 distinct solutions.
11. A cylinder is obtained by revolving a rectangle about the x -axis, the base of the rectangle lying on the x -axis and the entire rectangle lying in the region between the curve $y = \frac{x}{x^2 + 1}$ & the x -axis. Find the maximum possible volume of the cylinder.
12. The value of ' a ' for which $f(x) = x^3 + 3(a - 7)x^2 + 3(a^2 - 9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.
13. The function $f(x)$ defined for all real numbers x has the following properties
- $f(0) = 0$, $f(2) = 2$ and $f'(x) = k(2x - x^2)e^{-x}$ for some constant $k > 0$. Find
 - the intervals on which f is increasing and decreasing and any local maximum or minimum values.
 - the intervals on which the graph f is concave down and concave up.
 - the function $f(x)$ and plot its graph.
14. Use calculus to prove the inequality, $\sin x \geq 2x/\pi$ in $0 \leq x \leq \pi/2$.
Use this inequality to prove that, $\cos x \leq 1 - x^2/\pi$ in $0 \leq x \leq \pi/2$.
15. The circle $x^2 + y^2 = 1$ cuts the x -axis at P & Q . Another circle with centre at Q and variable radius intersects the first circle at R above the x -axis & the line segment PQ at S . Find the maximum area of the triangle QSR .

ANSWER KEY

TANGENT & NORMAL

EXERCISE # 1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. A | 4. C | 5. B | 6. C | 7. D |
| 8. C | 9. B | 10. C | 11. D | 12. D | 13. C | 14. C |
| 15. D | 16. C | 17. A | 18. D | 19. C | 20. A | 21. D |
| 22. C | 23. A | 24. D | 25. C | | | |

EXERCISE # 2

- | | |
|--|--|
| 1. $x + y - 1 = 0$ | 2. $x = 1$ when $t = 1$, $m \rightarrow \infty$; $5x - 4y = 1$ if $t \neq 1$, $t = 1/3$ |
| 3. $a = 1$, $b = \frac{-5}{2}$ | 4. $-\frac{82.7^3}{3}$ |
| 5. $x + 2y = \pi/2$ & $x + 2y = -3\pi/2$ | |
| 6. $a = -1/2$; $b = -3/4$; $c = 3$ | 7. 3, 12 |
| 8. $2e^{-x/2}$ | 9. (a) $n = -2$ |
| 10. (b) $\pm \frac{1}{2\sqrt{2}}$ | 11. $\theta = \tan^{-1} \frac{2}{C}$ |
| 12. $1/9 \pi \text{ m/min}$ | 13. (i) 6 km/h; (ii) 2 km/hr |
| 14. (4, 11) & (-4, -31/3) | 15. $3/8 \pi \text{ cm/min}$ |
| 16. $\frac{\sqrt{2}}{4\pi} \text{ cm/s}$ | 17. $200 \pi r^3 / (r + 5)^2 \text{ km}^2 / \text{h}$ |
| 19. (i) (a) 6.05, (b) $\frac{80}{27}$; (ii) $9.72 \pi \text{ cm}^3$ | 20. $1 + 36 \pi \text{ cu. cm/sec}$ |
| 21. $1/48 \pi \text{ cm/s}$ | |
| 22. $\frac{1}{4} \text{ cm/sec.}$ | 23. (a) $-\frac{1}{24\pi} \text{ m/min.}$, (b) $-\frac{5}{288\pi} \text{ m/min.}$ |
| 24. (a) $r = (1 + t)^{1/4}$, (b) $t = 80$ | |

EXERCISE # 3 (JM)

- | | | | | | | |
|-------|---------|-------|----------|-------|-------|-------|
| 1. 1 | 2. 2, 4 | 3. 1 | 4. 4 | 5. 1 | 6. 2 | 7. 4 |
| 8. 3 | 9. 3 | 10. 3 | 11. 4 | 12. 1 | 13. 2 | 14. 2 |
| 15. 2 | 16. 2 | 17. 1 | 18. 4.00 | | | |

EXERCISE # 4 (JA)

- | | | | |
|---|------|------|------|
| 1. $\sqrt{2}x + y - 2\sqrt{2} = 0$ or $\sqrt{2}x - y - 2\sqrt{2} = 0$ | 2. D | 3. D | 4. D |
| 5. A | | | |

EXERCISE # 5

- | | | | | | |
|---------|------------|---------------|----------|----------|---------|
| 1. A, C | 2. A, C | 3. A, B, C | 4. C, D | 5. A, B | 6. A, C |
| 7. A, C | 8. A, B, C | 9. A, B, C, D | 10. A, D | 11. A, B | |

EXERCISE # 6

- | | |
|---|------------|
| 1. $2\sqrt{3}x - y = 2(\sqrt{3} - 1)$ or $2\sqrt{3}x + y = 2(\sqrt{3} + 1)$ | 2. (0, 1) |
| 3. T: $x - 2y = 0$; N: $2x + y = 0$ | 4. 20 |
| 5. 1/16 | 6. $a = 1$ |
| 7. $\frac{66}{7}$ | |

MONOTONOCITY

EXERCISE # 1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. A | 5. A | 6. A | 7. B |
| 8. C | 9. B | 10. B | 11. C | 12. C | 13. C | 14. A |
| 15. A | 16. A | 17. D | 18. C | 19. A | 20. A | 21. D |
| 22. B | 23. B | 24. D | 25. A | 26. A | | |

EXERCISE # 2

- (a) I in $(2, \infty)$ & D in $(-\infty, 2)$

(b) I in $(1, \infty)$ & D in $(-\infty, 0) \cup (0, 1)$

(c) I in $(0, 2)$ & D in $(-\infty, 0) \cup (2, \infty)$

(d) I for $x > \frac{1}{2}$ or $-\frac{1}{2} < x < 0$ & D for $x < -\frac{1}{2}$ or $0 < x < \frac{1}{2}$
- $(-2, 0) \cup (2, \infty)$
- (a) I in $[0, 3\pi/4) \cup (7\pi/4, 2\pi]$ & D in $(3\pi/4, 7\pi/4)$

(b) I in $[0, \pi/6) \cup (\pi/2, 5\pi/6) \cup (3\pi/2, 2\pi]$ & D in $(\pi/6, \pi/2) \cup (5\pi/6, 3\pi/2]$

(c) I in $[0, \pi/2) \cup (3\pi/2, 2\pi]$ and D in $(\pi/2, 3\pi/2)$
- continuous but not diff. at $x = 1$
- (a) $(\pi/6) + (1/2)\ln 3, (\pi/3) - (1/2)\ln 3$

(b) Maximum at $x = 1$ and $f(-1) = 18$; Minimum at $x = 1/8$ and $f(1/8) = -9/4$

(c) 2 and -10
- $(b-a)^{3/4}$
- $a \geq 0$
- $(-1, 0) \cup (0, \infty)$
- $c = \frac{mb+na}{m+n}$ which lies between a & b
- $a = 3, b = 4$ and $m = 1$

EXERCISE # 3 (JM)

- | | | | | | | | | | | | | | |
|-----|---|-----|---|-----|-------|-----|---|-----|---|-----|---|-----|---|
| 1. | 4 | 2. | 2 | 3. | 3 | 4. | 2 | 5. | 1 | 6. | 4 | 7. | 1 |
| 8. | 2 | 9. | 4 | 10. | 2 | 11. | 1 | 12. | 2 | 13. | 4 | 14. | 2 |
| 15. | 2 | 16. | 1 | 17. | Bonus | 18. | 4 | 19. | 1 | | | | |

EXERCISE # 4 (JA)

SECTION-1

- | | | | | | | | | | |
|----|-----------------------|----|---|----|---|----|--------------|-----|---|
| 1. | (a) B ; (b) D ; (c) C | 2. | (a) A, (b) $\cos\left(\frac{1}{3}\cos^{-1}p\right)$ | 3. | A | 4. | (a) D; (b) C | | |
| 6. | D | 7. | B | 8. | A | 9. | C | 10. | A |

SECTION-2

- | | | | | | | | | | | | |
|-----|-------------------------|-----|--------------------------------------|-----|-------|-----|-----|-----|---|-----|---|
| 13. | (i) B, (ii) A, (iii) A; | 14. | (A) P,Q,R; (B) P,S; (C) R,S; (D) P,Q | | | | | | | | |
| 15. | (a) C, (b) A, B, C, D | 16. | B, C, D | 17. | B,C | 18. | 2 | 19. | C | | |
| 20. | B | 21. | A,B,C,D | 22. | B,C | 23. | B,C | 24. | D | 25. | D |
| 26. | D | 27. | B,D | 28. | A,B,D | | | | | | |

EXERCISE # 5

- | | | | | | | | | | | | | | |
|-----|---------|----|-------|-----|---------|-----|---------|-----|---------|----|-----|----|-----|
| 1. | A,D | 2. | A,C,D | 3. | A,B,C | 4. | B,C,D | 5. | B,C | 6. | B,C | 7. | B,C |
| 8. | A,C | 9. | B,D | 10. | A,B,C,D | 11. | A,B,C,D | 12. | A,B,C,D | | | | |
| 13. | A,B,C,D | | | | | | | | | | | | |

EXERCISE # 6

- | | | | | | | | |
|----|------------------------|----|--|----|---------------|----|-----|
| 1. | $(0, 1/3) \cup (1, 5)$ | 2. | $(6, \infty)$ | 3. | $[1, \infty)$ | 4. | -10 |
| 5. | $[-7, -1) \cup [2, 3]$ | 8. | $\uparrow$ in $(3, \infty)$ and $\downarrow$ in $(1, 3)$ | | | | |

MAXIMA - MINIMA

EXERCISE # 1

- | | | | | | | |
|--------|-------|-------|-------|-------|-------|-------|
| 1. A,C | 2. C | 3. B | 4. D | 5. C | 6. B | 7. C |
| 8. C | 9. C | 10. B | 11. C | 12. C | 13. C | 14. C |
| 15. B | 16. A | 17. C | 18. A | 19. A | 20. D | 21. C |
| 22. B | 23. B | 24. D | 25. C | 26. A | | |

EXERCISE # 2

- $f(x) = x^3 + x^2 - x + 2$
- max. at $x = 1$; $f(1) = 0$, min. at $x = 7/5$; $f(7/5) = -108/3125$
- (a) Max at $x = 2\pi$, Max value = 2π , Min. at $x = 0$, Min value = 0
(b) Max at $x = \pi/6$ & also at $x = 5\pi/6$ and Max value = $3/2$, Min at $x = \pi/2$, Min value = -3
- $a = \frac{1}{4}$; $b = -\frac{5}{4}$; $f(x) = \frac{1}{4}(x^2 - 5x + 8)$
- 6
- $75\sqrt{3}$ sq. units
- $r = \sqrt{\frac{2A}{\pi+4}}$, $s = \sqrt{\frac{2A}{\pi+4}}$
- $3x + 4y - 9 = 0$; $3x - 4y + 9 = 0$.
- $1/\pi$ cu m
- 110', 70'
- side 10', height 10'
- 32 sq. units
- $a = 1$, $b = 0$
- $6' \times 18'$
- 32
- 74

EXERCISE # 3 (JM)

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|----------|
| 1. 4 | 2. 3 | 3. 4 | 4. 4 | 5. 2 | 6. 4 | 7. 1 |
| 8. 4 | 9. 3 | 10. 4 | 11. 2 | 12. 3 | 13. 4 | 14. 3 |
| 15. 4 | 16. 3 | 17. 1 | 18. 4 | 19. 1 | 20. 3 | 21. 4 |
| 22. 3 | 23. 4 | 24. 3 | 25. 2 | 26. 1 | 27. 3 | 28. 3.00 |
| 29. 4 | | | | | | |

EXERCISE # 4 (JA)

SECTION-1

- | | | | | | |
|------|-----------------|------|------|------|------|
| 1. A | 2. (a) D; (b) A | 3. A | 4. D | 5. C | 6. D |
|------|-----------------|------|------|------|------|

SECTION-2

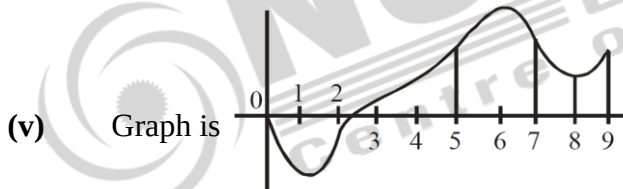
- | | | | |
|----------------------------|----------------------|------------------|-------------------------------------|
| 7. 2ab | 8. (a) (2, 1); (b) 5 | 10. $4\sqrt{65}$ | 11. (a) B, C; (b) A (c) 6 solutions |
| 12. (i) A, (ii) A, (iii) B | 13. (a) 0; (b) 7 | 14. 1 | |
| 15. 5 | 16. 9 | 17. A,C | 18. A,B |
| | | 19. 4 | 20. B, C |

EXERCISE # 5

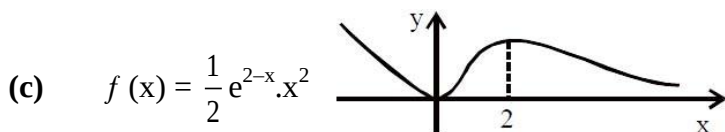
1. B,D 2. A,B,D 3. A,D 4. A,B,D 5. B,D 6. A,B,D 7. A,C
8. B,D 9. B,C,D 10. A,B,C,D 11. A,C 12. B,C
13. A,B,C 14. A,B,C,D 15. B,C,D 16. A,B,C

EXERCISE # 6

1. $P_{\max} = a \left(1 + \cos \sec \frac{\alpha}{2} \right)$ 2. $4\sqrt{2} \text{ m}$ 3. $|a - b|$
4. (a) $x = y = \frac{d}{\sqrt{2}}$, (b) $x = \frac{d}{\sqrt{3}}$, $y = \sqrt{\frac{2}{3}}d$ 5. $(0, 0)$
7. (a) f is continuous at $x = 0$; (b) $-2/e$;
(c) does not exist, does not exist; (d) pt. of inflection $x = 1$
8. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$, (b) no inflection point,
(c) maxima at $x = \pi/2$ and $-3\pi/2$ and no minima,
(d) $x = 3\pi/2$ and $x = -\pi/2$, (e) $-\pi \ln 2$
9. (i) I in $(1, 6) \cup (8, 9)$ and D in $(0, 1) \cup (6, 8)$;
(ii) L. Min. at $x = 1$ and $x = 8$; L. Max. $x = 6$
(iii) CU in $(0, 2) \cup (3, 5) \cup (7, 9)$ and CD in $(2, 3) \cup (5, 7)$;
(iv) $x = 2, 3, 5, 7$



10. $m \in \left(\frac{1}{32}, \frac{1}{16} \right)$ 11. $\pi/4$ 12. 320
13. (a) increasing in $(0, 2)$ and decreasing in $(-\infty, 0) \cup (2, \infty)$, local min. value = 0 and local max. value = 2
(b) concave up for $(-\infty, 2 - \sqrt{2}) \cup (2 + \sqrt{2}, \infty)$ and concave down in $(2 - \sqrt{2}, 2 + \sqrt{2})$



15. $\frac{4}{3\sqrt{3}}$

ADDITIONAL EXERCISE ON APPLICATION OF DERIVATIVE

[STRAIGHT OBJECTIVE TYPE]

- The minimum value of the function $f(x) = \frac{\tan\left(x + \frac{\pi}{6}\right)}{\tan x}$ is -
(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 3
- The least value of 'a' for which the equation, $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ has atleast one solution on the interval $(0, \pi/2)$ is :
(A) 3 (B) 5 (C) 7 (D) 9
- Let $g : [1, 6] \rightarrow [0, \infty)$ be a real valued differentiable function satisfying $g'(x) = \frac{2}{x + g(x)}$ and $g(1) = 0$, then the maximum value of g cannot exceed
(A) $\ln 2$ (B) $\ln 6$ (C) $6 \ln 2$ (D) $2 \ln 6$
- Let $f(x) = \left\{ b^2 + (a-1)b - \frac{1}{4} \right\} x + \int_0^x (\sin^4 t + \cos^4 t) dt$. If $f(x)$ be a non-decreasing function of $x \forall x \in \mathbb{R}$ and $\forall b \in \mathbb{R}$, then the number of positive integral values of parameter 'a' is -
(A) 1 (B) 2 (C) 3 (D) 0
- A closed vessel tapers to a point both at its top E and its bottom F and is fixed with EF vertical when the depth of the liquid in it is x cm, the volume of the liquid in it is, $x^2 (15 - x)$ cu. cm. The length EF is:
(A) 7.5 cm (B) 8 cm (C) 10 cm (D) 12 cm
- A horse runs along a circle with a speed of 20 km/hr . A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $\frac{1}{8}$ of the circle in km/hr is
(A) 20 (B) 40 (C) 30 (D) 60
- If $f(x)$ is a differentiable function such that $f'(\alpha) = f'(\beta) = 0$ & $f''(\alpha) f''(\beta) < 0$, $f(\alpha) = 3$, $f(\beta) = -1$, $(\alpha < \beta)$. If $f''(\beta) - f''(\alpha) < 0$, then minimum number of roots of $f'(x) = 0$ in $[\alpha, \beta]$ will be-
(A) 0 (B) 2 (C) 3 (D) 4

8. Number of solution(s) satisfying the equation, $3x^2 - 2x^3 = \log_2(x^2 + 1) - \log_2 x$ is :
(A) 1 (B) 2 (C) 3 (D) none
9. Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals
(A) $-\frac{2\sqrt{3}}{9}$ (B) $-\frac{2}{9}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{1}{3\sqrt{3}}$
10. Let $f(x) = x - \frac{1}{x}$ then which one of the following statement is true
(A) Function is invertible if defined from $\mathbb{R} - \{0\} \rightarrow \mathbb{R}$.
(B) $f(x_1) > f(x_2)$, $\forall x_1 > x_2$ and $x_1, x_2 \neq 0$.
(C) Graph of the function has exactly one asymptote.
(D) Function is one-one in every continuous interval $[a, b]$ defined on one side of origin.
11. If the function $f(x) = \frac{t + 3x - x^2}{x - 4}$, where 't' is a parameter has a minimum and a maximum then the range of values of 't' is
(A) (0, 4) (B) (0, ∞) (C) ($-\infty$, 4) (D) (4, ∞)
12. The equation $x \ln x + 1 = kx$ has -
(A) exactly one solution for $k = 2$ (B) exactly two solutions for $k = -2$
(C) exactly one solution for $k = 1$ (D) exactly two solutions for $k = -1$
13. P and Q are two points on a circle of centre C and radius α , the angle PCQ being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when
(A) $\sin \theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\sin \theta = \frac{\sqrt{5}-1}{2}$ (C) $\sin \theta = \frac{\sqrt{5}+1}{2}$ (D) $\sin \theta = \frac{\sqrt{5}-1}{4}$
14. The range of values of m for which the line $y = mx$ and the curve $y = \frac{x}{x^2 + 1}$ enclose a region, is
(A) $(-1, 1)$ (B) (0, 1) (C) $[0, 1]$ (D) $(1, \infty)$
15. For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be
(A) $1.25 C$ (B) $1.5 C$ (C) $1.75 C$ (D) $2 C$

16. Let $f(x) = x + \sqrt{x}$ on $[1, 4]$. The mean value theorem says that there must be some number 'c' between 1 and 4 so that $f'(c)$ is equal to the average slope of $f(x)$ on $[1, 4]$ the number 'c' must be
- (A) $\frac{5}{2}$ (B) $\frac{9}{4}$ (C) $\frac{11}{4}$ (D) 3
17. Read the following mathematical statements carefully :
- I. A differentiable function ' f ' with maximum at $x = c \Rightarrow f''(c) < 0$.
- II. Antiderivative of a periodic function is also a periodic function.
- III. If f has a period T then for any $a \in \mathbb{R}$, $\int_0^T f(x) dx = \int_0^T f(x+a) dx$
- IV. If $f(x)$ has a maxima at $x = c$, then ' f ' is increasing in $(c-h, c)$ and decreasing in $(c, c+h)$ as $h \rightarrow 0$ for $h > 0$.
- Now indicate the correct alternative.
- (A) exactly one statement is correct. (B) exactly two statements are correct.
(C) exactly three statements are correct. (D) All the four statements are correct.
18. If 'X' is a five digit number abcde, then maximum value of $\frac{X}{a+b+c+d+e}$ will be :-
- (A) 10000 (B) 11000 (C) 9000 (D) 9999
19. Consider $f(x) = \int_1^x \left(t + \frac{1}{t}\right) dt$ and $g(x) = f'(x)$ for $x \in \left[\frac{1}{2}, 3\right]$
- If P is a point on the curve $y = g(x)$ such that the tangent to this curve at P is parallel to a chord joining the points $\left(\frac{1}{2}, g\left(\frac{1}{2}\right)\right)$ and $(3, g(3))$ of the curve, then the coordinates of the point P
- (A) can't be found out (B) $\left(\frac{7}{4}, \frac{65}{28}\right)$
(C) (1, 2) (D) $\left(\sqrt{\frac{3}{2}}, \frac{5}{\sqrt{6}}\right)$
20. If f be a continuous function on $[0, 1]$, differentiable in $(0, 1)$ such that $f(1) = 0$, then there exists some $c \in (0, 1)$ such that
- (A) $cf'(c) - f(c) = 0$ (B) $f'(c) + cf(c) = 0$ (C) $f'(c) - cf(c) = 0$ (D) $cf'(c) + f(c) = 0$

21. The value of k for which the equation $e^{2x} = k\sqrt{x}$ has exactly one solution, is
 (A) $\frac{1}{\sqrt{e}}$ (B) $\sqrt{e}$ (C) $2\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$
22. Given that $f(x)$ is continuously differentiable on $a \leq x \leq b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?
 (i) $f(x)$ is bounded on $a \leq x \leq b$.
 (ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$.
 (iii) The maximum and minimum values of $f(x)$ on $a \leq x \leq b$ occur at points where $f'(c) = 0$.
 (iv) There is at least one point c with $a < c < b$ where $f'(c) > 0$.
 (v) There is at least one point d with $a < d < b$ where $f'(c) < 0$.
 (A) only (ii) and (iv) are true (B) all but (iii) are true
 (C) all but (v) are true (D) only (i), (ii) and (iv) are true
23. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then
 (A) f has exactly one stationary point (B) f must have no stationary point.
 (C) f must have exactly 2 stationary points (D) f has either 0 or 2 stationary points.
24. Let $f(x) = \begin{cases} -x^2 & \text{for } x < 0 \\ x^2 + 8 & \text{for } x \geq 0 \end{cases}$. Then x intercept of the line that is tangent to both branches of $f(x)$ is
 (A) zero (B) -1 (C) -2 (D) -4

[REASONING TYPE]

25. Consider the following statements
Statement-1 : The function $f(x) = x^3 + x$ is continuous, monotonic and has a range equal to $\mathbb{R}$. (set of real numbers)
because
Statement-2 : If $f(x)$ is continuous and monotonic then its range is equal to $\mathbb{R}$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

26. **Statement-1** : The number 'c' in the domain of function f such that either $f'(c) = 0$ or $f'(c)$ fails to exist imply that $f(x)$ has local extremum at $x = c$.
because
Statement-2 : If $f(x)$ has local extremum at $x = c$ then either $f'(c) = 0$ or $f'(c)$ does not exist.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
27. **Statement-1** : Suppose $f(x) = 2^x + 1$ and $g(x) = 4^{-x} + 2^{-x}$. The equation $f(x) = g(x)$ has exactly one root.
because
Statement-2 : If $f(x)$ and $g(x)$ are two differentiable functions defined for all $x \in \mathbb{R}$ and if $f(x)$ is strictly increasing and $g(x)$ is strictly decreasing for every $x \in \mathbb{R}$ then the equation $f(x) = g(x)$ must have exactly one root.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
28. **Statement-1** : If $f(x)$ is differentiable in $[0, 1]$ such that $f(0) = f(1) = 0$, then for any $\lambda \in \mathbb{R}$, there exists c such that $f'(c) = \lambda f(c)$, $0 < c < 1$.
because
Statement-2 : If $g(x)$ is differentiable in $[0, 1]$ where $g(0) = g(1)$, then there exists c such that $g'(c) = 0$, $0 < c < 1$.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

29. Let $f(x)$ be a twice differentiable function.
Statement-1 : If $a < b < c < d$ and $f(a) = 0$, $f(b) = 1$, $f(c) = -1$, $f(d) = 0$, then the minimum number of zeroes $g(x) = (f'(x))^2 + f(x)f''(x)$ in $[a, d]$ is 4.
because
Statement-2 : If $f(\alpha)f(\beta) < 0$, then $f(\gamma) = 0$ for some $\alpha < \gamma < \beta$ and if $f(\alpha) = f(\beta) = 0$ then $f'(\gamma) = 0$ for some $\alpha < \gamma < \beta$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
30. $f : \mathbb{R} \rightarrow \mathbb{R}$
Statement-1 : $f(x) = 12x^5 - 15x^4 + 20x^3 - 30x^2 + 60x + 1$ is monotonic and surjective on $\mathbb{R}$.
because
Statement-2 : A continuous function defined on $\mathbb{R}$, if strictly monotonic has its range $\mathbb{R}$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
31. **Statement-1** : For the function $f(x) = x^2 + 3x + 2$, LMVT is applicable in $[1, 2]$ and the value of c is $3/2$.
because
Statement-2 : If LMVT is known to be applicable for any quadratic polynomial in $[a, b]$ then c of LMVT is $\frac{a+b}{2}$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

32. Let $f(x) = \ln(2+x) - \frac{2x+2}{x+3}$.

Statement-1 : The equation $f(x) = 0$ has a unique solution in the domain of $f(x)$.

because

Statement-2 : If $f(x)$ is continuous in $[a, b]$ and is strictly monotonic in (a, b) then f has a unique root in (a, b)

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

33. Consider the functions $f(x) = x - \sin x$ and $g(x) = x^4$.

Statement-1 : $f(x)$ is increasing for $x > 0$ as well as $x < 0$ and $g(x)$ is increasing for $x > 0$ and decreasing for $x < 0$.

because

Statement-2 : If an odd function is known to be increasing on the interval $x > 0$ then it is increasing for $x < 0$ also and if an even function is increasing for $x > 0$ then it is decreasing for $x < 0$.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true

34. A function $y = f(x)$ is defined on $[0, 4]$ as

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ (x-2)^2 & \text{if } 1 < x \leq 3 \\ 1 & \text{if } 3 < x \leq 4 \end{cases}$$

For the function $y = f(x)$

Statement-1 : All the three conditions of Rolles Theorem are violated on $[0, 4]$ but still $f'(x)$ vanishes at a point in $(0, 4)$.

because

Statement-2 : The conditions for Rolles Theorem are sufficient but not necessary.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

35. **Statement-1** : $f(x) = \log_{e+x}(\pi + x)$ is strictly increasing for all $x > 0$.
because
Statement-2 : $\pi + x > e + x \forall x > 0$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
36. Consider the graph of the function $f(x) = x + \sqrt{|x|}$
Statement-1 : The graph of $y = f(x)$ has only one critical point
because
Statement-2 : $f'(x)$ vanishes only at one point
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
37. Consider the following statements.
Statement-1: The function $f(x) = \ln x$ is increasing in $(0, 10)$ and $g(x) = \frac{1}{x}$ is decreasing in $(0, 10)$
because
Statement-2: If a differentiable function increases in the interval (a, b) then its derivative function decreases in the interval (a, b) .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

38. Consider the following statements.

Statement-1 : The function $f(x) = x \sin x$ and $f'(x) = x \cos x + \sin x$ are both aperiodic.
because

Statement-2 : The derivative of differentiable aperiodic function is an aperiodic function.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT a correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

[COMPREHENSION TYPE]

Paragraph for question nos. 39 to 41

Let $f(x) = \left(1 + \frac{1}{x}\right)^x$ ($x > 0$) and $g(x) = \begin{cases} x \ln(1 + (1/x)) & \text{if } 0 < x \leq 1 \\ 0 & \text{if } x = 0 \end{cases}$

39. $\lim_{x \rightarrow 0^+} g(x)$

- (A) is equal to 0 (B) is equal to 1 (C) is equal to e (D) is non existent

40. The function $f(x)$

- (A) has a maxima but no minima (B) has a minima but no maxima
(C) has both a maxima and minima (D) is monotonic

41. $\lim_{n \rightarrow \infty} \left\{ f\left(\frac{1}{n}\right) \cdot f\left(\frac{2}{n}\right) \cdot f\left(\frac{3}{n}\right) \cdots f\left(\frac{n}{n}\right) \right\}^{1/n}$ equals

- (A) $\sqrt{2}e$ (B) $\sqrt{2}e$ (C) $2\sqrt{e}$ (D) $\sqrt{e}$

Paragraph for Question 42 & 43

Consider a real valued function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f\left(\frac{2x+3y}{5}\right) = \frac{2f(x)+3f(y)}{5} \forall x, y \in \mathbb{R}$
and $f'(0) = 2$; $f(0) = 1$.

42. Minimum distance of a point on graph of $y = f(x)$ from origin is less than or equal to -

- (A) $\frac{1}{\sqrt{2}}$ units (B) $\frac{1}{2}$ units (C) $\frac{1}{\sqrt{5}}$ units (D) $\frac{1}{\sqrt{7}}$ units

43. Which of the following is true about f ?

- (A) f is monotonic $\forall x \in \mathbb{R}$ (B) f is periodic function
(C) f is an odd function (D) f is a bijection

Paragraph for Question 44 & 45

Consider $f(x) = 2 - (4 - (x - 1)^2)^{1/2}$.

44. Interval(s) of monotonicity for $f(x)$ is/are -
 (A) $x \in (-1, 1)$ (B) $x \in (-1, 2)$ (C) $x \in (0, 3)$ (D) $x \in (1, 3)$
45. Which of the following line(s) can be tangent to the curve $y = f(x)$?
 (A) $y = 2x - 2\sqrt{5}$ (B) $y = 2x + 2\sqrt{5}$
 (C) $y = -3x + 5 + 2\sqrt{10}$ (D) $y = -3x + 5 - 2\sqrt{10}$

Paragraph for Question 46 to 47

Let $f(x) = \left| \sin x + \frac{2}{3 + \sin x} - 1 \right|$.

46. Number of integers in the range of $f(x)$ is equal to-
 (A) 0 (B) 1 (C) 2 (D) 3
47. If 'M' and 'm' denotes the number of local maxima and local minima of $f(x)$ on interval $(0, 10\pi)$, then 'M + m' is equal to
 (A) 10 (B) 15 (C) 20 (D) 5

Paragraph for question nos. 48 to 50

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in \mathbb{R}$.

48. For $a = 1$ if $y = f(x)$ is strictly increasing $\forall x \in \mathbb{R}$ then maximum range of values of b is
 (A) $\left(-\infty, \frac{1}{3}\right]$ (B) $\left(\frac{1}{3}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $(-\infty, \infty)$
49. For $b = 1$, if $y = f(x)$ is non monotonic then the sum of all the integral values of $a \in [1, 100]$, is
 (A) 4950 (B) 5049 (C) 5050 (D) 5047
50. If the sum of the base 2 logarithms of the roots of the cubic $f(x) = 0$ is 5 then the value of 'a' is
 (A) -64 (B) -8 (C) -128 (D) -256

[MULTIPLE OBJECTIVE TYPE]

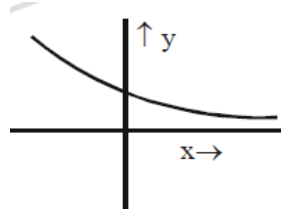
51. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$.
 Number of possible distinct intersection(s) of these graph is
 (A) 0 (B) 2 (C) 3 (D) 4

52. If $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{\lceil a^{|x|} \operatorname{sgn} x \rceil}$ for $a > 1$ and $x \in \mathbb{R}$, where $\{.\}$ & $\lceil . \rceil$ denote the fractional part and integral part functions respectively, then which of the following statements can hold good for the function $h(x)$, where $(\ln a) h(x) = (\ln f(x) + \ln g(x))$.
 (A) 'h' is even and increasing (B) 'h' is odd and decreasing
 (C) 'h' is even and decreasing (D) 'h' is odd and increasing.
53. On which of the following intervals, the function $x^{100} + \sin x - 1$ is strictly increasing.
 (A) $(-1, 1)$ (B) $(0, 1)$ (C) $(\pi/2, \pi)$ (D) $(0, \pi/2)$
54. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is correct?
 (A) $f(x)$ has minima but no maxima.
 (B) $f(x)$ increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$.
 (C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$.
 (D) $x = 3$ is the point of inflection.
55. $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true ?
 (A) $f(x)$ has a relative minimum on $(-3, 4)$
 (B) $f(x)$ has a minimum on $[-3, 4]$
 (C) $f(x)$ is concave upwards on $[-3, 4]$
 (D) if $f(3) = f(4)$ then $f(x)$ has a critical point on $[-3, 4]$
56. Consider the function $f(x) = x^2 - x \sin x - \cos x$ then the statements which holds good, are
 (A) $f(x) = k$ has no solution for $k < -1$.
 (B) f is increasing for $x < 0$ and decreasing for $x > 0$.
 (C) $\lim_{x \rightarrow \pm\infty} f(x) \rightarrow \infty$
 (D) The zeros of $f(x) = 0$ lie on the same side of the origin.
57. Assume that inverse of the function f is denoted by g . Then which of the following statement hold good?
 (A) If f is increasing then g is also increasing.
 (B) If f is decreasing then g is increasing.
 (C) The function f is injective.
 (D) The function g is onto.
58. For the function $f(x) = \ln(1 - \ln x)$ which of the following do not hold good?
 (A) increasing in $(0, 1)$ and decreasing in $(1, e)$
 (B) decreasing in $(0, 1)$ and increasing in $(1, e)$
 (C) $x = 1$ is the critical number for $f(x)$.
 (D) f has two asymptotes

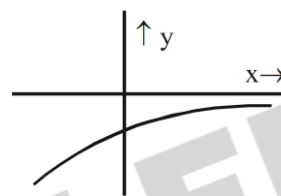
59. The function $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$
- (A) is continuous for all $x \in \mathbb{R}$
 (B) is continuous but not differentiable $\forall x \in \mathbb{R}$
 (C) is such that $f'(x)$ change its sign exactly twice
 (D) has two local maxima and two local minima.
60. Which of the following statements is/are TRUE ?
- (A) If f has domain $[0, \infty)$ and has no horizontal asymptotes then $\lim_{x \rightarrow \infty} f(x) = \infty$ or $\lim_{x \rightarrow \infty} f(x) = -\infty$.
 (B) If f is continuous on $[-1, 1]$ and $f(-1) = 4$ and $f(1) = 3$ then there exist a number r such that $|r| < 1$ and $f(r) = \pi$.
 (C) $\lim_{x \rightarrow \infty} \arcsin\left(\frac{x+1}{x}\right)$ does not exist.
 (D) For all values of $m \in \mathbb{R}$ the line $y - mx + m - 1 = 0$ cuts the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ orthogonally.
61. Consider the function $f(x) = \left[\cos\left(\tan^{-1}\left(\sin(\cot^{-1}x)\right)\right) \right]^2$. Which of the following is correct ?
- (A) range of f is $(0, 1)$ (B) f is even
 (C) $f'(0) = 0$ (D) the line $y = 1$ is asymptotes to the graph $y = f(x)$
62. Let $f(x) = ax^3 + bx^2 + cx + 1$ have extrema at $x = \alpha, \beta$ such that $\alpha\beta < 0$ and $f(\alpha) \cdot f(\beta) < 0$, then which of the following can be true for the equation $f(x) = 0$?
- (A) three equal roots
 (B) three distinct real roots
 (C) one positive root if $f(\alpha) < 0$ and $f(\beta) > 0$
 (D) one negative root if $f(\alpha) > 0$ and $f(\beta) < 0$.
63. A function f is defined by $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good?
- (A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$
 (B) Maximum value of f is π
 (C) There exists atleast one $c \in (0, 2\pi)$ s.t. $f'(c) = 0$.
 (D) Minimum value of f is $-\frac{\pi}{2}$.

64. Which of the following pair(s) of curves touch each other?
 (A) $y^2 = 4x$ and $x^2 + y^2 = 6x - 1$ (B) $xy = 4$ and $x^2 + y^2 = 8$
 (C) $y = 2x - 3$ and $y = x^3 - x + 1$ (D) $y = 6 + x - x^2$ and $y = \frac{x+2}{x-2}$
65. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(x)f'(x) < 0 \forall x \in \mathbb{R}$, then which of the following is/are correct -

(A) The possible graph of $f(x)$ may be



(B) The possible graph of $f(x)$ may be



- (C) $|f(x)|$ is always decreasing function
 (D) $|f(x)|$ is always increasing function

66. Consider a real valued continuous function $f(x)$ defined on the interval $[a, b]$. Which of the following statements does not hold(s) good?

- (A) If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x) dx \leq \int_a^b f^2(x) dx$
 (B) If $f(x)$ is increasing on $[a, b]$, then $f^2(x)$ is increasing on $[a, b]$.
 (C) If $f(x)$ is increasing on $[a, b]$, then $f(x) \geq 0$ on (a, b)
 (D) If $f(x)$ attains a minimum at $x = c$ where $a < c < b$, then $f'(c) = 0$.

67. If $f(x)$ is continuous and derivable $\forall x \in \mathbb{R}$ and $f'(c) = 0$ for exactly 2 real values of 'c' then the number of real and distinct values of 'd' for which $f(d) = 0$ can be

- (A) 1 (B) 2 (C) 3 (D) 4

68. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x + \sin x$. Which of the following is/are the correct statement(s)?

- (A) The function is strictly increasing at every point on $\mathbb{R}$ except at 'x' equal to an odd integral multiple of π where the derivative of $f(x)$ is zero and where the function f is not strictly increasing.
 (B) The function is bounded in every bounded interval but unbounded on whole real line.
 (C) The graph of the function $y = f(x)$ lies in the first and third quadrants only.
 (D) The graph of the function $y = f(x)$ cuts the line $y = x$ at infinitely many points.

[SUBJECTIVE TYPE]

69. A straight line is drawn through the origin and parallel to the tangent to a curve $\frac{x + \sqrt{a^2 - y^2}}{a} = \ln \left(\frac{a + \sqrt{a^2 - y^2}}{y} \right)$ at an arbitrary point M. Show that the locus of the point P of intersection of the straight line through the origin & the straight line parallel to the x-axis & passing through the point M is $x^2 + y^2 = a^2$.
70. If the tangent at the point (x_1, y_1) to the curve $x^3 + y^3 = a^3$ ($a \neq 0$) meets the curve again in (x_2, y_2) then show that $\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$.
71. The tangent at a variable point P of the curve $y = x^2 - x^3$ meets it again at Q. Show that the locus of the middle point of PQ is $y = 1 - 9x + 28x^2 - 28x^3$.
72. A curve is given by the equations $x = at^2$ & $y = at^3$. A variable pair of perpendicular lines through the origin 'O' meet the curve at P & Q. Show that the locus of the point of intersection of the tangents at P & Q is $4y^2 = 3ax - a^2$.
73. A and B are points of the parabola $y = x^2$. The tangents at A and B meet at C. The median of the triangle ABC from C has length 'm' units. Find the area of the triangle in terms of 'm'.
74. An open Can of oil is accidentally dropped into a lake; assume the oil spreads over the surface as a circular disc of uniform thickness whose radius increases steadily at the rate of 10 cm/sec. At the moment when the radius is 1 meter, the thickness of the oil slick is decreasing at the rate of 4 mm/sec, how fast is it decreasing when the radius is 2 meters.
75. (a) Find the set of all values of the parameter 'a' for which the function, $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$ increases for all $x \in \mathbb{R}$ and has no critical points for all $x \in \mathbb{R}$.
(b) Find the set of values of 'a' for which x the function $f(x) = (a^2 + a - 6)\cos 2x + (a - 2)x + \cos 1$ has no critical points.
76. Let $f(x) = \left(1 + \frac{1}{x}\right)^x$ and $g(x) = \left(1 + \frac{1}{x}\right)^{x+1}$, both f and g being defined for $x > 0$, then prove that f(x) is increasing and g(x) is decreasing.

77. Suppose that the function $f(x) = \log_c \frac{x-2}{x+2}$ is defined for all x in the interval $[a, b]$ is monotonic decreasing. Find the value of 'c' for which there exists 'a' and 'b' ($b > a > 2$) such that the range of the function is $[\log_c(b-1), \log_c(a-1)]$.
78. Prove that $\tan^2 x + 6 \operatorname{In} \sec x + 2 \cos x + 4 > 6 \sec x$ for $x \in \left(\frac{3\pi}{2}, 2\pi\right)$
79. Let f defined on $[0, 1]$ be a twice differentiable function such that, $|f''(x)| \leq 1$ for all $x \in [0, 1]$. If $f(0) = f(1)$, then show that, $|f'(x)| < 1$ for all $x \in [0, 1]$
80. For a given curved surface of a right circular cone when the volume is maximum, prove that the semi vertical angle is $\sin^{-1} \frac{1}{\sqrt{3}}$.
81. A window of fixed perimeter (including the base of the arch) is in the form of a rectangle surmounted by a semicircle. The semicircular portion is fitted with coloured glass while the rectangular part is fitted with clean glass. The clear glass transmits three times as much light per square meter as the coloured glass does. What is the ratio of the sides of the rectangle so that the window transmits the maximum light?
82. A trapezium ABCD is inscribed into a semicircle of radius l so that the base AD of the trapezium is a diameter and the vertices B & C lie on the circumference. Find the base angle θ of the trapezium ABCD which has the greatest perimeter.
83. From a fixed point A on the circumference of a circle of radius 'a', let the perpendicular AY fall on the tangent at a point P on the circle, prove that the greatest area which the ΔAPY can have is $3\sqrt{3} \frac{a^2}{8}$ sq. units.
84. Let α, β be real numbers with $0 \leq \alpha \leq \beta$ and $f(x) = x^2 - (\alpha + \beta)x + \alpha\beta$ such that $\int_{-1}^1 f(x) dx = 1$.
Find the maximum value $\int_0^\alpha f(x) dx$.
85. Show that for each $a > 0$ the function $e^{-ax} \cdot x^{a^2}$ has a maximum value say $F(a)$, and that $F(x)$ has a minimum value, $e^{-e/2}$.

86. Find the positive value of k for the value of the definite integral $\int_0^{\pi/2} |\cos x - kx| dx$ is minimised.
87. Which is the radius of the smallest circular disk large enough to cover every acute isosceles triangle of a given perimeter L ?
88. Find the magnitude of the vertex angle ' α ' of an isosceles triangle of the given area ' A ' such that the radius ' r ' of the circle inscribed into the triangle is the maximum.
89. Let $f(x) = e^{(p+1)x} - e^x$ for real number $p > 0$.
 (a) Find the value of $x = s_p$ for which $f(x)$ is minimum.
 (b) Let $g(t) = \int_t^{t+1} f(x) e^{t-x} dx$. Find the value if $t = t_p$, for which $g(t)$ minimum.
 (c) Use the fact that $1 + \frac{p}{2} \leq \frac{e^p - 1}{p} \leq 1 + \frac{p}{2} + p^2$ ($0 < p \leq 1$) find the value of $\lim_{p \rightarrow 0^+} (s_p - t_p)$.
90. Find the cosine of the angle at the vertex of an isosceles triangle having the greatest area for the given constant length l of the median drawn to its lateral side.

ANSWER KEY

1. D 2. D 3. D 4. B 5. C 6. B 7. D
8. A 9. A 10. D 11. C 12. C 13. B 14. B
15. B 16. B 17. A 18. A 19. D 20. D 21. C
22. D 23. D 24. B 25. C 26. D 27. C 28. A
29. A 30. C 31. A 32. C 33. A 34. A 35. D
36. D 37. C 38. C 39. A 40. D 41. D
42. A,B,C 43. A,D 44. A,D 45. A,D 46. C 47. C 48. C
49. B 50. D
51. B,C,D 52. D 53. B,C,D 54. B,C,D
55. B,C,D 56. A,C 57. A,C,D 58. A,B,C
59. A,B,D 60. B,C,D 61. B,C,D 62. B,C,D
63. C,D 64. A,B 65. A,B,C 66. A,B,C,D
67. A,B,C 68. B,C,D 73. $\frac{m\sqrt{m}}{\sqrt{2}}$ 74. 0.05 cm/sec
75. (a) $a < -(2 + \sqrt{5})$ or $a > \sqrt{5}$ (b) $a \in \left(-\frac{7}{2}, -\frac{5}{2}\right)$ 77. $0 < c < \frac{1}{9}$
81. $6/(6 + \pi)$ 82. $\theta = 60^\circ$ 84. $\frac{\sqrt{6}}{108}$
86. $k = \frac{2\sqrt{2}}{\pi} \cos\left(\frac{\pi}{2\sqrt{2}}\right)$ 87. $L/4$ 88. $\pi/3$
89. (a) $-\frac{\ln(p+1)}{p}$; (b) $-\frac{1}{p} \ln\left(\frac{(p+1)(e^p-1)}{p}\right)$; (c) $\frac{1}{2}$ 90. $\cos A = 0.8$

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018

AIR
29



Sarthak Behera

AIR
36



Shashank Agrawal

AIR
46



Shreya Pathak

AIR
48



Raaghav Raaj

AIR
2
SC



Aayush Kadam

AIR
18
OBC



Gunjan

Result Highlights 2017

AIR
12



Abhay Goyal

AIR
14



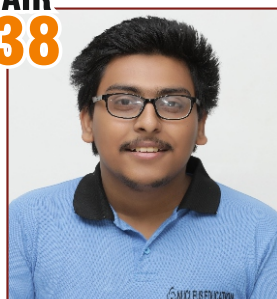
Piyush Tibarwal

AIR
20



Abhinav Kumar

AIR
38



Tushar Gautam

AIR
49



Saurav Sahoo

AIR
60



Pratyush

Mentored by Team

AIR
4



Kushal Babel
2014

AIR
15



Gourika Bang
2015

AIR
21



Amit Malav
2013

AIR
24



Suraj Sanjay Jog
2012

AIR
38



Chitraang Murdia
2014

AIR
58



Pulkit Tandon
2012

AIR
68



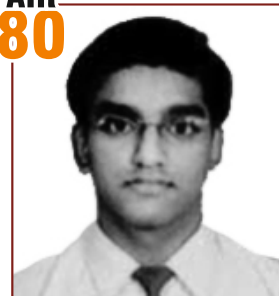
Kapil Shobhnani
2016

AIR
70



Utkarsh Sinha
2015

AIR
80



S John S
2015